

MTH111-General Mathematics I

DEPARTMENT OF MATHEMATICS

FEDERAL UNIVERSITY LAFIA

Purpose

Introduction

This is the digitized lecture notes of **General Mathematics I** of the Department of Mathematics, Federal University Lafia. MTH 111 is a first semester 3-credit course for first year undergraduate students of Science. The course is broadly divided in six modules. Each module begins with a clear statement of learning objectives, outcomes and an introduction. This is followed by relevant definitions, remarks and examples. Students are encouraged to solve each worked example as an unseen. Those who successfully do so will match their solution against the one provided. But those who fail to do so can after an attempt go through the solution provided. Students are expected to explore the notes and exercises, individually and in groups, allowing the lectures to be delivered essentially in tutorial mode. Relevant questions can be raised in class thereafter. The notes is complemented with solved problems; past examination questions with model answers and marking schemes. Group assignments and laboratory sessions are part of the learning process. Exercises have been introduced for practice and also to provide a variety of related problems illustrating applications.

1 Elementary Set Theory and Number Systems

1.1 Objectives

In this module:

- We define a set and list some examples. Standard notation used in set theory is then introduced with a collection of some commonly used sets.

- The basic set operations are defined and explored. Set operations are used to convert one or more sets into new sets, resulting in the algebra of sets. Venn diagrams are introduced to visualise operations on sets.
- Relations are introduced as subsets of a Cartesian product and the domain, range and inverse of a relation defined and explored. The reflexive, symmetric and transitive properties of relations are discussed and the concept of a function is introduced through the study of relations.
- The number system is then explored from a set theoretic perspective.

1.2 Learning Outcomes

At the end of this module, students should be able to:

- Define a set and identify collections of objects that are sets and those which are not.
- List commonly used sets and explore problems using the algebra of sets.
- Identify different subsets of the real numbers and perform set operations with them.

1.3 Learning Activities

Students should:

- Explore notes, power point presentations and exercises, individually and in groups;
- Engage in discussions with lecturers and their peers using forums and other interactive facilities on the Moodle site;
- Take and pass on-line and other self assessment tests.
- Solve relevant questions from past MTH111 Examinations.

1.4 Sets and Set Notation-Introduction

The theory of sets is an important tool in Mathematics. The study of sets which began in the nineteenth century was championed by George Cantor (1845-1918). It has assumed a

central role in every branch of Mathematics today. For example, algebra is concerned with sets of numbers and operations on these sets whereas analysis deals with sets of functions.

A set is a well defined collection of objects. A collection of objects is well defined if given any object, one is able to decide whether it belongs to the collection or not. Every object that belongs to a set is called a member or an element of the set.

For example, the collection of all traders in Lafia forms a set and any trader in Lafia is a member of this set. A teacher in Lafia is however not a member of this set. Another example of a set is the collection of all students in Nigerian Universities. A student who is not in any of the Nigerian Universities is not a member of this set. A teacher in Lafia is also not a member of this set. We use upper case letters such as $A, B, C, \dots$ to denote sets whereas the lower case letters such as $a, b, c, \dots$ are used to denote members of a set. The statement $x \in A$ means " x is a member of, or an element of a set A ; or x belongs to the set A ". The statement " $y \notin A$ " means that y is not a member of, or is not an element of A ". It also means y does not belong to the set A .

1.4.1 Methods of describing Sets

Roster method:

The most common method of describing a set is by listing its members. This is conventionally done by enclosing them in a pair of braces, $\{\}$. In this convention, each element is separated by a comma. For example, the set X whose elements are the months of the year that begin with the letter A is $X = \{April, August\}$. The set of integers whose squares is less than 20 can be written as $A = \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$

The Set-Builder Notation:

The set-builder notation (or the rule method) is a method of specifying a set in which rather than listing all the elements of a set, we specify the set by writing within braces a formal description that describes or identifies the members of the set. As part of this formal description, we use a variable (such as x) to represent any one of the members of the set. The following convention is used in a set-builder notation: Name of Set = $\{x|x \text{ is word description of the set}\}$ or Name of Set = $\{x:x \text{ is a word description of the set}\}$. Note that the slash (/) or the colon (:) is read as "such that".

Now considering the example we mentioned earlier, that is the set X whose elements are the months of the year that begin with the letter A, we can describe the set using the set-builder notation as follows: $X = \{x : x \text{ is a month of the year that begins with the letter A}\}$. The statement is read: X is the set of all x such that x is a month of the year that begins with the letter A.

Consider the set whose elements are the days in a week. Such a set can be described using the set-builder notation as: $A = \{x|x \text{ is a day of the week}\}$. If we use the roster notation discussed earlier the set can be written as: $A = \{\text{Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday}\}$.

Exercise 1 Using the roster method, describe the following sets:

1. $A = \{x : x \text{ is a month of the year}\}.$
2. $B = \{x : x^2 - 3x + 2 = 0\}.$
3. $C = \{x : x \text{ is a continent}\}.$
4. The set of integers between 1 and 50 divisible by 9.

Exercise 2 Use the set-builder notation to describe the following sets.

1. $A = \{a, e, i, o, u\}.$
2. $B = \{-3, -2, -1, 0, 1, 2, 3, 4, 5, 6\}.$

1.4.2 Types of Sets

A Finite Set is a set that has a finite number of elements. The following are examples of finite sets,

$A = \{1, 2, 3, 4, 5, 6\}$ is a set with 6 elements.

$B = \{-6, -5, -4, \dots, 4, 5, 6\}$ is a set with 13 elements.

$C = \{a, b, x, y, z\}$ is a set with 5 elements.

Definition 1.1 *An Infinite Set is a set that has an infinite number of elements. The following are examples of infinite sets.*

$$X = \{1, 2, 3, \dots\}; \quad Y = \{\dots, -4, -3, -2, -1, 0, 1, 2, 3, 4, \dots\}$$

The three dots indicate that the established pattern of writing down the members of the set will continue indefinitely.

Definition 1.2 *The Empty Set or void or Null Set. A set that has no element in it is called the empty or the null set. The empty or null set is denoted by the symbol $\emptyset$ or $\{\}$. We must exercise caution not to write $\{\emptyset\}$ to represent the null set. $\{\emptyset\}$ is a set with one element (the empty set), and is therefore non empty.*

One way to describe the elements of the empty set is to take "the set of traders who live on the moon". This is the empty set because no trader lives on the moon. Another way to describe it is to take " the set of real numbers whose square is negative". It is important to note that there is only one empty set but it can be described in many different ways. That is why we talk about THE empty set and not AN empty set.

Definition 1.3 *The Singleton Set or the Unit Set.* Any set that has exactly one element is called a singleton set or a unit set.

Consider the set of all months with less than 30 days. This set has only one element, namely, February. It can be listed as follows $\{February\}$. The following are also singleton sets

1. $A = \{garri\}$.
2. $B = \{2\}$.

Definition 1.4 : A set X is said to be a subset of set Y if every element of set X is an element of set Y .

We use the notation $X \subseteq Y$ to describe X is a subset of Y ; so that the symbol $\subseteq$ should be read as “is a subset of”. We also say that X is contained in Y when X is a subset of Y .

The set X is said to be a proper subset of Y if every element of set X is an element of set Y , and also the set Y has at least one other element that is not in X . We denote this by $X \subset Y$. This symbol is read as “ X is a proper subset of Y ”. We can write $Y \supset X$ to mean the same thing which can be read as Y contains X .

Given the set $X = \{-3, -2, -1, 0, 1, 2, 3\}$ and $Y = \{-4, -3, -2, -1, 0, 1, 2, 3, 4, 5\}$, set $X \subset Y$ since every element of the set X is in Y and in addition the set Y has some elements that are not in X .

Definition 1.5 : Two sets are said to be equal if they have exactly the same elements

Equivalently two sets A and B are equal if $A \subseteq B$ and $B \subseteq A$. Let $X = \{a, b, f, g\}$ and $Y = \{f, a, b, g\}$ then the set X is equal to set Y and we write $X = Y$.

The order in which the elements of a set are written is not important. Elements can also be repeated without having any effect. The set $A = \{\text{all the vowels in English Alphabet}\}$ and the set $B = \{e, o, a, u, i\}$ are equal since both sets have the same elements. To prove that $A = B$ it suffices to show that $A \subseteq B$ and $B \subseteq A$.

Let $A = \{a, b, c, d\}$ and $B = \{a, b, f, d, e, c, g, h\}$. Every element in set A , i.e. each of, a, b, c and d is also contained in set B . Notice further that the two sets are not equal; so $A \subset B$. The set $X = \{\text{All students taking MTH 111}\}$ is a subset of the set $Y = \{\text{All FULafia students}\}$. In particular, it should be noted that every set is a subset of itself and the empty set is a subset of any given set. X is a proper subset of Y as there are FULafia students not reading MTH 111.

Definition 1.6 : *Family of Sets or Class of Sets:* A Set whose members or elements are sets is referred to as a family of sets or a class of sets. This is so to avoid using the confusing phrase - "a set of sets".

Definition 1.7 : The Power Set: Given a set A , the power set of A denoted $\rho(A)$ is the family of all subsets of A .

For example the power set of the set $\{x, y\}$ is $\rho(\{x, y\}) = \{\emptyset, \{x\}, \{y\}, \{x, y\}\}$. Generally, for all integers $n \geq 0$, if a set X has n elements, then the power set $\rho(X)$ has 2^n elements. This result can be proved by considering the number of subsets containing no elements, 1 element, 2 elements and so on until we get to subsets containing n elements.

Definition 1.8 : Cardinality of a Set: The cardinality of a set or the cardinal number of any set A is defined as the number of elements in set A . This is denoted by the symbol $n(A)$, read as "the number of elements in set A ." The set $X = \{1, 2, 4\}$ for example has cardinality or cardinal number 3. That is $n(X) = 3$. $A = \{x, y, z, \text{Moses}, \text{Yam}\}$ has cardinality $n(A) = 5$

The set $Y = \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$ has cardinal number $n(Y) = 9$. We note that all the examples above are finite sets.

Definition 1.9 : Equivalent Sets: Two sets A and B are said to be equivalent if they can be put into one to one correspondence with each other.

Consider the two sets $A = \{2, 4, 6\}$ and $B = \{a, b, c\}$. By making 2 correspond to a , 4 to b , and 6 to c , we have established a one to one correspondence between sets A and B . The 2 sets are therefore equivalent. They are however not equal. As finite sets they both have exactly three elements. If we now consider the sets $A = \{1, 2, 3, \dots\}$ and $B = \{2, 4, 6, \dots\}$, we see that a one to one correspondence is established between the sets A and B , using a correspondence between n and $2n$. The two infinite sets are equivalent and have the same number of elements. The two sets are not equal.

Definition 1.10 The Universal set: Whenever we are discussing sets, we employ the use of a background set called the universal set. This universal set contains all the elements that are under consideration.

Thus the set $A = \{2, 3\}$ can be viewed from the perspective that its universal set U is given by $U = \{1, 2, 3, \dots\}$. Here all elements under consideration will be the natural numbers which is the background set. Every element of $A = \{2, 3\}$ is a natural number. If we consider the set of negative integers $B = \{-1, -2, -3, \dots\}$ and the set of positive integers, $C = \{1, 2, 3, \dots\}$, then a suitable universal set is the set of integers, $U = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$.

1.4.3 Set operations

Given two sets A and B we can combine them to form a third set in several different ways. We illustrate this with definitions and examples below.

Definition 1.11 : *Union of two Sets:* The union of two sets A and B denoted $A \cup B$ (read as “ A union B ”) is the collection of all elements that are either in A or B or in both A and B . Union is called a binary operation as it combines 2 sets at a time to give a third set. Using mathematical notation, we write $A \cup B = \{x : x \in A \text{ or } x \in B\}$. This “or” is called the inclusive or and prescribes that x belongs to A or B or both A and B .

We explore $A \cup B$ for given sets.

If $A = \{1, 2, 5, 6\}$ and $B = \{0, 1, 2, 3, 5\}$, then $A \cup B = \{1, 2, 5, 6\} \cup \{0, 1, 2, 3, 5\} = \{0, 1, 2, 3, 5, 6\}$

If $A = \{\text{Orange, Apple, John}\}$ and $B = \{\text{Orange, Yam, Goat, Rice}\}$ then $A \cup B = \{\text{Orange, Apple, John, Yam, Goat, Rice}\}$.

Given three sets A , B , and C and the binary operation of union, we can form $(A \cup B)$ and then combine the new set with C to get $(A \cup B) \cup C$. We can also first form $B \cup C$ and then combine it with A to get $A \cup (B \cup C)$.

We explore some simple examples to confirm that $(A \cup B) \cup C = A \cup (B \cup C)$. This result allows us to write $A \cup B \cup C$ meaningfully.

Remark 1 If $A = A_1, B = A_2$ and $C = A_3$ we get: $A_1 \cup A_2 \cup A_3 = \bigcup_{i=1}^3 A_i$. More generally if $A_1, A_2, \dots, A_n$ are n sets, then their union is the set of all objects which belong to at least one of them and it is denoted by $A_1 \cup A_2 \cup \dots \cup A_n = \bigcup_{i=1}^n A_i$. The union of infinite sets, $A_1, A_2, \dots, A_n \dots$ is given by $A_1 \cup A_2 \cup \dots \cup A_n \cup \dots = \bigcup_{i=1}^{\infty} A_i$.

Here are some basic properties on the union of sets which we shall jointly explore.

1. $A \cup \phi = A$
2. $A \cup B = B \cup A$ Commutativity law for union
3. $A \cup (B \cup C) = (A \cup B) \cup C$ Associativity law for union
4. $A \cup A = A$ Idempotent law.
5. $A \cup U = U$ Universal Union
6. $A \subset B$ if and only if $A \cup B = B$

The proof of one of these basic properties is given and the others are discussed.

We show that $A \cup B = B \cup A$.

Proof: Take $x \in A \cup B$ and show that $x \in B \cup A$.

$x \in A \cup B$ Arbitrary choice

$\Rightarrow x \in A$ or $x \in B$, from definition of union

$\Rightarrow x \in B$ or $x \in A$, definition of union

$\Rightarrow x \in B \cup A$. Thus $A \cup B \subseteq B \cup A$.

Similarly we can show that $B \cup A \subseteq A \cup B$ and then conclude that $A \cup B = B \cup A$.

Definition 1.12 Intersection of Sets: The intersection of two sets A and B , denoted as $A \cap B$ (read as "A intersection B") is the set of all elements that are in both sets A and B . Mathematically we write $A \cap B = \{x : x \in A \text{ and } x \in B\}$.

Now if $A = \{-1, 0, 2, 3, 4\}$ and $B = \{0, 2, 4, 6, 8\}$ then $A \cap B = \{0, 2, 4\}$.

The intersection of n sets $A_1, A_2, \dots, A_n$ is the set of all objects which belong to every one of them, and is denoted by $A_1 \cap A_2 \cap \dots \cap A_n = \bigcap_{i=1}^n A_i$. If the number of sets is infinite, that is,

$A_1, A_2, \dots, A_n, \dots$ then we write $A_1 \cap A_2 \cap \dots \cap A_n \cap \dots = \bigcap_{i=1}^{\infty} A_i$

The basic properties about union operation, as well as their proofs are similar to the basic properties of intersection operation. This is called the duality property, and the basic dual results are given below.

1. $A \cap \phi = \phi$ Null intersection.
2. $A \cap B = B \cap A$ Commutativity law for intersection.
3. $A \cap (B \cap C) = (A \cap B) \cap C$ Associativity law for intersection.
4. $A \cap U = A$ Universal Intersection.
5. $A \cap A = A$ Idempotent law.
6. $A \subset B$ if and only if $A \cap B = A$.

There are also laws which involve both union, $\cup$ and intersection, $\cap$. They show that union is distributive over intersection and the dual.

Theorem 1 Thus for any subsets A, B , and C of some universal set U ,

1. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
2. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

These identities are called the distributive laws. We shall prove the first identity by showing that

$$[A \cap (B \cup C)] \subseteq [(A \cap B) \cup (A \cap C)] \text{ and conversely, i.e.}$$

$$[(A \cap B) \cup (A \cap C)] \subseteq [A \cap (B \cup C)].$$

Proof: First we show, $[A \cap (B \cup C)] \subseteq [(A \cap B) \cup (A \cap C)]$

Take $x \in [A \cap (B \cup C)]$ and show that $x \in [(A \cap B) \cup (A \cap C)]$.

$x \in [A \cap (B \cup C)]$ Arbitrary choice

$\Rightarrow x \in A$ and $x \in (B \cup C)$, definition of intersection

$\Rightarrow x \in A$ and $x \in B$ or $x \in C$, definition of union

$\Rightarrow x \in A$ and $x \in B$ or $x \in A$ and $x \in C$

$\Rightarrow x \in (A \cap B)$ or $x \in (A \cap C)$ definition of intersection

$\Rightarrow x \in [(A \cap B) \cup (A \cap C)]$, definition of union

Thus $[A \cap (B \cup C)] \subseteq [(A \cap B) \cup (A \cap C)] \dots (i)$

Conversely,

let $x \in (A \cap B) \cup (A \cap C)$ and show that $x \in A \cap (B \cup C)$.

$x \in (A \cap B) \cup (A \cap C)$ Arbitrary choice

$\Rightarrow x \in (A \cap B)$ or $x \in (A \cap C)$ definition of union

$\Rightarrow x \in A$ and $x \in B$ or $x \in A$ and $x \in C$ definition of intersection

$\Rightarrow x \in A$ and $x \in B$ or $x \in C$ definition of union

$\Rightarrow x \in A$ and $x \in (B \cup C)$.

$\Rightarrow x \in A \cap (B \cup C)$. Thus $[(A \cap B) \cup (A \cap C)] \subseteq [A \cap (B \cup C)] \dots (ii)$

From the two results (i) and (ii) above, equality follows.

Theorem 2 *The general distributive laws are:*

$$(i) A \cup (B_1 \cap B_2 \cap \dots \cap B_n) = (A \cup B_1) \cap (A \cup B_2) \cap \dots \cap (A \cup B_n).$$

$$(ii) A \cap (B_1 \cup B_2 \cup \dots \cup B_n) = (A \cap B_1) \cup (A \cap B_2) \cup \dots \cup (A \cap B_n).$$

These can be proved by the principle of mathematical induction (which will be discussed later in the course).

Definition 1.13 : Disjoint Sets: Two sets are called disjoint (or mutually exclusive) if and only if they have no elements in common. Symbolically, A and B are disjoint if and only if $A \cap B = \phi$.

Let $A = \{3, 7, 9\}$ and $B = \{4, 8, 10\}$. Then A and B are disjoint sets since $A \cap B = \phi$ (they have no element in common).

In general, given n sets $A_1, A_2, \dots, A_n$ they are said to be mutually disjoint (pairwise disjoint or non overlapping) if and only if no two different sets A_i and A_j have any elements in common. More precisely, for all $i, j = 1, 2, \dots, n$, $A_i \cap A_j = \phi$, where $i \neq j$.

Let $A_1 = \{2, 4, 7, 8\}$, $A_2 = \{1, 3, 6\}$, and $A_3 = \{5, 9\}$. A_1 , A_2 and A_3 are mutually disjoint since A_1 and A_2 have no elements in common, A_1 and A_3 have no elements in common, and A_2 and A_3 have no elements in common.

Definition 1.14 : Partition of Sets: A collection of non empty sets $\{A_1, A_2, \dots, A_n\}$ is a partition of set A if and only if $A = A_1 \cup A_2 \cup \dots \cup A_n$ and $A_1, A_2, \dots, A_n$ are mutually disjoint.

Let $A = \{-3, -2, -1, 0, 1, 2, 3, 4, 5, 6, 7\}$, $A_1 = \{-3, -2, -1\}$, $A_2 = \{0, 1, 2\}$, $A_3 = \{3, 4\}$ and $A_4 = \{5, 6, 7\}$.

The family of sets $\{A_1, A_2, A_3, A_4\}$ is a partition of A since $A = A_1 \cup A_2 \cup A_3 \cup A_4$ and the sets A_1, A_2, A_3, A_4 are mutually disjoint.

Definition 1.15 : Complementation: The complement of set A with respect to the universal set U , denoted by A' or A^c (read as “ A complement”) is the set of all elements in the universal set that are not in A . Thus if the universal set $U = \{1, 2, 4, 8, 9, 10, 11\}$ and $A = \{2, 4, 8, 9\}$ then $A^c = \{1, 10, 11\}$.

If the universal set $U = \{\text{all men}\}$ and $Y = \{\text{all men who are married}\}$, then $Y^c = \{\text{all men who are not married}\}$

Theorem 3 The following are some basic facts about complementation. Let A and B be subsets of the universal set U then

1. $(A')' = A$.
2. $\emptyset' = U$ and $U' = \emptyset$.
3. $A \cap A' = \emptyset$ and $A \cup A' = U$.
4. $A \subset B$ if and only if $B' \subset A'$.
5. $(A \cup B)' = A' \cap B'$.
6. $(A \cap B)' = A' \cup B'$.

Statements (5) and (6) are called De-Morgan's laws. These laws are very important in the algebra of sets and can be generalised.

We now prove the De Morgan's law for any pair of sets; that is, if A and B are any two sets, $(A \cup B)' = A' \cap B'$.

Proof: Take $y \in (A \cup B)'$ Arbitrary choice

Then $y \notin (A \cup B)$ definition of complement

$\Rightarrow y \notin A$ or $y \notin B$ definition of union

$\Rightarrow y \in A'$ and $y \in B'$ definition of intersection

$\Rightarrow y \in (A' \cap B')$

Thus $(A \cup B)' \subseteq (A' \cap B')$

Conversely,

Let $y \in A' \cap B'$

$\Rightarrow y \in A'$ and $y \in B'$ definition of intersection

$\Rightarrow y \notin A$ or $y \notin B$ definition of union

$\Rightarrow y \notin (A \cup B) \Rightarrow y \in (A \cup B)'$

Thus $(A' \cap B') \subseteq (A \cup B)'$ and the result follows.

Generally, if given n sets, $A_1, A_2, \dots, A_n$, a finite collection of n sets, then

$$(A_1 \cup A_2 \cup \dots \cup A_n)' = A_1' \cap A_2' \cap \dots \cap A_n'. \text{ and } (A_1 \cap A_2 \cap \dots \cap A_n)' = A_1' \cup A_2' \cup \dots \cup A_n'.$$

Definition 1.16 : Difference of two sets: The difference between sets A and B denoted as $A - B$ (read as "A minus B ") means the set of all elements that belong to set A , but not to set B . Symbolically, $A - B = \{x : x \in A \text{ but } x \notin B\}$ and we can write $A - B = A \cap B'$.

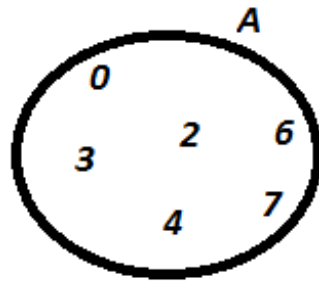
If $A = \{2, 4, 5, 8\}$ and $B = \{2, 5\}$, then $A - B = \{4, 8\}$.

Similarly, consider the universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. If $A = \{1, 2, 4, 5, 8, 9, 10\}$ and $B = \{3, 5, 6, 7, 8\}$, then the set $A - B = \{1, 2, 4, 9, 10\}$ and $A \cap B' = \{1, 2, 4, 9, 10\}$ since $B' = \{1, 2, 4, 9, 10\}$.

Note that $A - B = A \cap B'$ and observe that this is an illustration and not a proof.

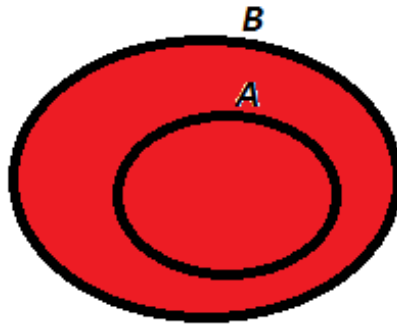
1.4.4 Representation of Sets with Venn diagrams

One important way of visualizing the relationship between subsets and the corresponding universal set is by Venn diagram. Venn diagram is therefore made up of circles or ellipses which are drawn in a rectangle to represent set S . For example, the set $A = \{0, 2, 3, 4, 6, 7\}$ can be represented by a Venn diagram below.



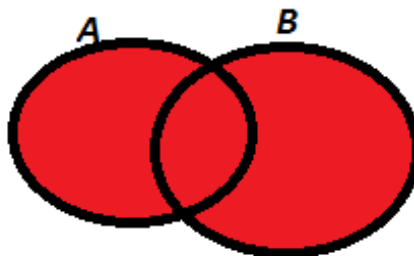
(1)

Suppose, A is a subset of B , i.e. $A \subset B$, then a Venn diagram for this relationship is as shown in the shaded figure.



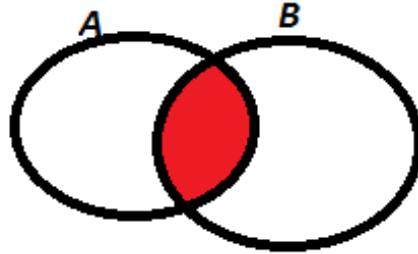
(2)

Suppose that A and B are two sets then their union and intersection can respectively be display by the shaded Venn diagrams shown below.



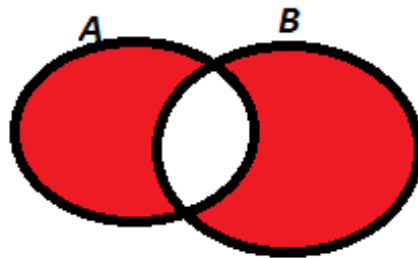
(3)

and



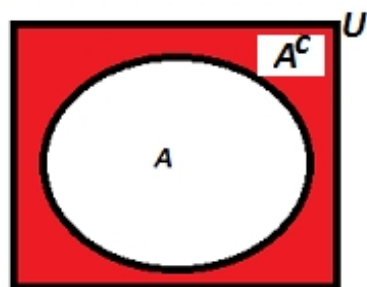
(4)

If sets A and B have no element in common, then the relationship can be shown by the unshaded portion of the Venn diagram.



(5)

The complement A^c of a set A with reference to a universal set U can be represented by the Venn diagram below.



(6)

Definition 1.17 : *Symmetric difference of Sets:* If A and B are subsets of the universal set U, then the symmetric difference (or Boolean sum) of A and B is the set $A + B$ (or $A \Delta B$) defined by $A \Delta B = (A - B) \cup (B - A)$.

An alternative way of defining symmetric difference is $A\Delta B = \{x : x \in A \text{ or } x \in B, \text{ but not both}\}$

If $A = \{1, 2, 4, 5, 8, 9, 10\}$ and $B = \{3, 5, 6, 7, 8\}$ then the symmetric difference between the two sets is

$$A + B = (A - B) \cup (B - A) = \{1, 2, 4, 9, 10\} \cup \{3, 6, 7\} = \{1, 2, 3, 4, 6, 7, 9, 10\}.$$

Theorem 4 *Two interesting identities related to De Morgan's laws for any sets A , B and C are listed herewith:*

$$(i) \quad A - (B \cup C) = (A - B) \cap (A - C); (ii) \quad A - (B \cap C) = (A - B) \cup (A - C).$$

We prove the first result.

Proof: Let $x \in A - (B \cup C)$ Arbitrary choice.

Then $x \in A$ but $x \notin (B \cup C)$.

Consequently

$x \notin B$ or C , so that $x \in A - B$ and $x \in A - C$.

But this implies that $x \in (A - B) \cap (A - C)$; hence $A - (B \cup C) \subseteq (A - B) \cap (A - C)$.

Similarly, if $x \in (A - B) \cap (A - C)$ then $x \in A - B$ and $x \in A - C$ which

implies that $x \in A$ but not B and $x \in A$ but not C . It follows that

$x \in A$ but $x \notin B$ or C . Therefore, $(A - B) \cap (A - C) \subseteq A - (B \cup C)$.

Thus $A - (B \cup C) = (A - B) \cap (A - C)$.

Definition 1.18 : ***Cartesian product of sets:** The Cartesian product (also called the cross product) of two sets A and B , denoted by $A \times B$, is defined as $A \times B = \{(x, y) : x \in A, y \in B\}$.*

Suppose that $A = \{Musa, Pam, Neshi\}$ and $B = \{a, b, c\}$ then $A \times B = \{(musa, a), (musa, b), (musa, c), (pam, a), (pam, b), (pam, c), (neshi, a), (neshi, b), (neshi, c)\}$

If $A = \{1, 2\}$ and $B = \{2, 4, 5\}$. Find $A \times B$ and $B \times A$

$$A \times B = \{(1, 2), (1, 4), (1, 5), (2, 2), (2, 4), (2, 5)\}$$

$$B \times A = \{(2, 1), (2, 2), (4, 1), (4, 2), (5, 1), (5, 2)\}$$

Definition 1.19 : ***Relation:** A relation is any subset of the Cartesian product $A \times B$ of any two sets A and B .*

If $\mathfrak{R}$ is a relation in $A \times B$ (i.e. $\mathfrak{R} \subset A \times B$) we say $\mathfrak{R}$ is a relation from A to B . If $A = B$, we simply say $\mathfrak{R}$ is a relation in A . To give an example, let A be an arbitrary set and consider

the subset $\mathfrak{R}$ of $A \times A$ consisting of all pairs (x, y) such that $x = y$. Here the relation $\mathfrak{R}$ is the relation of equality between elements of A .

If $\mathfrak{R}$ is a relation, then $(x, y) \in \mathfrak{R}$ will sometimes be written as $x\mathfrak{R}y$.

The domain of a relation $\mathfrak{R}$ is the set $Dom(\mathfrak{R}) = \{x : (x, y) \in \mathfrak{R} \text{ for some } y\}$.

The range of $\mathfrak{R}$ is the set $Ran(\mathfrak{R}) = \{y : (x, y) \in \mathfrak{R} \text{ for some } x\}$.

The inverse of $\mathfrak{R}$ is the set $\mathfrak{R}^{-1} = \{(y, x) : (x, y) \in \mathfrak{R}\}$. Thus $\mathfrak{R}^{-1}$ is obtained by reversing each pair in $\mathfrak{R}$.

Reflexive, symmetric and transitive relations

Let A be a set. A relation $\mathfrak{R}$ in A is said to be

1. Reflexive if $x\mathfrak{R}x$ for all $x \in A$;
2. Symmetric if $x\mathfrak{R}y$ implies that $y\mathfrak{R}x$ for all $x, y \in A$;
3. Transitive if $x\mathfrak{R}y$ and $y\mathfrak{R}z$ implies that $x\mathfrak{R}z$ for all $x, y, z \in A$.

Definition 1.20 : For any set A , an equivalence relation $\mathfrak{R}$ on A is a relation $\mathfrak{R} \subset A \times A$ satisfying (1), (2) and (3) above.

Definition 1.21 : Image: If $\mathfrak{R}$ is a relation and E is a set, we define the image of E under $\mathfrak{R}$ to be the set $\mathfrak{R}(E) = \{y : (x, y) \in \mathfrak{R} \text{ for some } x \in E\}$

Definition 1.22 : Single-valued Relation: A relation $\mathfrak{R}$ is single-valued if $(x, y) \in \mathfrak{R}$ and $(x, z) \in \mathfrak{R}$ imply that $y = z$; that is, if no two distinct members of $\mathfrak{R}$ can have the same first element.

If $\mathfrak{R}$ and $\mathfrak{R}^{-1}$ are single-valued, we say that $\mathfrak{R}$ is a one-to-one relation.

Definition 1.23 : Function: Let A and B be sets. A function from A to B is a single valued relation $\mathfrak{R}$ such that $Dom(\mathfrak{R}) \subset A$ and $Ran(\mathfrak{R}) \subset B$.

From now on, we shall adopt the well-established convention of using small letters f, g , for functions in the place of $\mathfrak{R}$ but it is important to bear in mind all the time that a function is a subset of the Cartesian product of two sets.

Definition 1.24 : Surjective or Onto function: Let f be a function having A as its domain and having its range contained in B . Let $x \in A$; then there is a unique $y \in B$ such that $x f y$. It is convenient to denote this unique element by the symbol $f(x)$ and call it the image of x under f or the value of f at x .

If $Dom(f) = A$ and $Ran(f) \subset B$, we shall say that f is a function on A and B and write $f : A \rightarrow B$. If $Ran(f) = B$, we say that f is a function from A onto B . or f is a surjective function.

Example 1.1 : Suppose that the set $U = \{-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ is a universal set, and A, B and C are subsets of the universal set U given by $A = \{-5, -4, 0, 1, 2, 6, 9, 10\}$, $\{-3, 0, 2, 4, 5, 6, 7, 8, \}$ and $C = \{-5, -4, -3, -2, -1, 0, 1, 2, 3\}$.

Find

1. $A \cup B$
2. $A \cup C$
3. $(A \cup B) \cap C$
4. $A - B$.
5. $(A - B) \cap C$.
6. A^c .
7. B^c .
8. $A \cap B^c$.
9. $(A \cup B)^c \cap A^c$

Solution:

1. Given that $U = \{-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and $A = \{-5, -4, 0, 1, 2, 6, 9, 10\}$, $\{-3, 0, 2, 4, 5, 6, 7, 8, \}$ and $C = \{-5, -4, -3, -2, -1, 0, 1, 2, 3\}$. Then $A \cup B = \{-5, -4, -3, 0, 1, 2, 4, 5, 6, 7, 8, 9, 10\}$.
2. $A \cup C = \{-5, -4, -3, -2, -1, 0, 1, 2, 3, 6, 9, 10\}$.
3. $(A \cup B) \cap C = \{-5, -4, -3, 0, 1, 2, 4, 5, 6, 7, 8, 9, 10\} \cap \{-5, -4, -3, -2, -1, 0, 1, 2, 3\} = \{-5, -4, -3, 0, 1, 2\}$
4. $A - B = \{-5, -4, 0, 1, 2, 6, 9, 10\} - \{-3, 0, 2, 4, 5, 6, 7, 8, \} = \{-5, -4, 1, 9, 10\}$.
5. $(A - B) \cap C = \{-5, -4, 1, 9, 10\} \cap \{-5, -4, -3, -2, -1, 0, 1, 2, 3\} = \{-5, -4, 1\}$.
6. $A^c = U - A = \{-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\} - \{-5, -4, 0, 1, 2, 6, 9, 10\} = \{-3, -2, -1, 3, 4, 5, 7, 8\}$.
7. $B^c = \{-5, -4, -2, -1, 1, 3, 9, 10\}$.
8. $A \cap B^c = \{-5, -4, 0, 1, 2, 6, 9, 10\} \cap \{-5, -4, -2, -1, 1, 3, 9, 10\} = \{-5, -4, 1, 9, 10\}$.
9. $(A \cup B)^c \cap A^c = \{-2, -1, 3\} \cap \{-3, -2, -1, 3, 4, 5, 7, 8\} = \{-2, -1, 3\}$.

Example 1.2 : Given that $X = \{2, 3\}$ and $Y = \{3, 5, 6\}$, Find $X \times Y$ and $Y \times X$.

Solution: Then $X \times Y = \{(2, 3), (2, 5), (2, 6), (3, 3), (3, 5), (3, 6)\}$
and $Y \times X = \{(3, 2), (3, 3), (5, 2), (5, 3), (6, 2), (6, 3)\}$

Example 1.3 : If $A = \{1, 2, 3, 4\}$. Find $\rho(A)$ and cardinality of A .

Solution: Given that $A = \{1, 2, 3, 4\}$.

Then, $\rho(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{4\}, \{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}, \{1, 2, 3\}, \{1, 2, 4\}, \{1, 3, 4\}, \{2, 3, 4\}, \{1, 2, 3, 4\}\}$ and the cardinality of A is given as $n(A) = 4$.

Example 1.4 : Prove each of the following 2 set identities: $(A \cup B) - C = (A - C) \cup (B - C)$ and $A \cup B = (A - B) \cup (B - A) \cup (A \cap B)$.

Proof: Let sets A , B and C be given. Then

$(A \cup B) - C = (A \cup B) \cap C'$ By alternate representation of set difference.

$= C' \cap (A \cup B)$ By commutative law for $\cap$

$= (C' \cap A) \cup (C' \cap B)$ By distributive law.

$= (A \cap C') \cup (B \cap C')$ By Commutative law for $\cap$

$= (A - C) \cup (B - C)$ By alternate representation of set difference.

For the second result, we show as follows:

$A \cup B = (A \cup B) \cap U$ By definition of intersection (where U is the universal set).

$= (A \cup B) \cap [(A' \cup B') \cup (A \cap B)]$ By definition of universal set

$= [(A \cup B) \cap U] \cap [((A' \cup B') \cap U) \cup (A \cap B)]$ By the definition of intersection.

$= [(A \cup B) \cap (B \cup B')] \cap [((A' \cap B') \cap (A \cup A')) \cup (A \cap B)]$ definition of universal set

$= [(A \cap B') \cup B] \cap [((A \cap B') \cup A') \cup (A \cap B)]$ By the distributive law.

$= [(A \cap B') \cup B] \cap [((A \cap B') \cup A')] \cup (A \cap B)$

$$\begin{aligned}
&= [(A \cap B') \cup (B \cap A')] \cup (A \cap B) \\
&= (A - B) \cup (B - A) \cup (A \cap B)
\end{aligned}$$

Example 1.5 : Simplify each of the 2 set- theoretic expression as far as possible, stating the set laws used

$$(A \cap B')' \cup B = A \cup B \text{ and } A \cap B \cap C \cup Y' \cup A' \cap C \cup B' \cap C \cup C \cap Y$$

Solution: $(A \cap B')' \cup B = A' \cup (B')' \cup B$ De Morgan's law.

$$= A' \cup B \cup B \quad \text{Double complement}$$

$$= A' \cup B \quad \text{Idempotence.}$$

The second result is given by:

$$\begin{aligned}
&A \cap B \cap C \cup Y' \cup A' \cap C \cup B' \cap C \cup C \cap Y \\
&= A \cap B \cap C \cup Y' \cup (A' \cup B' \cup Y) \cap C \quad \text{Distributivity of } \cap \text{ over } \cup \\
&= (A \cap B \cap Y' \cup (A' \cup B' \cup Y)) \cap C \quad \text{Distributivity of } \cap \text{ over } \cup \\
&= (A \cap B \cap Y' \cup (A' \cup B' \cup Y'')) \cap C \quad \text{Double complement} \\
&= (A \cap B \cap Y' \cup (A \cap B \cap Y')') \cap C \quad \text{De Morgan's law} \\
&= U \cap C \\
&= C
\end{aligned}$$

1.4.5 Application of Venn diagrams on Sets

We now illustrate the application of set operations with Venn diagrams.

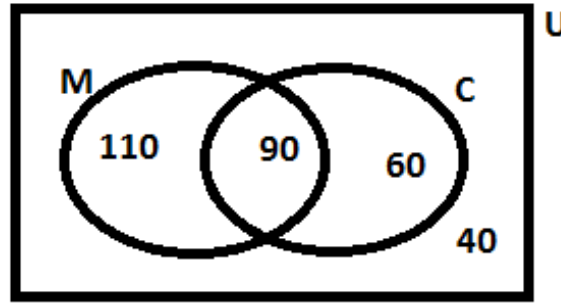
Example 1.6 : In a class of 300 students, 200 register Mathematics and 150 register Computer Science. (a) How many students registered both courses if only 40 students register neither Mathematics nor Computer Science. (b) How many students registered at least Mathematics or Computer Science.

Solution: (a) Let M and C represent Mathematics and Computer Science respectively.

Then $n(M \cup C)^c = 40$ is the number of students that register neither course. Therefore the number of students that either of the courses = $300 - 40 = 260$. Suppose x is the number of students that register both Mathematics and Computer science. Then $n(M \cap C^c) =$

$200 - x$ is the number of students that register Mathematics without Computer Science and

$n(C \cap M^c) = 150 - x$ is the number of students that register Computer Science without Mathematics. Thus the number of students that registered the two courses is given by $(200 - x) + x + (150 - x) = 260 \Rightarrow 350 - x = 260 \Rightarrow x = 90$ is the number of students that register both courses. (b) Thus $n(M \cap C^c) = 200 - 90 = 110$ is the number of students that register Mathematics only and $n(C \cap M^c) = 150 - 90 = 60$ is the number of students that register Computer Science only. Therefore the number of students that register at least one of the courses is $110 + 90 + 60$. The solution is illustrated with the Venn diagrams below.

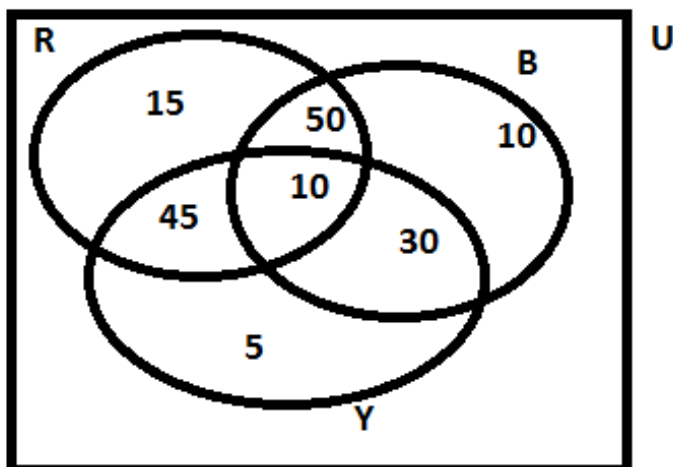


(7)

In a restaurant, the food seller prepared rice, beans and yams in a day. 120 customers were served rice, 100 were served beans and 90 yams. 60 were served rice and beans, 55 rice and yams, and 40 beans and yams. If 10 customers were served rice and beans and yams, how many people were served at least one type of the three food on that day?

Solution: Let R, B and Y represent rice, beans and yams that were served to the customers. Then $n(R) = 120$, $n(B) = 100$, and $n(Y) = 90$ represent the people that were served the three food type. Also $n(R \cap B) = 60$, $n(R \cap Y) = 55$, $n(B \cap Y) = 40$ and $n(R \cap B \cap Y) = 10$ represent the customers that were served two and the three meals respectively. Customers that were served rice and beans only $n(R \cap B \cap Y^c) = 60 - 10 = 50$; Customers that were served rice and yams only $n(R \cap B^c \cap Y) = 55 - 10 = 45$;

Customers that were served beans and yams only $n(R^c \cap B \cap Y) = 40 - 10 = 30$. Customers that were served rice only $n(R \cap B^c \cap Y^c) = 120 - 50 - 45 - 10 = 15$; Customers that were served beans only $n(R^c \cap B \cap Y^c) = 100 - 50 - 30 - 10 = 10$ and Customers that were served yams only $n(R^c \cap B^c \cap Y) = 90 - 45 - 30 - 10 = 5$. Therefore the number of customers that were served food that day is $15 + 50 + 10 + 45 + 30 + 5 + 10 = 165$. The solution is illustrated with the Venn diagrams below.



Exercise 3

- Let $U = \{-8, -7, -6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ be a universal set and let $A = \{x \in Z : -2 < x \leq 3\}$, $B = \{x \in Z : -8 \leq x \leq 5\}$, and $C = \{x \in Z : 0 < x \leq 12\}$ where Z is an integer. Find if possible (i) $A \cup B$ (ii) $B \cup C$. (iii) A^c (iv) B^c (v) C^c (vi) $(A \cup B) - C$. (vii) $(A - B) \cap (A - C)$ (viii) $(A \cup B)^c \cap (A - B)$. (ix) $(A^c \cup B^c)^c \cap C^c$. (x) $\rho(A)$. (xi) $\rho(B)$.
- Let $X = \{\text{John}, \text{Mary}\}$ and $Y = \{2, 4, 6\}$. Find (i) $X \times Y$ (ii) $Y \times Y$ (iii) $Y \times X$.
- Suppose that A , B , C and D are subsets of the universal set U . Prove that: (i) $(A - B) \cap B = \emptyset$; (ii) $B - A = B \cap A'$; (iii) $(A - B) \cap (C - D) = A \cap C - B \cup D$; (iv) $A - (B - C) = (A - B) \cup (A \cap C)$; (v) $A - (A \cap B) = A - B$; (vi) $A \Delta (B \Delta C) = (A \Delta B) \Delta C$.
- Simplify each set-theoretic expression as far as possible, naming the set laws used: (i) $[(A \cap B)' \cup (A' \cup B)] \cup [(A \cup B)' \cup (A' \cap B)]$; (ii) $[(A \cap B') \cup (A' \cup B)]' \cup [(A \cap B) \cap (A \cap B')']$.
- If A and B are subsets of the universal set U , show that (i) $A \cup (A \cap B) = A$; (ii) $A \cap (A \cup B) = A$.
- Show that the following statements about sets A and B are equivalent: (i) $A \subseteq B$ (ii) $A \cap B = A$; (iii) $A \cup B = B$.
- In class 3 of a Public primary school in Lafia, 25 pupils can correctly recite the multiplication table, 30 can correctly recite the English alphabets, and 10 can correctly recite both. If every pupil in the class can recite atleast one of the two correctly, how many pupils are in the class?

8. In a Language class of 40 students, 25 speak Hausa, 21 speak Yoruba and 16 speak Igbo and each of the students speaks at least one of the three Languages. If 8 speak Hausa and Igbo, 11 speak Hausa and Yoruba and 6 speak Igbo and Yoruba. (a) Evaluate the number of students that speak the three Languages and (b) Illustrate your solution with a Venn diagram.
9. In a survey of 290 Newspaper readers, 181 of them read ThisDay, 142 the Guardian, and 117 read the Punch and each read at least one of the three papers. If 75 read ThisDay and the Guardian, 60 read ThisDay and the Punch and 54 read the Guardian and the Punch. (a) Illustrate the above information in a Venn diagram. (b) How many readers read: (i) all three papers; (ii) exactly two of the papers; (iii) exactly one of the papers and (iv) the Guardian paper alone.
10. 1000 people were randomly selected from a large population. The following information was recorded: 449 were males, 96 were left-handed, 209 were at least 1.5metres tall, 10 had all three of the above characteristics. How many had exactly two of these three characteristics if there were 501 right-handed females under 1.5metres?

1.5 The Number System

The number system as we know it today is a result of gradual development as indicated in the presentation below.

1.5.1 Natural numbers

The natural numbers are numbers that occur commonly and obviously in nature.

This set of numbers, which was first used in counting, is denoted by N : Thus $N = \{1, 2, 3, \dots\}$. The three dots after the first few members of the set means the pattern of adding 1 to the preceeding number to get the next one continues indefinitely. The symbols used to describe the members of N have varied with the times. For example, the Romans used I, II, III, IV, ...

In mathematical expressions, lowercase, italicized letters represent unknown or unspecified numbers and the most common for the natural numbers is n .

Addition of natural numbers

If a and b are natural numbers, then their sum $a + b$ is also a natural number, for example $5 + 3 = 8$. For this reason the set of natural numbers is said to be closed under the operation of addition.

Multiplication of natural numbers

Given any two natural numbers a and b , their product, $a \times b$, $a.b$ or ab is also a natural number.

For instance $2 \times 3 = 6$. This therefore means that the natural numbers are closed with respect to multiplication.

It should be observed that the order of the terms in the products just like the summands has no influence on the result; for example $5 \times 4 = 4 \times 5 = 20$ and $6 \times 8 = 8 \times 6 = 48$. Since this property of the summands or products holds for all natural numbers, one can write that $a + b = b + a$ or $ab = ba$ and these are known as the commutative law of addition and multiplication, respectively.

It should be noted that the difference, $a - b$ between any two natural numbers a and b , is not always a natural number, e.g. $3 - 7 = -4$. In other words, the closure property on subtraction $-$, of the natural numbers does not hold.

The natural numbers can be **ordered** in the sense that given any two natural numbers n_1 and n_2 , exactly one of the following relations holds: $n_1 < n_2$ i.e n_1 is less than n_2 , for example $10 < 35$, OR $n_1 = n_2$ i.e n_1 is equal to n_2 for example $48 = 48$ OR $n_1 > n_2$ i.e n_1 is greater than n_2 , for example $95 > 40$.

These order relations have a property called transitivity; for example if $n_1 < n_2$ and $n_2 < n_3$ then $n_1 < n_3$. Also if $n_4 > n_5$ and $n_5 > n_6$ then $n_4 > n_6$.

This relationship orders the natural numbers linearly. The illustration of this linear order is the number line otherwise called the number ray.

On this number ray, the natural numbers are represented by a set of isolated (discrete) points. The fact that n_1 is less than n_2 means that the point on the number ray corresponding to n_1 lies to the left of the point corresponding to n_2 .

The set $\mathbf{N}$ is called a countable set. (sometimes the term denumerable is used). Any other set which can be put in one to one correspondence with N is also called countable.

Countability refers to the fact that, even though there might be an infinite number of elements in the set, those elements can be (COUNTED). Essentially that is equivalent to putting them in an orderly arranged list or in one to one correspondence with the elements of N .

1.5.2 Integers

Integers arose to permit solutions of equations such as $x + b = a$, where a and b are any natural numbers. The integers therefore consist of the numbers $\{0, 1, 2, 3, \dots\}$ and their negatives $\{-1, -2, -3, \dots\}$. The set of all integers is usually denoted by $\mathbf{Z}$ which stands for *Zahlen* (German name for "number").

Integers can be added, subtracted, multiplied, and ordered (or compared). The ordering is given by $\dots < -2 < -1 < 0 < 1 < 2 < \dots$

Introducing the negative integers makes it possible to solve all equations of the form $a + x = b$ (where a and b are natural numbers (constants) for the unknown x . If x is constrained to a natural numbers, only some of these equations are solvable.

We call an integer positive if it is greater than zero; zero itself is neither considered to be positive nor negative.

The order is compatible with the algebraic operations in the following way:

1. If $a < b$ and $c < d$, then $a + c < b + d$.
2. If $a < b$ and $c > 0$, then $ac < bc$.

1.5.3 Rational numbers

In mathematics, a rational number (or informally a fraction) is a ratio of two integers, usually written as $\frac{a}{b}$, where b is not zero.

The operations of rational numbers are guided by the following rules:

1. $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$, and
2. $\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$.

Two rational numbers $\frac{a}{b}$ and $\frac{c}{d}$ are equal if and only if $ad = bc$.

The set of rational numbers is denoted by Q . Rational numbers such as $\frac{3}{4}, -\frac{7}{2}, \frac{1}{5}, \dots$, arose from the need to find solutions of equations such as $bx = a$ for all integers a and b where $b \neq 0$. This leads to the operation of division or inverse of multiplication, and we write $x = \frac{a}{b}$ or $a \div b$ (called the quotient of a and b) where a is the numerator and b is the denominator. Each rational number can be written in many forms, for example $\frac{3}{6} = \frac{2}{4} = \frac{1}{2}$. The simplest form is when a and b are coprime; i.e. they have no common factors, and every rational number has a simplest form of this type. The decimal expansion of a rational number is either finite or eventually periodic, and this property characterises rational numbers.

It should be noted that the set of integers is a subset of the set of rational numbers, since the integers correspond to rational numbers $\frac{a}{b}$ where $b = 1$. The set of rational numbers is closed under the operations of addition, multiplication, subtraction and division, so long as division by zero is excluded. If r and t are rational numbers such that $r < t$, then there exists a (infinitely many) rational number s such that $r < s < t$. This is true no matter how small the difference between r and t , as long as the two are not equal.

1.5.4 Irrational numbers

An irrational number is any number that is not a rational number, i.e. a number that cannot be expressed as $\frac{a}{b}$ with a and b integers and b not zero. The irrational numbers are precisely those numbers whose decimal expansion never ends and never enters a periodic pattern. Some irrational numbers are algebraic numbers such as $\sqrt{2}$ (the square root of two) and $\sqrt[3]{3}$ (the cube root of 3) but others are not; for instance e and π .

Example 1.7 : Prove that $\log_2 3$ and $\sqrt{2}$ are irrational numbers.

Proof: Take $\log_2 3$. The argument we present is one by contradiction (*reductio ad absurdum*).

We suppose $\log_2 3$ is rational. Then for some positive integers m, n we can write $\log_2 3 = \frac{m}{n}$. Consequently $2^{\frac{m}{n}} = 3$. So $2^m = 3^n$. But 2^m is even (because at least one of its prime factors is 2) while 3^n is odd (because none of its prime factors is 2 (they are 3)). That is impossible. This therefore implies that $\log_2 3$ is not rational. When a number is not rational we say it is irrational.

Proof: We now consider $\sqrt{2}$. The discovery of irrational numbers is usually attributed to Pythagoras or one of his followers, who produced a (most likely geometrical) proof of the irrationality of the square root of 2.

One proof of this irrationality is by *reductio ad absurdum*-by the method of contradiction.

The proposition is proved by assuming the opposite and showing that, it leads to a conclusion which is false. We conclude therefore that the proposition must be true.

Assume that $\sqrt{2}$ is a rational number. Meaning that there exists integers a and b so that $\frac{a}{b} = \sqrt{2}$.

Then $\sqrt{2}$ can be written as an irreducible fraction $\frac{a}{b}$ (the fraction is simplified as much as possible) such that a and b are coprime (i.e. a and b have no common factors other than 1).

Squaring both sides implies that $(\frac{a}{b})^2 = 2$.

It follows that $\frac{a^2}{b^2} = 2$ or $a^2 = 2b^2$. Therefore a^2 is even because it is equal to $2b^2$ which is obviously even.

It follows that a must be even. (Odd numbers have odd squares and even numbers have even squares.)

Because a is even, there exists an integer k such that $a = 2k$

We substitute $a = 2k$ in the equation $a^2 = 2b^2$ giving $(2k)^2 = 2b^2$ and we get $4k^2 = 2b^2$ or $b^2 = 2k^2$.

It then follows that b^2 is also even which means a and b are both even, contradicting the fact that $\frac{a}{b}$ is irreducible.

Since we have found a contradiction from the assumption that $\sqrt{2}$ is a rational number, Our assumption is therefore false.

We have proved that $\sqrt{2}$ is irrational. This proof can be generalized.

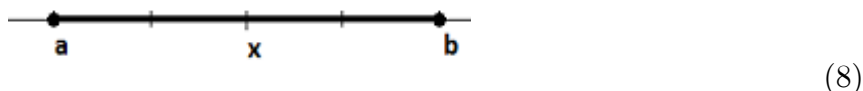
1.5.5 Real Numbers

The real numbers $\mathbb{R}$ is made up of the rational and irrational numbers, i.e. $\mathbb{R} = \mathbb{Q} \cup \mathbb{Q}^c$.

If x and z are real numbers such that $x < z$, then there is always a real number y such that $x < y < z$. In working with real numbers, it is often convenient to have a geometrical representation before us or in our minds. This representation of numbers by points takes the form of a line called the **number line** or **ideal ruler**. To do this, we draw a line L which is assumed to extend indefinitely in both directions. A reference point is chosen arbitrarily on L and labeled 0. The point 0 divides the line L into two equal half-lines, the half to the right of 0 is designated positive and the half to the left of 0 is designated negative. The point 0 is called the origin. Suppose now that P is any point on L distinct from 0 and suppose that the distance from 0 to P equals the real number x (which must be positive since distance is always nonnegative), then we take P to represent the number x or the number $-x$ according as P lies on the negative or positive half of L . In either case, given the point P on the line L , the real number x or $-x$ is uniquely determined.

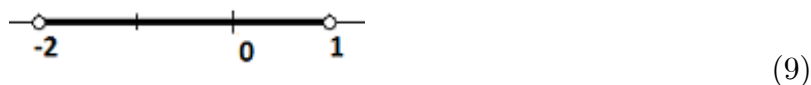
Conversely, given a real number $x > 0$, we shall assume (at least from intuition) that there is one and only one point P lying to the right of 0 such that the distance of P from 0 is x . If $x < 0$, then P is assumed to lie to the left of 0. In this way we obtain a one to one correspondence between the set of all points on L and the set of all real numbers (the point 0 corresponding to the real number zero). In view of this correspondence, no distinction need to be made between the line L and the real number system $\mathbb{R}$ and for this reason we often use point and real number interchangeably.

We shall denote the set of real numbers by $\mathbb{R}$. There are some sets of real numbers called intervals. An interval is the set of all points between given points a and b . If an interval contains the points a and b then such an interval is called a closed interval denoted by $[a, b]$. More formally, if $a \in \mathbb{R}$ and $b \in \mathbb{R}$, with $a < b$, then $[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$ is called a closed interval. This interval can be represented geometrically as



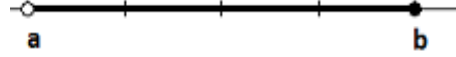
The shaded part constitutes the closed interval $[a, b]$. For example, the set $[1, 2] = \{x \in \mathbb{R} : 1 \leq x \leq 2\}$ is a closed interval. Its graphical representation can be shown.

On the other hand if the interval does not contain its end points a and b , then such an interval is called an open interval. We write $(a, b) = \{x \in \mathbb{R} : a < x < b\}$ to indicate that this interval is open. Graphically an open interval of the form (a, b) can be represented by the following example: The open interval $(-2, 1) = \{x \in \mathbb{R} : -2 < x < 1\}$ and is the set of real numbers lying between -2 and 1 . Its graphical representation is given below.



Intervals that include only one of the end points are called half-open intervals (sometimes

referred to as half-closed). We write $[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$ and $(a, b] = \{x \in \mathbb{R} : a < x \leq b\}$ to denote these intervals respectively. A half open interval of the form $(a, b]$ can be represented as follows:



(10)

The lower end of the interval is open whereas the upper end is closed.

Sets of the form $[a, \infty) = \{x \in \mathbb{R} : x \geq a\}$ are referred to as unbounded intervals. Other unbounded intervals are $(a, \infty) = \{x \in \mathbb{R} : x > a\}$; $(-\infty, b] = \{x \in \mathbb{R} : x \leq b\}$; $(-\infty, b) = \{x \in \mathbb{R} : x < b\}$ and $(-\infty, \infty) = \{x \in \mathbb{R} : -\infty < x < \infty\}$

1.5.6 Elementary operations on set of real numbers

Elementary set operations such as the union of sets, intersection of sets, and complementation of a set can be performed on sets of real numbers. We illustrate these with some examples.

Example 1.8 : Let $A = \{x \in \mathbb{R} : -2 \leq x \leq 20\}$; $B = \{x \in \mathbb{R} : 0 \leq x \leq 10\}$; and $C = \{x \in \mathbb{R} : -15 < x \leq 5\}$. Find if possible (i) $A \cup B$, (ii) $A \cap B$, (iii) C'

Solution:

$$A \cup B = \{x \in \mathbb{R} : -2 \leq x \leq 20\} \cup \{x \in \mathbb{R} : 0 \leq x \leq 10\} = \{x \in \mathbb{R} : -2 \leq x \leq 20\}.$$

$$A \cap B = \{x \in \mathbb{R} : -2 \leq x \leq 20\} \cap \{x \in \mathbb{R} : 0 \leq x \leq 10\} = \{x \in \mathbb{R} : 0 \leq x \leq 10\}.$$

In a similar way, the complement of the set C denoted as C' or C^c is given by $C' = (-\infty, -15] \cup (5, \infty)$.

We can now find more complicated combinations: $(A \cup B)' \cap C$ and $A \cap C'$ as follows:

Since $A \cup B = \{x \in \mathbb{R} : -2 \leq x \leq 20\}$ it follows that $(A \cup B)' = \{x \in \mathbb{R} : -\infty < x < -2\} \cup \{x \in \mathbb{R} : 20 < x < \infty\} = (-\infty, -2) \cup (20, \infty)$.

Now using alternative notation $A' = (-\infty, -2) \cup (20, \infty)$ and $B' = (-\infty, 0) \cup (10, \infty)$ giving $A' \cap B' = (-\infty, -2) \cup (20, \infty)$

$(A \cup B)' = A' \cap B'$ as expected from De Morgan's laws.

Similarly $(A \cup B)' = (-\infty, -2) \cup (20, \infty)$. Therefore $(A \cup B)' \cap C = \{(-\infty, -2) \cup (20, \infty)\} \cap (-15, 5]$

$(A \cup B)' \cap C = \{(-\infty, 0) \cap (-15, 5]\} \cup \{(20, \infty) \cap (-15, 5]\}$ by distributive laws

$(A \cup B)' \cap C = (-15, -2)$. This can be written in another form as $(A \cup B)' \cap C = \{x \in \mathbb{R} : -15 < x < -2\}$.

To determine $A \cap C'$, recall that $C' = (-\infty, -15] \cup (5, \infty)$. Therefore, $A \cap C' = [-2, 20] \cap \{(-\infty, -15] \cup (5, \infty)\}$

$$= \{[-2, 20] \cap (-\infty, -15)\} \cup \{[-2, 20] \cap (5, \infty)\} = [-2, -15] \cup (5, 20).$$

1. If $A = \{x \in \mathbb{R} : -12 \leq x < 30\}$, $B = \{x \in \mathbb{R} : -5 \leq x \leq 35\}$ and $C' = \{x \in \mathbb{R} : -5 < x < 13\}$. Determine: A' ; B' ; C' ; $(A \cap B)'$; $(A \cup B)'$; $(C \cup B) \cap A'$; $(A \cap B) \cup C'$; and $A' \cap (B' \cup C)$
2. Suppose N is the set of natural numbers and that the sets E, F are defined as $E = \{x \in N \cup \{0\} : 0 \leq x \leq 35\}$ and $F = \{x \in N : 2 \leq x \leq 100\}$. Determine: $(E \cap G) \cup F$; $(E' \cup F') \cap E$; $E \setminus \{0\}$ and $(E - G) \cup (E \Delta G)$
3. Given that Z , the set of integers is the universal set, and the sets X, Y and W are defined as $X = \{x \in Z : -15 \leq x \leq 70\}$, $Y = \{x \in Z : -10 < x < 10\}$ and $W = \{x \in Z : -17 < x < 23\}$. Find: X' ; Y' ; W' ; $(X \cup Y)'$; $(X \cup W) \cap Y'$; $X - Y$ and $(X \Delta Y) - W$.

2 Mathematical induction and Binomial theorem

2.1 Learning Objectives

In this module:

- We introduce the principle of mathematical induction and consolidate its use as a method of proof.
- We expose students to the expansion of expressions of the form $(a+x)^n$ in several ways specifying the best option for use when n is a large natural number.
- We demonstrate the use of the binomial expansions of expressions with fractional and negative powers.

2.2 Learning Outcomes

At the end of this module, students should be able to:

- Identify when and how to use mathematical induction as a method of proof.
- Expand expressions using the binomial theorem for positive, negative and fractional indices.

2.3 Learning Activities

Students should:

- Explore notes, power point presentations and exercises, individually and in groups.
- Engage in discussions with lecturers and their peers using forums and other interactive facilities on the Moodle site.
- Take and pass online and other self assessment tests.
- Solve relevant questions from past MTH111 Examinations.

2.4 Mathematical Induction

Introduction

It is a common knowledge that whereas experimental scientist search for evidence to support a theory, mathematicians search for a proof to validate a statement. The Principle of Mathematical Induction is one of the famous ways of establishing mathematical proofs. Mathematical induction is a technique for proving propositions or statements about the natural numbers. It also provides a uniform framework for studying the natural numbers. We will now formally state the Principle of Mathematical Induction.

2.4.1 The Principle of Mathematical Induction

The principle of mathematical induction is used to prove that a proposition $P(n)$ is true for all positive integers n , or for all positive integers $n \geq k_0$.

We do so in the following way: We first prove that $P(1)$ is true. We then assume $P(k)$ is true, and use the proof and the assumption to show that $P(k + 1)$ is true. It follows then that $P(n)$ is true for every positive integer n .

The first step (i.e prove that $P(1)$ is true) is usually called the basis step and the second step assumes $P(k)$ is true, and then show that $P(k + 1)$ is true) is called the inductive step.

What the inductive step does is to assert that if the process has reached some stage, then it will go one stage further, and the result follows.

Example 2.1 : *Use the principle of Mathematical induction to show that*

$$\sum_{r=1}^n r = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

Solution: Let $P(n)$ be the statement that

$$\sum_{r=1}^n r = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

Basis step: Since $\text{RHS} = \sum_{r=1}^1 r = 1 = \frac{1(1+1)}{2} = \text{LHS}$. This means that $P(1)$ is true.

Inductive step: Assume that the $P(k)$ is true and proceed to prove that $P(k+1)$ is true.

By the inductive hypothesis

$$\sum_{r=1}^k r = 1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2}$$

We now show that

$$\sum_{r=1}^{k+1} r = 1 + 2 + 3 + \dots + k + (k+1) = \frac{(k+1)(k+2)}{2}.$$

$$\sum_{r=1}^{k+1} r = 1+2+3+\dots+k+(k+1) = \sum_{r=1}^k r + (k+1) = \frac{k(k+1)}{2} + (k+1) = \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+1)(k+2)}{2}$$

Since the basis step and the inductive step have been verified, the principle of mathematical induction tells us that $P(n)$ is true for all positive integers n .

Example 2.2 :

Use the principle of Mathematical induction to show that

$$1.2 + 2.3 + 3.4 + \dots + n(n+1) = \frac{n(n+1)(n+2)}{3}$$

and that

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$$

Solution: Let $P(n)$ be the proposition:

$$\sum_{r=1}^n r(r+1) = 1.2 + 2.3 + 3.4 + \dots + n(n+1) = \frac{n(n+1)(n+2)}{3}$$

Basis Step: Note that $\text{LHS} = \sum_{r=1}^1 r(r+1) = 1 \cdot 2 = \frac{1(1+1)(1+2)}{3} = \frac{1(2)(3)}{3} = 2 = \text{LHS}$; and $P(1)$ is true.

Inductive Step: Assume that the $P(k)$ is true and proceed to prove that $P(k+1)$ is true.

$P(k)$ is true means $\sum_{r=1}^k r(r+1) = 1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \dots + k(k+1) = \frac{k(k+1)(k+2)}{3}$.

$$\begin{aligned} \sum_{r=1}^{k+1} r(r+1) &= \sum_{r=1}^k r(r+1) + (k+1)(k+2) = \frac{k(k+1)(k+2)}{3} + (k+1)(k+2) \\ &= \frac{k(k+1)(k+2) + 3(k+1)(k+2)}{3} = \frac{(k+1)(k+2)(k+3)}{3} \end{aligned}$$

Since the basis step and the inductive step have been verified, the principle of mathematical induction tells us that $P(n)$ is true for all positive integers.

We prove that $\sum_{n=1}^n r^3 = 1^3 + 2^3 + 3^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$

Basis Step: When $n = 1$, we see that $\text{LHS} = \sum_{n=1}^1 r^3 = 1^3 = \left[\frac{1(1+1)}{2} \right]^2 = 1 = \text{RHS}$. This means $P(1)$ is true.

Inductive Step: We assume that $P(k)$ is true, that is:

$$\sum_{n=1}^k r^3 = 1^3 + 2^3 + 3^3 + \dots + k^3 = \left[\frac{k(k+1)}{2} \right]^2$$

We show that $P(k+1)$ is true in the following way:

$$\begin{aligned} \sum_{n=1}^{k+1} r^3 &= 1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3 = \sum_{n=1}^k r^3 + (k+1)^3 = \left[\frac{k(k+1)}{2} \right]^2 + (k+1)^3 \\ &= \frac{(k(k+1))^2 + 4(k+1)^3}{4} = \frac{(k+1)^2 [k^2 + 4(k+1)^3]}{4} = \frac{(k+1)^2 [k^2 + 4k + 4]}{4} = \left[\frac{(k+1)(k+2)}{2} \right]^2 \end{aligned}$$

This shows that if $P(k)$ is true implies $P(k+1)$ is true. Therefore $P(n)$ must be true for all n .

Example 2.3 : Use induction to show that $4^n - 1$ is divisible by 3 for $n = 1, 2, 3, \dots$

Basis Step: If $n = 1$, $4^n - 1 = 4^1 - 1 = 3$ which is divisible by 3.

Inductive Step: Assume that the statement is true for $n = k$. That is $4^k - 1$ is divisible by 3. We now show that $4^{k+1} - 1$ is divisible by 3. To relate the $(k+1)$ th case to the k th case, we write $4^{k+1} - 1 = 4 \cdot 4^k - 1 = 4^k - 1 + 3 \cdot 4^k$.

By assumption, $4^k - 1$ is divisible by 3 and since $3 \cdot 4^k$ is divisible by 3, the sum $(4^k - 1) + 3 \cdot 4^k = 4^{k+1} - 1$, is divisible by 3.

Since the basis step and the inductive step have been verified, the principle of Mathematical induction tells us that $4^n - 1$ is divisible by 3 for $n = 1, 2, 3, \dots$

Exercise 4

Establish the identity by induction:

$$\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}.$$

1. Establish the identity by induction

$$\frac{1^2}{1 \times 3} + \frac{2^2}{3 \times 5} + \dots + \frac{n^2}{(2n-1)(2n+1)} = \frac{n(n+1)}{2(2n+1)}$$

2. Establish the identity by induction:

$$a + ar + \dots + ar^{n-1} = \frac{a(1-r^n)}{1-r}$$

3. Establish the identity by induction:

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

4. Establish the identity by induction:

$$\frac{1}{1 \cdot 3} + \frac{1}{2 \cdot 4} + \dots + \frac{1}{n(n+2)} = \frac{3}{4} - \frac{2n+3}{2(n+1)(n+2)} \text{ for } n \geq 2$$

2.5 Binomial theorem

Introduction

Expansion of expressions of the form $(a+x)^n$ where $n = 1, 2, 3, \dots$ can easily be done, but when n becomes very large, negative or fractional, such expansions cannot easily be done. Pascal (1623-1662) in a search of a way to expand such expressions with large and

fractional n produced what is known as the Pascal triangle. This he derived by observing the pattern formed in the expansion

$$(1+x)^n, n > 0 \quad (1.4)$$

Binomial	Expansion				Coefficients			
$(1+x)^0$	1				1			
$(1+x)^1$	1	x			1	1		
$(1+x)^2$	1	$2x$	x^2		1	2	1	
$(1+x)^3$	1	$3x$	$3x^2$	x^3	1	3	3	1

It is interesting to note that in the coefficients of the various expansions which form the Pascal triangle, the next line is generated from the previous. In all cases, the first and the last terms is 1.

To carry out expansion using the Pascal triangle for large n will not be easy, thus we need a more general form for such an expansion. To achieve that, we desire a formula for the coefficients of x^r

The expansion $(1+x)^n$ is then given by:

$$(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + {}^nC_3x^3 + {}^nC_4x^4 + \dots + {}^nC_rx^r + \dots + {}^nC_{n-1}x^{n-1} + {}^nC_nx^n$$

The coefficients in the expansion of $(1+x)^n$ gives the pattern

$$1, n, \frac{n(n-1)}{1*2}, \frac{n(n-1)(n-2)}{1*2*3}, \dots, \frac{n(n-1)(n-2)\dots(n-r+1)}{r(r-1)(r-2)\dots 3.2.1}, \dots, n, 1$$

where $\frac{n(n-1)(n-2)\dots(n-r+1)}{r(r-1)(r-2)\dots 3.2.1} = {}^nC_r = \frac{n!}{r!(n-r)!}$ is the number of combinations of n objects taken r at a time.

The two statements above together form the binomial theorem for $(1+x)^n$.

The binomial theorem for the general expansion of $(a+x)^n$ where n is a positive, negative or fractional power is stated as follows:

$$(a+x)^n = {}^nC_0a^n + {}^nC_1xa^{n-1} + {}^nC_2x^2a^{n-2} + \dots + {}^nC_rx^ra^{n-r} + \dots + {}^nC_nx^n$$

2.5.1 The Binomial expansion for positive number

We prove the theorem using the principle of mathematical induction.

Let $P(n)$ be the proposition.

$$(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + {}^nC_3x^3 + {}^nC_4x^4 + \dots + {}^nC_rx^r + \dots + {}^nC_{n-1}x^{n-1} + \dots + {}^nC_nx^n$$

$P(1)$ is true since $\text{LHS} = (1+x)^1 = 1 + {}^1C_1x = \text{RHS}$.

We assume $P(k)$ is true:

$$(1+x)^k = {}^kC_0 + {}^kC_1x + {}^kC_2x^2 + \dots + {}^kC_rx^r + \dots + {}^kC_{k-1}x^{k-1} + {}^kC_kx^k$$

We show this implies $P(k+1)$ is true:

LHS =

$$(1+x)^{k+1} = (1+x)(1+x)^k = (1+x)({}^kC_0 + {}^kC_1x + {}^kC_2x^2 + \dots + {}^kC_rx^r + \dots + {}^kC_{k-1}x^{k-1} + {}^kC_kx^k)$$

$$= 1 + ({}^kC_0 + {}^kC_1)x + ({}^kC_1 + {}^kC_2)x^2 + \dots + ({}^kC_r + {}^kC_{r+1})x^{r+1} + \dots + ({}^kC_{k-1} + {}^kC_k)x^k + {}^kC_kx^k$$

and more generally,

$${}^{k+1}C_r = {}^kC_r + {}^kC_{r-1}$$

Hence the result.

Example 2.4 : Expand $(x-2y)^5$ by the binomial theorem.

Solution: $(x-2y)^5 = x^5 + {}^5C_1x^4(-2y) + {}^5C_2x^3(-2y)^2 + {}^5C_3x^2(-2y)^3 + {}^5C_4x(-2y)^4 + {}^5C_5(-2y)^5$

$$= x^5 - 10x^4y + 40x^3y^2 - 80x^2y^3 + 80xy^4 - 32y^5$$

Example 2.5 : Find the coefficient of x^6 in the expansion $(\frac{1}{x^2} - x)^{18}$.

Solution: Given the expansion $(x+y)^n$, the coefficient of x^r is given by ${}^nC_rx^{n-r}y^r$

Thus we have ${}^nC_r(\frac{1}{x^2})^{n-r}(x)^r$ with $n = 18$ using the general term

$${}^{18}C_r(\frac{1}{x^2})^{18-r}(x)^r = {}^{18}C_rx^{-2(18-r)}x^r = {}^{18}C_r(x^r)^{-36+2r+r} = {}^{18}C_rx^{3r-36}$$

We now want $3r - 36 = 6$ and this condition gives $r = 14$.

Thus ${}^{18}C_r(x)^{3r-36} = {}^{18}C_{14}x^{42-36} = {}^{18}C_{14}x^6 = 3060x^6$, thus the coefficient is 3060.

Example 2.6 : Find the term independent of y in the expansion $(\frac{y^3}{x^2} + \frac{x}{3y})^8$

Solution: The procedure is essentially the same as in the previous example.

We take the general term with $n = 8$

$${}^nC_r x^{n-r} y^r = {}^nC_r \left(\frac{y^3}{x^2}\right)^{n-r} \left(\frac{x}{3y}\right)^r = {}^8C_r \left(\frac{y^3}{x^2}\right)^{8-r} \left(\frac{x}{3y}\right)^r$$

For the term to be independent of y , we want the powers of r in the numerator to be equal to the power of r in the denominator: $24 - 3r = r$, which gives $r = 6$.

For $r = 6$ we get the term independent of y in the expansion as

$${}^8C_6 \frac{x^2}{3^6} = \frac{8!}{6!2!} \frac{x^2}{3^6} = \frac{28x^2}{729}$$

Example 2.7 : Find the value of k if the coefficients of y^2 in the expansion of $(1+ky)^4(2-y)^3$ has value 6

Solution: $(1+ky)^4 = 1 + 4(ky) + 6(ky)^2 + 4(ky)^3 + (ky)^4$

$$(2-y)^3 = 2^3 + 3 * 2^2(-y) + 3 * 2(-y)^2 + (-y)^3 = 8 - 12y + 6y^2 - y^3$$

Thus

$$(1+ky)^4(2-y)^3 = (1 + 4ky + 6k^2y^2 + 4k^3y^3 + k^4y^4)(8 - 12y + 6y^2 - y^3)$$

Put the coefficient of y^2 equal to 6 to get $6 - 48k + 48k^2 = 6$; $k(k-1) = 0$ and $k = 0, k = 1$.

Example 2.8 : Expand $(x+3y)^5$. Hence evaluate $(1.03)^5$ correct to 3 decimal places.

$$(x+3y)^5 = x^5 + 5x^4(3y) + 10x^3(3y)^2 + 10x^2(3y)^3 + 5x(3y)^4 + (3y)^5 = x^5 + 15x^4y + 90x^3y^2 + 270x^2y^3 + 405xy^4 + 243y^5$$

To evaluate $(1.03)^5$ we substitute $x = 1$, and $y = 0.01$ to get:

$$(1.03)^5 = 1 + 15(0.01) + 90(0.01)^2 + 270(0.01)^3 + 405(0.01)^4 + 243(0.01)^5 = 1 + 0.15 + 0.009 + 0.00027 + 0.000027 = 1.159297$$

2.5.2

The Binomial expansion for negative and fractional numbers

It can be verified that for $-1 < \frac{n}{2} < 1$,

$$(1+x)^{\frac{n}{2}} = 1 + \frac{n}{2}x + \frac{\frac{n}{2}(\frac{n}{2}-1)}{2!}x^2 + \frac{\frac{n}{2}(\frac{n}{2}-1)(\frac{n}{2}-2)}{3!}x^3 + \dots$$

which is also a binomial theorem.

Example 2.9 : If y is so small that its fourth and higher powers can be neglected, show

that $\sqrt{\left(\frac{1+2y}{1-2y}\right)^n} = 1 + 2ny + 2n^2y^2 + \frac{4}{3}n(n^2 + 2)y^3$. By letting $y = \frac{1}{7}$ and $n = 1$, prove that $\sqrt{5} = \frac{2295}{1029}$

Solution: $\sqrt{\left(\frac{1+2y}{1-2y}\right)^n} = (1 + 2y)^{\frac{n}{2}}(1 - 2y)^{-\frac{n}{2}}$

Expanding by the binomial theorem as given

$$\begin{aligned}(1 + 2y)^{\frac{n}{2}} &= 1 + \frac{n}{2}(2y) + \frac{\frac{n}{2}(\frac{n}{2}-1)}{2!}(2y)^2 + \frac{\frac{n}{2}(\frac{n}{2}-1)(\frac{n}{2}-2)}{3!}(2y)^3 + \dots \\ &= 1 + ny + \frac{4n(n-2)}{8}y^2 + \frac{8n(n-2)(n-4)}{48}y^3 + \dots = 1 + ny + \frac{1}{2}(n^2 - 2n)y^2 + \frac{1}{6}(n^3 - 6n^2 + 8n)y^3 + \dots\end{aligned}$$

$$\begin{aligned}\text{Also } (1 - 2y)^{-\frac{n}{2}} &= 1 - \frac{n}{2}(-2y) + \frac{(-\frac{n}{2})(-\frac{n}{2}-1)}{2!}(-2y)^2 + \frac{(-\frac{n}{2})(-\frac{n}{2}-1)(-\frac{n}{2}-2)}{3!}(-2y)^3 + \dots \\ &= 1 + ny + \frac{n}{2} \frac{(\frac{n}{2} + 2)}{2!} 4y^2 + \frac{n}{2} \frac{(\frac{n}{2} + 2)(\frac{n}{2} + 4)}{3!} 8y^3 + \dots = 1 + ny + (n^2 + 2n)y^2 + \frac{1}{6}(n^3 + 6n^2 + 8n)y^3 + \dots\end{aligned}$$

The product is $(1 + 2y)^{\frac{n}{2}}(1 - 2y)^{-\frac{n}{2}} = (1 + ny + \frac{1}{2}(n^2 - 2n)y^2 + \frac{1}{6}(n^3 - 6n^2 + 8n)y^3 + \dots)(1 + ny + (n^2 + 2n)y^2 + \frac{1}{6}(n^3 + 6n^2 + 8n)y^3 + \dots)$

which gives

$$= 1 + 2ny + 2n^2y^2 + \frac{4}{3}n(n^2 + 2)y^3 + \dots$$

for $n = 1, y = \frac{1}{7}$ then

$$\sqrt{\frac{(1+\frac{2}{7})}{(1-\frac{2}{7})}} = 1 + \frac{2}{7} + \frac{2}{49} + \frac{4}{343}.$$

$$\sqrt{\frac{\frac{9}{7}}{\frac{5}{7}}} = \sqrt{\frac{9}{5}} = \frac{343 + 98 + 14 + 4}{343} = \frac{459}{343}$$

Thus we get the result $\sqrt{5} = \frac{2295}{1029}$.

$$\frac{3}{\sqrt{5}} = \frac{459}{343}, \quad \frac{3\sqrt{5}}{5} = \frac{459}{343}, \quad \sqrt{5} = \frac{459}{343} \times \frac{5}{3}$$

Example 2.10 : Find the expansion of $\sqrt{1 - 3x}$. Hence evaluate $\sqrt{0.7}$ correct to four decimal places.

Solution: $(1 - 3x)^n = 1 + n(-3x) + \frac{n(n-1)}{2!}(-3x)^2 + \frac{n(n-1)(n-2)}{3!}(-3x)^3 + \dots$

and

$$\sqrt{1 - 3x} = (1 - 3x)^{\frac{1}{2}} = 1 - \frac{1}{2}(3x) + \frac{\frac{1}{2}(\frac{1}{2} - 1)}{2!}9x^2 - \frac{\frac{1}{2}(\frac{1}{2} - 1)(\frac{1}{2} - 2)}{3!}27x^3 + \dots = 1 - \frac{3}{2}x - \frac{9}{8}x^2 - \frac{81}{48}x^3 + \dots$$

We write $\sqrt{0.7} = (0.7)^{\frac{1}{2}} = (1 - 3(0.1))^{\frac{1}{2}}$ with $x = 0.1$.

Substituting gives:

$$\sqrt{0.7} = (0.7)^{\frac{1}{2}} = (1 - 3(0.1))^{\frac{1}{2}} = 1 - \frac{3}{2}(0.1) - \frac{9}{8}(0.1)^2 - \frac{81}{48}(0.1)^3 + \dots = 1 - 0.15 - 0.01125 = 0.8388$$

Exercise 5

1. Expand $(1 + 3x)^{\frac{1}{3}}$. Hence evaluate $3\sqrt{1.06}$ correct to 4 decimal places.
2. Obtain the first 4 terms of the expansion $(1 + 2x)^{12}$ in ascending powers of x . Hence find the value of $(0.998)^{12}$ correct to four decimal places.
3. Show that $\sqrt{9 + y^2} \approx 3 + \frac{1}{6}y^2 - \frac{1}{316}y^4$. For what values of x is the expansion valid?
4. Find the term independent of x in the expansion $\left(\frac{x^5}{y^4} + \frac{y^3}{2x^x}\right)^{10}$.
5. Expand $\sqrt{\frac{1+2x}{1-2x}}$ in ascending powers of x up to and including the term in x^2 . By substituting $x = \frac{1}{100}$. Show that $\sqrt{5} \approx 7\frac{7}{50}$.
6. If x is so small that its fourth and higher powers can be neglected, show that $\sqrt{\left(\frac{1+x}{1-x}\right)^3} = 1 + 3x + \frac{9}{2}x^2 + 38x^3$.
7. Find the coefficients of y^9 in the expansion $\left(\frac{1}{y^3} - y\right)^{18}$.
8. Expand $(1 + x)^{-1} + (1 + 2x)^{-\frac{1}{2}}$ as far as terms in x^3 . State the range of values for which the expansion is valid.
9. Find the expansion of $\left(\frac{1+\frac{1}{2}y}{1-\frac{1}{2}y}\right)^{\frac{3}{2}}$ in ascending powers of y as far as the term in y^3 .
10. If x is so small that terms in x^n , $n \geq 3$ can be neglected and $\left(\frac{3+ax}{3+bx}\right) = (1 - x)^{\frac{1}{3}}$. Find the values of a and b . Hence evaluate $(0.96)^{\frac{1}{3}}$.
11. Find b if the coefficients of x in the expansion of $(1 + bx)^8 (1 + 3x)^4 - (1 + x)^3 (1 + 2x)^4$ is zero. What is the coefficient of x^3 .

3 Real Sequences and Series

3.1 Objectives

In this module:

- We introduce the concepts of sequences and series and explain the relationship between them.
- We discuss with examples the Arithmetic and Geometric progressions as special cases.
- We end the module with discussions on the Arithmetic mean, the Geometric mean and the Sum to infinity of a geometric progression.

3.2 Learning Outcomes

At the end of this module students should be able to:

- Define a sequence and identify different types of sequences.
- Define a series and identify various types of series.
- Identify an Arithmetic Progression (AP) and find the sum of n terms of any AP.
- Identify a Geometric Progression (GP) and find the sum of n terms, or the sum to infinity of any GP.
- Find the Arithmetic Mean and the Geometric Mean of progressions.

3.3 Learning Activities

Students should:

- Explore notes and exercises, individually and in groups;
- Engage in discussions with lecturers and their peers using forums and other interactive facilities on the Moodle site.
- Solve relevant questions from past MTH111 Examinations.
- Take and pass on-line and other self assessment tests.

3.4 Sequences

Introduction:

Definition 3.1 : A sequence $\{u_n\}_{n=1}^k = \{u_1, u_2, u_3, \dots, u_k\}$ or $\{u_n\}_{n=1}^\infty = \{u_1, u_2, u_3, \dots, u_n, \dots\}$ is a function whose domain is a subset of natural numbers.

In other words, it is a set of numbers or quantities $u_1, u_2, \dots, u_n, \dots$ each of which is obtained by a prescribed rule, that is $u_n = f(n)$. Specific terms can be obtained by substituting $n = 1, 2, 3, \dots$, in the general term u_n .

For example: $1, 3, 5, 7, \dots$, is a sequence in which $u_n = 2n - 1$. The fifth term of the sequence is $u_5 = 2(5) - 1 = 9$.

Similarly $2, 8, 32, \dots$, is a sequence in which $u_n = 2 \times 4^{n-1}$. Thus $u_5 = 2 \times 4^4 = 2 \times 256 = 512$.

$1^2, -2^2, 3^2, -4^2, \dots$, is a sequence in which $u_n = (-1)^{n-1}n^2$. Thus $u_5 = (-1)^{(5-1)} \cdot 5^2 = 25$.

When the pattern in which the terms of a sequence can be obtained is not so easily recognised, the form of the general term u_n is more difficult to establish, as we see in the following example; $-1, 7, -15, 25, \dots$, is a sequence in which $u_n = (-1)^n[n(n+3) - 3]$. Thus $u_5 = (-1)^5[5(5+3) - 3] = -1.37 = -37$.

Definition 3.2 : Finite Sequence: A sequence that contains only a finite number of terms is called a finite sequence. Finite sequences have domains that are proper finite subsets of N . In general $\{u_n\}_{n=1}^k = \{u_1, u_2, u_3, \dots, u_k\}$.

In particular $\{1, 3, 5, 7, 9\} = \{2n - 1\}_{n=1}^5$ and $\{(-1)^{n-1}n^2\}_{n=1}^5 = \{1^2, -2^2, 3^2, -4^2, 5^2\} = \{1, -4, 9, -16, 25\}$. In these two examples the domain of the function $f(n)$ is the set of five first natural numbers $\{1, 2, 3, 4, 5\}$.

Definition 3.3 : Infinite Sequence: A sequence is said to be infinite if the terms of the sequence are not finite. In general $\{u_n\}_{n=1}^\infty = \{u_1, u_2, u_3, \dots, u_n, \dots\}$. Infinite sequences have domain N or any infinite subset of N that is equivalent to N .

In particular $\{1, 3, 5, 7, \dots\} = \{2n - 1\}_{n=1}^\infty$ and $\{(-1)^{n-1}n^2\}_{n=1}^\infty = \{1^2, -2^2, 3^2, -4^2, 5^2, \dots\} = \{1, -4, 9, -16, 25, \dots\}$ are examples of infinite sequences. So are reciprocals of natural numbers $\{1, \frac{1}{2}, \frac{1}{3}, \dots\} = \{\frac{1}{n}\}_{n=1}^\infty$.

Infinite sequences are more interesting than finite ones and lead to the definition of series. When successive terms of an infinite sequence get closer and closer to a real number L , we say L is the limit of the sequence. Thus $\frac{1}{n}$ gets closer and closer to 0 as n gets larger and larger and we say the sequence $\{\frac{1}{n}\}_{n=1}^\infty$ has limit 0. The sequence is said to converge. Some sequences do not converge. For example the terms in the sequence $\{1, 3, 5, 7, \dots\} = \{2n - 1\}_{n=1}^\infty$ gets larger and larger as n increases and the sequence does not converge. We say such sequences diverge. This means that a sequence that is not convergent is said to diverge.

3.5 Series

Definition 3.4 : Given a sequence $\{u_n\}_{n=1}^{\infty}$ we can form a new sequence $\{S_n\}_{n=1}^{\infty}$ by taking sum to n terms of the given sequence.

We let $S_1 = \sum_{n=1}^1 u_n = u_1$; $S_2 = \sum_{n=1}^2 u_n = u_1 + u_2$; and generally $S_k = \sum_{n=1}^k u_n = u_1 + u_2 + u_3 + \dots + u_k$.

S_n is called the n th partial sum of the sequence and the infinite sum $\sum_{n=1}^{\infty} u_n = u_1 + u_2 + u_3 + \dots + u_n + u_{n+1} + \dots$ is called a series.

The sum S of an infinite series is given by the limit of S_n as n tends to ∞ . When this sum exists, we say the infinite series is convergent. When this sum does not exist the series is said to be nonconvergent. Suppose that for any given positive number k , $S_n > k$ for all n or $S_n < -k$ then the series is said to diverge. Thus $1 + 3 + 5 + 7 = 16$. is the fourth partial sum of $\sum_{n=1}^k (2n - 1)$ and is a finite series. The infinite series $\sum_{n=1}^{\infty} (2n - 1)$ is not convergent.

We shall be looking at some special cases of infinite series which are convergent in the next section.

3.5.1 Arithmetic Progression (AP)

We define an Arithmetic Progression (AP) as a finite sequence of terms in which each term other than the first is obtained from the preceding one by adding a constant number. Thus if $u_1 = a$, say, $u_2 = u_1 + d = a + d$, similarly $u_3 = u_2 + d = (a + d) + d = a + 2d$ and in general $u_n = u_{n-1} + d = a + (n - 1)d$. We see that the difference between any two consecutive terms is the constant d ; $u_k - u_{k-1} = d$. This is why the constant d , is called the common difference of the AP. For an AP with n terms we call the n th term the last term and it is denoted by l . In order to find out whether a given sequence is an AP, we confirm that the difference between consecutive terms is constant.

Example 3.1 : If the third term of an Arithmetic Progression is 10 and the seventh term is 34. Find the first term and the common difference of the AP.

Solution: Let a be the first term and d the common difference of the AP. Then the third and seventh terms are $a + 2d$ and $a + 6d$ respectively.

Thus $a + 2d = 10$ and $a + 6d = 34$. Solving for a and d we obtain $a = -2$ and $d = 6$.

Example 3.2 : If the 9th term of an AP is three times the 5th term, find a relationship between a and d . Prove that the 8th term is five times the 4th term.

Solution: Let a be the first term and d the common difference of the AP. Then the 9th and 5th terms are therefore $a + 8d$ and $a + 4d$ respectively.

$$\text{Thus } a + 8d = 3(a + 4d) \Rightarrow a + 8d = 3a + 12d \Rightarrow -2a = 4d \Rightarrow a = -2d$$

This is the relationship we seek.

To prove that the 8th term is five times the 4th term, we have that the 8th and the 4th terms are $a + 7d$ and $a + 3d$ respectively. But $a = -2d$, thus the 8th term is $5d$. Similarly the 4th term is d .

The 8th term is five times the 4th term.

Definition 3.5 : *Arithmetic Series:* The Arithmetic series is defined as the sum of the terms of an AP.

Example 3.3 : Consider the sum of the first n positive integers, that is $S_n = \sum_{r=1}^n r = 1 + 2 + 3 + \dots + n$.

Solution: Writing the sum as S_n with the smallest term first we have $S_n = 1 + 2 + 3 + \dots + n$.

Rewriting the sum as S_n with the largest term first we have $S_n = n + (n - 1) + (n - 2) + \dots + 1$.

Adding the 2 equations term by term we have $2S_n = (n + 1) + (n + 1) + \dots + (n + 1)$ — (n times)

$$= n(n + 1). \text{ Thus } S_n = \frac{n}{2}(n + 1)$$

This result can also be proved using the principle of induction.

Example 3.4 : Consider the sum of the first n odd positive integers.

Solution: It can be written as $S_n = 1 + 3 + 5 + \dots + (2n - 1) = \sum_{r=1}^n (2r - 1)$

Using the method we introduced above we rewrite it as $S_n = (2n - 1) + (2n - 3) + (2n - 5) + \dots + 1$

Adding the 2 equations we get $2S_n = 2n + 2n + 2n + \dots + 2n$ (n times)

$$= n(2n) = 2n^2. \text{ Thus } S_n = n^2.$$

In general, when S_n denotes the sum of n terms of an Arithmetic Progression,

$S_n = a + (a + d) + (a + 2d) + \dots + [a + (n - 2)d] + [a + (n - 1)d]$ and we can rewrite it as:

$$S_n = [a + (n - 1)d] + [a + (n - 2)d] + [a + (n - 3)d] + \dots + (a + d) + a$$

Adding the 2 equations we get, $2S_n = [2a + (n-1)d] + [2a + (n-1)d] + [2a + (n-1)d] + \dots + [2a + (n-1)d]$

$= n[2a + (n-1)d]$ (n times). Thus $S_n = \frac{n}{2}[2a + (n-1)d]$.

Denoting the n th term of an AP with n terms, by l we can write the result in the form $S_n = \frac{n}{2}[a + a + (n-1)d] = \frac{n}{2}(a + l)$, where $l = a + (n-1)d$.

This result can also be established using the principle of induction.

Example 3.5 : *The 10th term of a certain AP is -29 and the 20th term is -69 . Find the sum of the first 20 terms of the AP.*

Solution: The 10th and the 20th terms are $a+9d$ and $a+19d$ respectively. Thus $a+9d = -29$ and $a + 19d = -69$

Solving for a and d , we obtain $a = 7$ and $d = -4$.

Now $S_{20} = \frac{20}{2}[2(7) + (20-1)(-4)] = 10[14 - 76] = -620$

Example 3.6 : *The 9th term of an AP is 24 and the sum of the first 9 terms is 126 . Find an expression for the n th term and the sum to n terms.*

Solution: The 9th term is $a + 8d$. Thus $a + 8d = 24$

The sum of the first 9 terms is $S_9 = \frac{9}{2}[2a + (9-1)d] = 126 \Rightarrow \frac{9}{2}[a + a + 8d] = 126$

Substituting for $a + 8d$ we have $9(a + 24) = 252 \Rightarrow 9a + 216 = 252 \Rightarrow a = 4$

Substituting for a we have $d = \frac{5}{2}$

Thus the expression for the n th term is $l = a + (n-1)d = 4 + (n-1)\frac{5}{2} = \frac{5n+3}{2}$,

and the expression for sum to n terms is $S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}[8 + \frac{5n-5}{2}] = \frac{5n^2+11n}{4}$

Definition 3.6 : *The Arithmetic Mean (AM): Let x, y and z be three consecutive terms of an AP. We call y the Arithmetic mean between x and z . Now the common difference of the AP is given by $d = y - x = z - y$. Therefore $2y = x + z$. Hence $y = \frac{x+z}{2}$.*

Also, in a finite Arithmetic Progression, $u_1 = a, u_2 = a + d, u_3 = a + 2d, \dots, u_{n-1} = a + (n-2)d, u_n = a + (n-1)d$, the terms in between the first term u_1 and the last u_n (i.e $u_2, u_3, \dots, u_{n-1}$) are called the Arithmetic means between u_1 and u_n .

For example, in the finite AP $2, 7, 12, 17, 22, 27, 32$, the numbers $7, 12, 17, 22, 27$ are the five Arithmetic means between 2 and 32 .

Example 3.7 : *Insert three Arithmetic means between 25 and 9 .*

Solution: In this case, we have $u_1 = 25, u_2, u_3, u_4, u_5 = 9$. Since $u_5 = a + 4d$, we have

$$9 = 25 + 4d \Rightarrow d = -4. \text{ Thus } u_2 = a + d = 25 - 4 = 21$$

$$u_3 = a + 2d = 25 - 8 = 17$$

$$u_4 = a + 3d = 25 - 12 = 13$$

Therefore, the AP is 25, 21, 17, 13, 9.

Example 3.8 : p, q, r are three consecutive terms of an AP whose sum is 21. The ratio $p : r = 6 : 1$. Find p, q and r .

Solution: By assumption, the sum of the three terms of the AP is 21. That is

$$p + q + r = 21.$$

$$p : r = 6 : 1 \Rightarrow \frac{p}{r} = \frac{6}{1} \text{ or } p = 6r$$

$$\text{But } q = \frac{p+r}{2}$$

Since p, q, r are three consecutive terms of an AP.

$$\text{Substituting we have } q = \frac{6r+r}{2} = \frac{7r}{2} \text{ and}$$

$$6r + \frac{7r}{2} + r = 21 \Rightarrow 12r + 7r + 2r = 42 \Rightarrow r = 2$$

$$\therefore p = 6r = 6(2) = 12 \text{ and } q = \frac{7r}{2} = \frac{7(2)}{2} = 7$$

$$\therefore (p, q, r) = (12, 7, 2).$$

3.5.2 Geometric Progression(GP)

Definition 3.7 : A Geometric Progression (GP) is a sequence of numbers in which each term other than the first is obtained from the preceding one by the multiplication of a non-zero constant number called the common ratio.

In other words, it is a progression in which the ratio between any two consecutive terms is constant (i.e the same). The common ratio is usually denoted by r . The first term of a GP is usually denoted by a . Hence a typical GP takes the following form: $a, ar, ar^2, \dots, ar^{n-1}$. Consequently the n th term of a GP is given by $u_n = l = ar^{n-1}$.

Example 3.9 : The third term of a Geometric Progression is 10, and the sixth term is 80. Find the common ratio and the first term of the GP.

Solution: Let a and r denote the first term and the common ratio respectively of the GP. Then $ar^2 = 10$ and $ar^5 = 80$. Eliminating a , we have $r^3 = 8 \Rightarrow r = 2$. Hence $4a = 10 \Rightarrow a = \frac{10}{4} = 2\frac{1}{2}$.

Definition 3.8 : Geometric Series: A Geometric series is defined as the sum of the terms of a Geometric Progression. The sum of n terms of a GP is easily obtained by the following argument.

A typical Geometric series is given by $S_n = a + ar + ar^2 + \dots + ar^{n-2} + ar^{n-1}$

Multiplying each term in the equation above by r we get $rS_n = ar + ar^2 + ar^3 + \dots + ar^{n-1} + ar^n$

Subtracting the second equation from the first equation we have

$$S_n - rS_n = a - ar^n \Rightarrow (1 - r)S_n = a(1 - r^n) \Rightarrow S_n = \frac{a(1-r^n)}{1-r}.$$

Hence $S_n = \frac{a(1-r^n)}{1-r}$, for $-1 < r < 1$ or $S_n = \frac{a(r^n-1)}{r-1}$, for $|r| > 1$.

Or simply,

$$S_n = \frac{a(1-r^n)}{1-r}, \text{ if } r < 1 \text{ and } S_n = \frac{a(r^n-1)}{r-1}, \text{ if } r > 1.$$

Example 3.10 : Find the sum of the first eight terms of the GP $5 + 15 + 45 + \dots$

Solution: In this case $a = 5$, and $r = 3$. Hence $S_n = \frac{a(r^n-1)}{r-1}$, since $r > 1$

$$\text{That is } S_8 = \frac{5(3^8-1)}{3-1} = \frac{5(6561-1)}{2} = 16400.$$

Example 3.11 : In a GP, the sum of the second and third terms is 9; and the seventh term is eight times the fourth term. Find the first term, the common ratio and the fifth term.

Solution:

Let a and r denote the first term and the common ratio respectively. Then

$$ar + ar^2 = 9 \Rightarrow ar(1 + r) = 9 \text{ and } ar^6 = 8ar^3 \Rightarrow r^3 = 8 \Rightarrow r = 2.$$

Substitute $r = 2$ in the first equation to have $6a = 9 \Rightarrow a = \frac{3}{2}$.

Thus the first term $a = \frac{3}{2}$ and the common ratio $r = 2$. Therefore the fifth term is ar^4 . That is $\frac{3}{2}(2^4) = 24$.

Definition 3.9 : Geometric Mean (GM): Let x, y and z be three consecutive terms of a GP with common ratio r . We call y the geometric mean between x and z . Thus $r = \frac{y}{x} = \frac{z}{y}$. Hence $y^2 = xz$ and so $y = \sqrt{xz}$

In a finite Geometric Progression, $u_1 = a, u_2 = ar, u_3 = ar^2, \dots, u_{n-1} = ar^{n-2}, u_n = ar^{n-1}$, the terms in between u_1 and u_n (i.e $u_2, u_3, \dots, u_{n-1}$) are called the geometric means between u_1 and u_n . For instance, in the finite GP $2, 6, 18, 54, 162$, the numbers $6, 18, 54$ are the three Geometric means between 2 and 162 .

Example 3.12 : Insert four geometric means between -2 and $\frac{1}{16384}$.

Solution: In this case, we have $u_1 = -2, u_2, u_3, u_4, u_5, u_6 = \frac{1}{16384}$.

Since $u_6 = ar^5$, we have that

$$\frac{1}{16384} = -2r^5 \Rightarrow r^5 = -\frac{1}{2(16384)} \text{ i.e } r^5 = -\frac{1}{2^{15}} \Rightarrow r = -\frac{1}{2^3} = -\frac{1}{8}.$$

$$\text{Hence } u_2 = ar = -2\left(-\frac{1}{8}\right) = \frac{1}{4}$$

$$u_3 = ar^2 = -2\left(-\frac{1}{8}\right)^2 = -\frac{1}{32}$$

$$u_4 = ar^3 = -2\left(-\frac{1}{8}\right)^3 = \frac{1}{256}$$

$$u = ar^4 = -2\left(-\frac{1}{8}\right)^4 = -\frac{1}{2048}$$

Thus the four geometric means are $\frac{1}{4}, -\frac{1}{32}, \frac{1}{256}, -\frac{1}{2048}$.

Hence the GP is $-2, \frac{1}{4}, -\frac{1}{32}, \frac{1}{256}, -\frac{1}{2048}, \frac{1}{16384}$.

Example 3.13 : *The third term of a geometric progression is 36 and the sixth term is $\frac{243}{2}$. Find the first term, the common ratio and the sum of the first eight terms.*

Solution: Let a be the first term and r the common ratio of the GP. Then we have that

$$u_3 = ar^2 = 36 \text{ and } u_6 = ar^5 = \frac{243}{2}.$$

Solving for a and r , we have $a = 16$ and $r = \frac{3}{2}$. Thus the sum of the first eight terms is

$$S_8 = \frac{16\left[\left(\frac{3}{2}\right)^8 - 1\right]}{\frac{3}{2} - 1} = 788\frac{1}{8}.$$

Example 3.14 : *The fifth, ninth and the sixteenth terms of an AP are consecutive terms of a GP. The sum of the first twelve terms of the AP is 492. Find the first term of the AP and the common ratio of the GP.*

Solution: Let a be the first term and d the common difference of the AP. Then we have that

$$u_5 = a + 4d, u_9 = a + 8d \text{ and } u_{16} = a + 15d.$$

Since these form three consecutive terms of a GP, then the common ratio r is given by

$$r = \frac{a+8d}{a+4d} = \frac{a+15d}{a+8d}.$$

$$\Rightarrow (a + 8d)^2 = (a + 4d)(a + 15d)$$

$$\Rightarrow a^2 + 16ad + 64d^2 = a^2 + 19ad + 60d^2$$

$$\Rightarrow 64d^2 - 60d^2 = 19ad - 16ad$$

$$\Rightarrow 4d^2 = 3ad$$

$$\Rightarrow d = \frac{3a}{4}.$$

The sum of the first twelve terms is 492. That is

$$S_{12} = \frac{12}{2} [2a + 11d] = 492, \Rightarrow 6 [2a + 11d] = 492 \Rightarrow 2a + 11d = 82$$

Substitute for d to get $2a + 11\left(\frac{3a}{4}\right) = 82 \Rightarrow 8a + 33a = 328 \Rightarrow 41a = 328 \Rightarrow a = 8$.

Substitute for a to get $d = \frac{3 \times 8}{4} = 6$.

Substituting for a and d in the first equation in order to find r we have

$$r = \frac{8+8(6)}{8+4(6)} = \frac{8+48}{8+24} = \frac{56}{32} = \frac{7}{4}.$$

Thus the first term of the AP is 8 and the common ratio of the GP is $\frac{7}{4}$.

Exercise

1. If $x + 1, x + 3$ and $x + 7$ are consecutive terms of a GP, find the value(s) of x .
2. The third term of a GP is five and the eighth term is 160. Find the first term and the common ratio of the GP.
3. The first term of a GP is 18 and the sum to infinity is 20. Find the common ratio and the sum of the first 6 terms.
4. Prove that the sum to n terms of the GP $1 + \frac{1}{a} + \frac{1}{a^2} + \frac{1}{a^3} + \dots + \frac{1}{a^{n-1}}$ is equal to $\left(\frac{a^n - 1}{a - 1}\right) a^{1-n}$.
5. The first three terms of a GP are also the first, ninth and eleventh terms respectively of an AP. Given that the terms of the GP are all different, find the common ratio r , if the sum to infinity of the GP is 8. Find also the first term and the common difference of the AP.
6. The coefficients of y^n, y^{n+1}, y^{n+2} in the binomial expansion of $(1 + y)^{14}$ are in AP. Find the possible values of n .
7. The sum of the first q terms of an AP is Q and the sum of the first r terms of the AP is R . Show that the common difference of the AP is $\frac{2(rQ - qR)}{qr(q - r)}$.
8. The p th, q th and r th terms of an AP are consecutive terms of a GP. Prove that the ratio of the first term of the AP to its common difference is $1 + \frac{q^2 - pr}{r + p - 2q}$. Hence given that $3(q^2 - pr) = r + p - 2q$ and that the 12th term of the AP is 74, find the common difference of the AP.
9. Show that the sum to n terms of the series $\log a + \log ax + \log ax^2 + \dots$ is $n \log a + \frac{1}{2}n(n - 1) \log x$.
10. If the sum of n terms of the series $1 + 8 + 15 + 22 + \dots$ is 5500, then find n .

11. The sum of the second and fourth terms of an AP is 15, and the sum of the fifth and sixth terms is 25. Find the first term and the common difference of the AP.
12. The sum of the first two terms of a GP is 3, and the sum of the second and third terms is -6 . Find the first term and the common ratio.

3.5.3 Infinite Series

In the work so far on Arithmetic Progressions and Geometric Progressions, we have been concerned with a limited number of terms of the series. That is, we have found S_n , where n is a finite number. But in some situations, the series goes on for ever and the problem then arises to finding the sum S_n as n tends to infinity.

With an AP, $S_n = \frac{n}{2} [2a + (n - 1)d]$. Therefore with finite values for a and d , as n increases, so does the value of S_n . Thus if n tends to infinity, then S_n tends to infinity in a positive or negative sense depending on the series.

On the other hand, the sum of n terms of a GP is given by $S_n = \frac{a(1-r^n)}{1-r}$ for $|r| < 1$. Recall that for $|r| < 1$, r^n gets smaller and smaller as n gets larger and larger. In the limiting situation, therefore, when n approaches infinity, r^n tends to zero. This limiting value of S_n is called the sum to infinity of the infinite GP. Then the sum of the GP becomes $S = S_\infty = \lim_{n \rightarrow \infty} S_n = \frac{a}{1-r}$.

4 Quadratic Equations and Remainder Theorem

4.1 Learning Objectives

In this module:

- We introduce various methods of solving quadratic equations and explore the properties of the roots.
- We explore quadratic inequalities and their solutions.
- We introduce the remainder theorem and highlight its relationship with solutions of quadratic equations.

4.2 Learning Outcomes

At the end of this module students should be able to:

- Find the roots of quadratic equations using different methods as appropriate.
- Construct quadratic equations based on given root properties.
- Solve simple problems involving properties of roots of quadratic equations.
- Discuss and solve quadratic inequalities.
- Decide when and how to use the remainder theorem.

4.3 Learning activities

Students should:

- Explore notes, power point presentations and exercises, individually and in groups.
- Engage in discussions with lecturers and their peers using forums and other interactive facilities on the Moodle site.
- Take and pass online and other self assessment tests.
- Solve relevant questions from past MTH111 Examinations.

4.4 Quadratic Equations

Definition 4.1 : Quadratic Equation: Any equation of the form $ax^2 + bx + c = 0$ where $a \neq 0$, b and c are constants is called a quadratic equation. Examples include $3x^2 + 2x + 4 = 0$, $x^2 - 6x = 0$ and $x^2 - 7 = 0$

Definition 4.2 : Solutions to Quadratic Equations: A number p is said to be a solution or a root of a quadratic equation if by replacing x by p the LHS and the RHS of the quadratic equation are equal. For example, $x = 3$, is the solution or a root of $x^2 - 5x + 6 = 0$ because when 3 is substituted in the LHS, we have $3^2 - 5(3) + 6 = 9 - 15 + 6 = 0 = \text{RHS}$.

When p_1 and p_2 are two distinct roots of a quadratic equation $ax^2 + bx + c = 0$, we can write $ax^2 + bx + c = (x - p_1)(x - p_2)$.

The factorisation of the quadratic equation $ax^2 + bx + c = 0$ is simple in the special cases when $b = 0$ or $c = 0$.

If $c = 0$, we have $ax^2 + bx = 0 \Rightarrow x(ax + b) = 0 \Rightarrow x = 0$ or $ax + b = 0$.

∴ The solutions of the equation are $x = 0$ and $x = \frac{-b}{a}$.

Similarly If $b = 0$, we have $ax^2 + c = 0$.

$$\Rightarrow x^2 + \frac{c}{a} = 0 \Rightarrow x^2 = \pm \sqrt{\frac{-c}{a}} \text{ and our solutions are } x = -\sqrt{\frac{-c}{a}} \text{ and } x = \sqrt{\frac{-c}{a}}.$$

Example 4.1 : Solve the equations: $2x^2 - 14x = 0$ and $3x^2 - 9 = 0$.

Solution: $2x^2 - 14x = 2x(x - 7)$

$$\therefore 2x^2 - 14x = 0 \Rightarrow 2x(x - 7) = 0 \Rightarrow 2x = 0 \text{ and } x - 7 = 0.$$

$$\Rightarrow x = 0 \text{ and } x = 7.$$

$$\text{Similarly } 3x^2 - 9 = 0 \Rightarrow 3x^2 = 9 \Rightarrow x = \pm\sqrt{3} \Rightarrow x = -\sqrt{3} \text{ and } x = +\sqrt{3}.$$

we note that the quadratic equations each have 2 distinct solutions.

Definition 4.3 : *The **Discriminant**: Given any quadratic equation $ax^2 + bx + c = 0$, the quantity $D = b^2 - 4ac$ is called the discriminant of the equation. It is used to determine the nature of roots of the quadratic equation.*

We can always give the general solution to the quadratic equation $ax^2 + bx + c = 0$ by completing the square in the following way:

$$ax^2 + bx + c = a\{x^2 + \frac{b}{a}x + \frac{c}{a}\} = a\{(x + \frac{b}{2a})^2 + \frac{c}{a} - \frac{b^2}{4a^2}\} = a\{(x + \frac{b}{2a})^2 + \frac{b^2 - 4ac}{4a^2}\}$$

$$\text{Hence if } ax^2 + bx + c = 0 \text{ then } \{(x + \frac{b}{2a})^2 + \frac{b^2 - 4ac}{4a^2}\} = 0 \text{ as we are given } a \neq 0.$$

Thus,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-b \pm \sqrt{D}}{2a}$$

When the discriminant $b^2 - 4ac = D$ is a perfect square d^2 , that is $D = d^2$ and $x = \frac{-b \pm \sqrt{D}}{2a} = \frac{-b \pm d}{2a}$.

then $(x - \frac{-b-d}{2a})$ and $(x - \frac{-b+d}{2a})$ are the factors of the quadratic equation.

The solutions of the equation are therefore $x = \frac{-b-d}{2a}$ and $x = \frac{-b+d}{2a}$.

Example 4.2 : Let us use this scheme to factorise $2x^2 - 7x + 3 = 0$.

Solution: The discriminant, $b^2 - 4ac = 7^2 - 4(2)(3) = 25 = D$ is a perfect square and d is 5.

Our solutions are $x = \frac{2}{4} = \frac{1}{2}$ and $x = \frac{12}{4} = 3$.

Exercise 6 Solve the following equations.

1. $x^2 - 5x - 6 = 0$
2. $3x^2 - 7x = 0$
3. $4x^2 - 8x + 4 = 0$
4. $x(x - 3) = x^2 - 6$
5. $3 - x - 2x^2 = 0$
6. $x^2 - 2ax + a^2 = 0$
7. $\frac{1}{x^2} - 1 = \frac{1}{x} - 1$
8. $x(1 - x) = x(2x - 1)$
9. $(x - 2)(x + 3) = (x - 2)(4 - x)$
10. $\frac{1}{x+1} + \frac{2}{x+2} = 1$

4.4.1 Solution of Quadratic Equations without Rational Roots

When a quadratic equation $ax^2 + bx + c = 0$ does not have rational roots (and this we know is when the discriminant $b^2 - 4ac$ is not a perfect square), We can still use the method of completing the square to obtain the solution.

Example 4.3 : Solve the equation $2x^2 - 5x + 1 = 0$

Solution: We now give a special case of the method of completing the square.

We divide the equation by 2 , the coefficient of x^2 and we obtain $x^2 - \frac{5}{2}x + \frac{1}{2} = 0 \Rightarrow x^2 - \frac{5}{2}x = -\frac{1}{2}$

Add the square of half of the coefficient of x in the last expression above; i.e. $(\frac{1}{2} \times \frac{5}{2})^2$ to both sides to make the LHS a perfect square, we have:

$$x^2 - \frac{5}{2}x + \left(\frac{5}{4}\right)^2 = -\frac{1}{2} + \left(\frac{5}{4}\right)^2 \Rightarrow \left(x - \frac{5}{4}\right)^2 = \frac{17}{16} \Rightarrow x - \frac{5}{4} = \pm\sqrt{\frac{17}{16}}.$$

$$\Rightarrow x = \frac{5}{4} \pm \sqrt{\frac{17}{16}} \Rightarrow x = \frac{5}{4} + \frac{\sqrt{17}}{4} \text{ and } x = \frac{5}{4} - \frac{\sqrt{17}}{4}.$$

Generally, for $ax^2 + bx + c = 0$, we can always follow this same procedure.

Thus dividing the equation by a , the coefficient of x^2 and we obtain $x^2 + \frac{b}{a}x + \frac{c}{a} = 0$.

$$\Rightarrow x^2 + \frac{b}{a}x = -\frac{c}{a}.$$

Add the square of half of the coefficient of x in the last expression above to both sides to make the LHS a perfect square, $x^2 + \frac{b}{a}x + \left(\frac{1}{2} \times \frac{b}{a}\right)^2 = -\frac{c}{a} + \left(\frac{1}{2} \times \frac{b}{a}\right)^2$.

$$\Rightarrow \left(x + \frac{b}{a}\right)^2 = \frac{-c}{a} + \left(\frac{b}{2a}\right)^2 \Rightarrow \left(x + \frac{b}{a}\right)^2 = \frac{-4ac+b^2}{4a^2} = \frac{b^2-4ac}{4a^2}.$$

$$\text{i.e } \left(x + \frac{b}{a}\right)^2 = \frac{b^2-4ac}{4a^2}.$$

$$\Rightarrow \left(x + \frac{b}{a}\right) = \pm \sqrt{\frac{b^2-4ac}{4a^2}} = \frac{\sqrt{b^2-4ac}}{2a} \Rightarrow x = \frac{-b}{2a} \pm \frac{\sqrt{b^2-4ac}}{2a}.$$

$$\therefore x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

This is called the quadratic formula.

4.4.2 Nature of the Roots of Quadratic Equations

Using the quadratic formula to solve the equation $ax^2 + bx + c = 0$, we see that $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$ and $x = \frac{-b - \sqrt{b^2-4ac}}{2a}$

Therefore in general, a quadratic equation has two solutions (called the roots).

If $b^2 - 4ac$ is positive, then $\sqrt{b^2 - 4ac}$ can be evaluated and the equation will have two real and distinct (different) roots. This includes the case when it is a perfect square.

If $b^2 - 4ac$ is zero, then the equation can be satisfied by $x = \frac{-b}{2a}$ and we say it has repeated roots or equal roots

If $b^2 - 4ac$ is negative, $\sqrt{b^2 - 4ac}$ has no real value and we say the equation has complex roots

In summary, the equation $ax^2 + bx + c = 0$ has two real and distinct roots if $b^2 - 4ac > 0$; has equal roots if $b^2 - 4ac = 0$ and it has complex roots if $b^2 - 4ac < 0$.

Example 4.4 : Determine the nature of roots of the equations.

1. $4x^2 - 7x + 3 = 0$

2. $x^2 + ax + a^2 = 0$

3. $x^2 - px - q^2 = 0$.

Solution:

1. Given $4x^2 - 7x + 3 = 0$, $\therefore b^2 - 4ac = (-7)^2 - 4(4)(3) = 1$ i.e $b^2 - 4ac = 1 > 0$, So the equation has two distinct roots. More over, 1 is a perfect square so the equation also has rational roots.

2. $x^2 + ax + a^2 = 0 \therefore b^2 - 4ac = a^2 - 4(1)(a^2) = -3a^2$ since a^2 is positive for all real and non zero a , $\therefore -3a^2 < 0$. i.e $b^2 - 4ac < 0$ so the equation has no real roots for any real and non zero value of a It has complex roots.

3. $x^2 - px^2 - q^2 = 0$; $b^2 - 4ac = (-p)^2 - 4(1)(-q^2) = p^2 + 4q^2$. As p^2 and q^2 are both positive $b^2 - 4ac > 0$ therefore, the equation has two real and distinct roots

Example 4.5 : Find the value of k if $2x^2 - kx + 8 = 0$ has equal roots.

Solution: For the roots of $2x^2 - kx + 8 = 0$ to be equal, the discriminant must be zero.

i.e $b^2 - 4ac = 0 \Rightarrow (-k)^2 - 4(2)(8) = 0 \Rightarrow k^2 - 64 = 0 \Rightarrow k = \pm 8$.

Exercise 7 Find the roots of the following equations.

1. $2x^2 - 6x + 4 = 0$.

2. $x^2 + 4x - 8 = 0$.

3. $2x^2 + 7x + 3 = 0$

4. $x^2 - 2x + a = 0$.

5. $x^2 - 2ax + b = 0$.

6. $ax^2 - bx + c = 0$.

Exercise 8 Determine the nature of roots of the following equations and hence find them.

1. $x^2 - 6x + 9 = 0$.

2. $2x^2 - 5x + 3 = 0$.

3. $x^2 - 6x + 10 = 0$.

4. $3x^2 + 4x + 2 = 0$.

5. $4x^2 - 12x + 9 = 0$.

6. $4x^2 - 12x - 9 = 0$.

7. For what value of k is $9x^2 + kx + 16$ a perfect square?

8. The roots of $3x^2 + kx + 12 = 0$ are equal, find k .

9. Prove that $kx^2 - 2x - (k - 2) = 0$ has real roots for any value of k .

10. Show that the roots of $ax^2 + (a + b)x + b = 0$ are real for any values of a and b .

11. Find the relationship between p and q if the roots of $px^2 + qx + 1 = 0$ are equal.

12. If $f = \frac{x^2+1}{2x-1}$ is real and $p = 3 \left(\frac{x^2+1}{2x-1} \right)$, prove that $p^2 = -3(p + 3) \geq 0$.

4.4.3 Relationship between the Roots and Coefficients of a Quadratic Equation

Let α and β be the roots of the quadratic equation $ax^2 + bx + c = 0$.

The equations $(x - \alpha)(x - \beta) = 0$ and $ax^2 + bx + c = 0$ have the same solution.

$\therefore x^2 - (\alpha + \beta)x + \alpha\beta = 0$ and $x^2 + \frac{b}{a}x + \frac{c}{a} = 0$ have the same solution. As the LHS and the RHS of both equations have the same coefficients of x^2 , it follows that the coefficients of x and the constant terms are equal

Hence $x^2 - (\alpha + \beta)x + \alpha\beta \equiv x^2 + \frac{b}{a}x + \frac{c}{a} = 0$.

$\therefore \alpha + \beta = -\frac{b}{a}$ and $\alpha\beta = \frac{c}{a}$ and the equation may be written as

$x^2 - (\text{sum of roots})x + \text{product of roots} = 0$.

So if $2x^2 - 3x + 6 = 0$ then the sum $\alpha + \beta = -(\frac{-3}{2}) = \frac{3}{2}$ and the product $\alpha\beta = \frac{6}{2} = 3$.

Also if a quadratic equation has roots whose sum is 7 and whose product is 10, then the equation can be written as $x^2 - 7x + 10 = 0$.

Symmetry of the roots: An expression involving the roots α, β of a quadratic equation is said to be symmetric if α is substituted for β and β for α , the expression remains the same.

For example, $\frac{1}{\alpha} + \frac{1}{\beta}$ becomes $\frac{1}{\beta} + \frac{1}{\alpha} = \frac{1}{\alpha} + \frac{1}{\beta}$ when we replace α by β and β by α in the expression. However $\alpha^2 - \beta$, this becomes $\beta^2 - \alpha$ which is not the same as the original expression and is therefore not symmetric.

Example 4.6 : If α and β are the roots of the quadratic equation $2x^2 - 7x + 4 = 0$, calculate the values of (i) $\alpha^2 + \beta^2$ (ii) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ (iii) $\alpha - \beta$ (iv) $\alpha^3 + \beta^3$.

Solution: From the given equation, the sum of the roots, $\alpha + \beta = -(\frac{-7}{2}) = \frac{7}{2}$, and the product of the roots, $\alpha\beta = \frac{4}{2} = 2$. Thus, (i) $(\alpha + \beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta \Rightarrow \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$.

But $\alpha + \beta = \frac{7}{2}$ and $\alpha\beta = 2$. $\therefore \alpha^2 + \beta^2 = (\frac{7}{2})^2 - 2(2) = \frac{49}{4} - 4 = \frac{33}{4}$.

(ii) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{\frac{33}{4}}{2} = \frac{33}{8}$ using the results of (i) above.

(iii) Observe that $(\alpha - \beta)^2 = \alpha^2 + \beta^2 - 2\alpha\beta = (\alpha + \beta)^2 - 2\alpha\beta - 2\alpha\beta$.

i.e $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$.

$\therefore \alpha - \beta = \pm \sqrt{(\alpha + \beta)^2 - 4\alpha\beta} = \pm \sqrt{(\frac{7}{2})^2 - 4(2)} = \pm \sqrt{\frac{17}{4}} = \pm \frac{\sqrt{17}}{2}$

(iv) Consider $(\alpha + \beta)^3 = \alpha^3 + 3\alpha^2\beta + 3\alpha\beta^2 + \beta^3 = \alpha^3 + \beta^3 + 3\alpha\beta(\alpha + \beta)$.

$\therefore (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = \alpha^3 + \beta^3$.

or $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = (\alpha + \beta)[(\alpha + \beta)^2 - 3\alpha\beta]$.

$$= \frac{7}{2} \left[\left(\frac{7}{2} \right)^2 - 3(2) \right] = \frac{175}{4}.$$

Example 4.7 : Write the equation of the quadratic function whose roots are α^2, β^2 where α, β are the roots of the quadratic equation $2x^2 + 7x + 4 = 0$.

Solution: $\alpha + \beta = -\frac{7}{2}$, $\alpha\beta = \frac{4}{2} = 2$ and the equation may be written as $x^2 - (\text{sum of roots})x + \text{product of roots} = 0$.

Since α^2, β^2 are the roots of the equation, sum of roots $= \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \left(-\frac{7}{2}\right)^2 - 2(2) = \frac{33}{4}$.

Product of roots $= \alpha^2\beta^2 = (\alpha\beta)^2 = (2)^2 = 4$.

$\therefore$ The equation is $x^2 - \frac{33}{4}x + 4 = 0$ or $4x^2 - 33x + 16 = 0$.

Example 4.8 : If α, β are the roots of the quadratic equation $x^2 - 3x - 2 = 0$, find the quadratic equation whose roots are α^3, β^3 .

Solution: Since α, β are the roots of $x^2 - 3x - 2 = 0$, $\alpha + \beta = -\left(\frac{-3}{1}\right) = 3$ and $\alpha\beta = \frac{-2}{1} = -2$.

Now sum of roots of desired equation $\alpha^3 + \beta^3 = (\alpha + \beta) [(\alpha + \beta)^2 - 3\alpha\beta] = 3(3^2 - 3(-2)) = 45$.

Product of roots of desired equation $= \alpha^3\beta^3 = (\alpha\beta)^3 = (-2)^3 = -8$.

$\therefore$ The equation whose roots are α^3, β^3 is $x^2 - 45x - 8 = 0$

Example 4.9 : (i) Find the range of values of k for which the equation $x^2 - 2x - k = 0$ has real roots (ii) If the roots of this equation differ by one, find the value of k .

(i) If $x^2 - 2x - k = 0$ has real roots, then $b^2 - 4ac \geq 0 \Rightarrow (-2)^2 - 4(1)(-k) = 0 \Rightarrow 4 + 4k \geq 0 \Rightarrow k \geq -1$.

(ii) Let one root of the equation be α , then the other root is $\alpha + 1$. Sum of roots $= \alpha + \alpha + 1 = 2\alpha + 1 = -(-2) \Rightarrow 2\alpha + 1 = 2 \Rightarrow \alpha = \frac{1}{2}$. product of roots $= \alpha(\alpha + 1) = -k$, since $\alpha = \frac{1}{2}$, then $k = -\frac{3}{4}$.

- Write down the sum and the product of the roots of the following equations: (a) $x^2 - 3x + 2 = 0$ (b) $4x^2 + 7x - 3 = 0$ (c) $x(x - 3) = x + 4$ (d) $\frac{x-1}{2} = \frac{3}{x+2}$ (e) $x^2 - kx + k^2 = 0$ (f) $ax^2 - x(a + x) - a = 0$.
- Write down the sums, the products and the equations whose roots are (a) 3, 4 (b) $-2, \frac{1}{2}$ (c) $\frac{1}{3}, \frac{-2}{5}$ (d) $-\frac{1}{4}, 0$ (e) a, a^2 (f) $-(k + 1), k^2 - 3$ (g) $-\frac{b}{a}, \frac{c^2}{b}$

3. The roots of the equation $2x^2 - 4x + 5 = 0$ are α and β . Find the value of (a) $\frac{1}{\alpha} + \frac{1}{\beta}$ (b) $(\alpha + 1)(\beta + 1)$ (c) $\alpha^2 + \beta^2$ (d) $\alpha^2\beta + \alpha\beta^2$ (e) $(\alpha - \beta)^2$ (f) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ (g) $\frac{1}{\alpha+1} + \frac{1}{\beta+1}$ (h) $\frac{1}{2\alpha+\beta} + \frac{1}{\alpha+2\beta}$ (i) $\frac{1}{\alpha^2+1} + \frac{1}{\beta^2+1}$ (j) $\alpha^2 - \beta^2$ (k) $\alpha^3 + \beta^3$ (l) $\alpha^3 - \beta^3$.
4. The roots of $x^2 - 2x + 3 = 0$ are α and β , find the equations whose roots are (a) $\alpha + 2, \beta + 2$ (b) $\frac{1}{\alpha}, \frac{1}{\beta}$ (c) α^2, β^2 (d) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ (e) $\alpha - \beta, \beta - \alpha$.
5. Write down and simplify the equation whose roots are the reciprocal of the roots of $3x^2 + 2x - 1 = 0$ without solving the given equation.
6. Write down and simplify the equation whose roots are the double of those of $4x^2 - 5x - 2 = 0$ without solving the equation.
7. Write down and simplify the equation whose roots are one less than those of $5x^2 + 3x - 2 = 0$.
8. Write down the equation whose roots are minus those of $2x^2 - 3x - 1 = 0$.
9. Find the value of k if the roots of $3x^2 + 5x - k$ differ by two.
10. Find the value of p if one root of $x^2 + px + 8 = 0$ is the square of the other.
11. If α and β are the roots of $ax^2 + bx + c = 0$, find the equation whose roots are $\frac{1}{\alpha}$ and $\frac{1}{\beta}$.
12. Find the relationship between a and c if one root of $ax^2 + bx + c = 0$ is the reciprocal of the other.
13. If one root of $ax^2 + bx + c = 0$ is the triple of the other, prove that $3b^2 - 16ac = 0$.
14. If α and β are the roots of the equation $x^2 + px + q = 0$, find the roots of the equation $x^2 + p^2x + p^2q = 0$ in terms of α and β .

4.4.4 The Curve of a Quadratic Function

Quadratic function: Any function f whose general form is $f(x) = ax^2 + bx + c$ where a, b and c are constants and $a \neq 0$ is called a quadratic function. Completing the square of the RHS of the expression gives

$$\begin{aligned}
 f(x) &= a \left[x^2 + \frac{b}{a}x + \frac{c}{a} \right] = a \left[x^2 + \frac{b}{a}x \right] + c = a \left[x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} - \frac{b^2}{4a^2} \right] + c \\
 &= a \left[x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} \right] + \frac{-b^2}{4a} + \frac{c}{a} = a \left[x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} \right] + \frac{-b^2 - 4ac}{4a} \\
 &= a \left[x + \frac{b}{2a} \right]^2 + \frac{4ac - b^2}{4a}
 \end{aligned}$$

Now whatever value x takes, $\frac{4ac - b^2}{4a}$ is a constant k say. Then $\left[x + \frac{b}{2a} \right]^2 \geq 0$ as it is a square quantity.

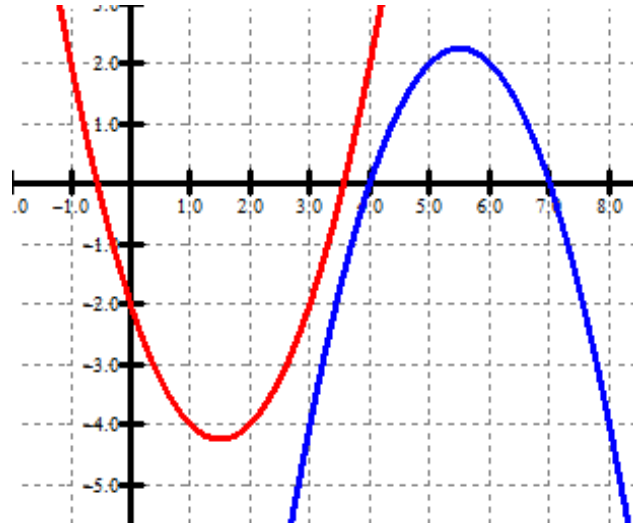
Therefore the function has a general form which is $f(x) = a(\text{zero or a +ve quantity}) + k$. So if a is positive, i.e $a \geq 0$, $f(x)$ has value at least k i.e $f(x)$ has its least value $\frac{4ac-b^2}{4a}$ when $x = -\frac{b}{2a}$. When a is negative, i.e $a < 0$ then $f(x)$ can never be greater than k . i.e $f(x)$ has the greatest value of $\frac{4ac-b^2}{4a}$ when $x = -\frac{b}{2a}$

Now, $x = -\frac{b}{2a}$ is the value of x corresponding to the greatest and least value of $f(x)$. Taking two values of x that are symmetric about $x = -\frac{b}{2a}$ i.e $-\frac{b}{2a} \pm k$ gives $f(-\frac{b}{2a} + k) = f(-\frac{b}{2a} - k) = ak^2 + \frac{4ac-b^2}{4a}$ i.e input of x symmetric about $x = -\frac{b}{2a}$ give the same output of $f(x)$.

From the analysis, we can deduce that when $f(x) = ax^2 + bx + c$, then

$f(x)$ has (i) a least value of $f(-\frac{b}{2a})$ if $a > 0$ (ii) a greatest value of $f(-\frac{b}{2a})$ if $a < 0$ and is symmetric in shape around the line $x = -\frac{b}{2a}$ which is called the axis of the curve.

The diagrams below represent the two alternative graphs of the quadratic function, where $a > 0$ (first red graph) and $a < 0$ (second blue graph) respectively.

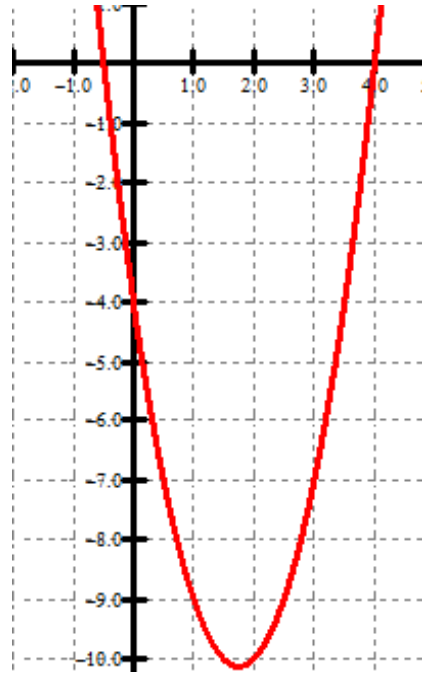


(11)

The curves representing any particular quadratic function can now be sketched using this information. For example, to sketch the curve representing $f(x) = 2x^2 - 7x - 4$, we proceed as follows:

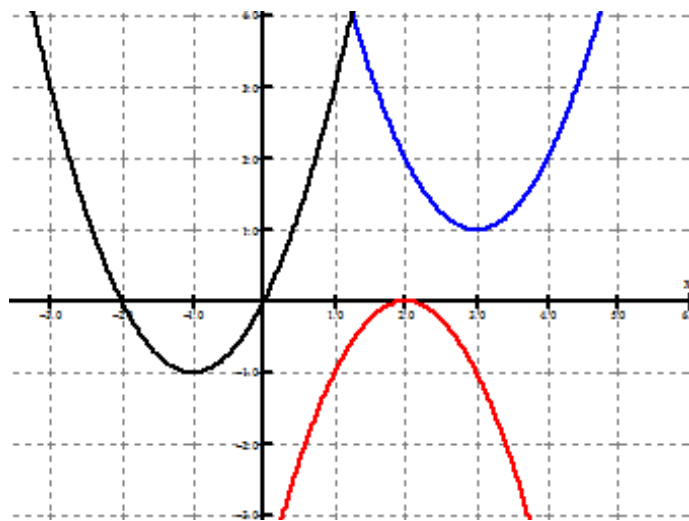
$$a = 2, b = -7 \text{ and } c = -4$$

$\therefore f(-\frac{b}{2a}) = f(\frac{7}{4}) = -\frac{81}{8}$ i.e $f(x)$ has the least value of $-\frac{81}{8}$, when $x = \frac{7}{4}$. To locate the curve accurately on the axes, we need one more pair of x and $f(x)$. $f(0)$ is easy to find i.e $f(0) = -4$. Again we need the points at which f crosses the x axis. This is the solution of $2x^2 - 7x - 4 = 0$ which is $x = \frac{1}{2}$ and $x = 4$ The curve is shown below:



(12)

For any quadratic function $f(x) = ax^2 + bx + c$, the point at which the curve cuts the x -axis is the solution of $f(x) = 0$ i.e $ax^2 + bx + c = 0$. If $ax^2 + bx + c = 0$ has real and distinct roots, the curve cuts the x -axis at two different points which are the roots of the quadratic. If the equation has equal roots, then the curve cuts (touches) the x -axis at 2 coincident points which means it touches the x -axis at the point $x = -\frac{b}{2a}$. When the roots are complex, the curve never touches the x -axis. Below are the possible diagrams of the curve of $f(x) = ax^2 + bx + c$; with black (first) depicting two different roots, red (second) depicting real repeated roots while blue (third) complex roots.



(13)

4.4.5 Quadratic Inequalities

Any inequality relationship that involves a quadratic function is called a quadratic inequality.

The range of values of x that satisfies the quadratic inequality $(x-2)(2x+1)>0$ can be found graphically as follows.

Let $f(x) = (x-2)(2x+1)$. The diagram below shows a sketch of $f(x)$ from which we see that $f(x) > 0$ (i.e the portion of the curve above the y -axis) are for values x greater than 2 and x less than $-\frac{1}{2}$. Therefore the range of values of x satisfying the inequality $(x-2)(2x+1) > 0$

is $\{x : x < -\frac{1}{2} \text{ or } x > 2\}$.

Alternatively, we can solve the inequality by analytic method.

Consider $(x-2)(2x+1) > 0$, $(x-2)$ and $(2x+1)$ must have the same sign.

Either both are negative or both are positive

$\Rightarrow (x-2) > 0$ and $(2x+1) > 0$ or $(x-2) < 0$ and $(2x+1) < 0$

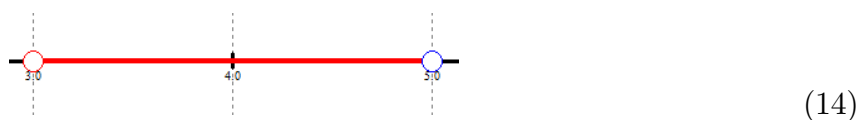
$\Rightarrow$ either $x > 2$ and $x > -\frac{1}{2}$ or $x < 2$ and $x < -\frac{1}{2}$, $\Rightarrow x > 2$ or $x < -\frac{1}{2}$

Example 4.10 : Find the range of values which are satisfied by the inequality $x^2 - 8x + 15 < 0$

Solution: The LHS of the inequality can be factorised as $(x-3)(x-5)$

Graphical method:

The sketch of $f(x) = (x-3)(x-5)$ is given below.



The portion of the curve satisfied by the inequality (portion below the x -axis) are for x between 3 and 5. Therefore the range of values satisfied by the inequality is $3 < x < 5$

Analytic method

Again, this can be solved analytically by observing that if $(x-3)(x-5) < 0$ then $x-3$ and $x-5$ must have different signs. Either $x-3 > 0$ and $x-5 < 0$ or $x-3 < 0$ and $x-5 > 0$.

$\Rightarrow$ either $x > 3$ and $x < 5$ or $x < 3$ and $x > 5$.

But the interval $x < 3$ and $x > 5$ has no solution because if x is greater than 3, it cannot be less than 5.

$\therefore$ The range is $x > 3$ and $x < 5 \Rightarrow x \in (3, 5)$ or $3 < x < 5$.

Exercise 9 Find the range(or ranges) of x satisfying the following inequalities

1. $(x + 2)(x - 2) > 0$.
2. $(x - 1)(x - 2) \geq 0$.
3. $(2x - 1)(x + 1) < 0$.
4. $x^2 - 4x > 5$.
5. $(x - 3)(x + 5) < 0$.
6. $5x^2 > 3x + 2$.
7. $(2 - x)(x + 4) < 0$.
8. $3x > x^2 + 2$.
9. $(3 - 2x)(x + 5) > 0$.
10. $(x - 1)(x + 2) < x(4 - x)$.

Example 4.11 : Find the range of values of k for which the equation $x^2 - kx + k + 3 = 0$ has real roots.

Solution: For $x^2 - kx + k + 3 = 0$ to have real roots, then $b^2 - 4ac \geq 0$.

$$\Rightarrow (-k)^2 - 4(1)(k + 3) \geq 0 \Rightarrow k^2 - 4k - 12 \geq 0.$$

From the graph of $f(k) \equiv k^2 - 4k - 12 \Rightarrow f(k) \equiv (k + 2)(k - 6)$.

From the diagram, we see that $k \leq -2$ and $k \geq 6$.

Example 4.12 : Find the set of values of p for which $f(x) \equiv x^2 - 3px + p > 0$ for all x .

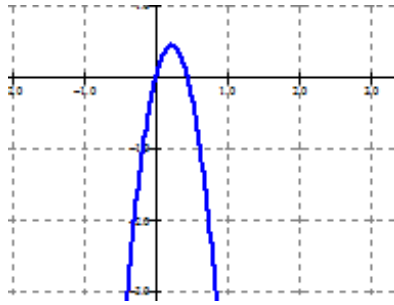
Solution: $f(x) \equiv x^2 - 3px + p = x^2 + 3px + \left(\frac{3p}{2}\right)^2 - \left(\frac{3p}{2}\right)^2 + p \equiv \left(x + \frac{3p}{2}\right)^2 + p - \frac{9p^2}{4}$.

$\therefore$ The least value of $f(x)$ is $p - \frac{9p^2}{4}$.

So for $f(x)$ to be greater than zero for all x , we must have that $p - \frac{9p^2}{4} > 0$.

$$\Rightarrow 4p - 9p^2 > 0 \Rightarrow p(4 - 9p) > 0.$$

The sketch of $f(p) = p(4 - 9p)$ is as below:



(15)

$$y = p(4 - 9p)$$

$$\Rightarrow p > 0 \text{ and } p < \frac{4}{9}.$$

Therefore, $f(x) > 0$ for all real x holds for the set of values of p such that $0 < p < \frac{4}{9}$.

Example 4.13 : Find the range of values of x for which $2x - 1 < x^2 - 4 < 12$

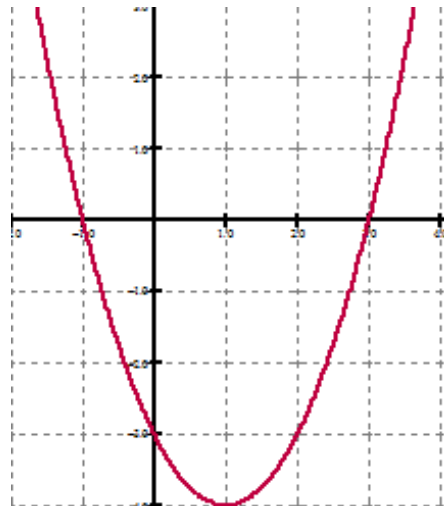
Solution: There are two inequality relationships here, viz

(a) $2x - 1 < x^2 - 4$ and (b) $x^2 - 4 < 12$.

For (a) $2x - 1 < x^2 - 4 \Rightarrow -x^2 + 2x + 3 < 0 \Rightarrow x^2 - 2x - 3 > 0 \Rightarrow (x - 3)(x + 1) > 0$.

Let $f(x) \equiv (x - 3)(x + 1)$. The graph of $f(x)$ is as drawn below.

$$y = (x - 3)(x + 1).$$



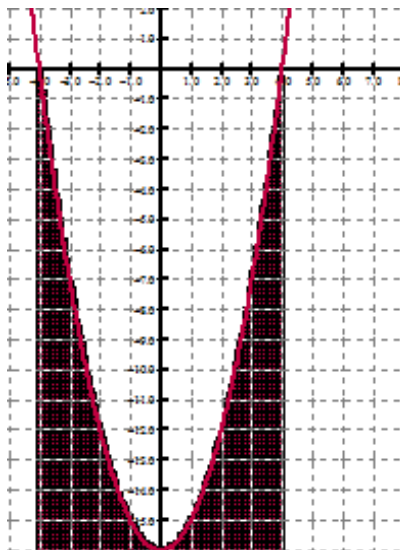
(16)

From the graph, we can see that the region satisfied by (a) is $x < -1$ and $x > 3$.

For (b) $x^2 - 4 < 12 \Rightarrow x^2 - 16 < 0 \Rightarrow (x + 4)(x - 4) < 0$.

Let $g(x) = (x + 4)(x - 4)$. The graph of g is as sketched below:

$$y = (x + 4)(x - 4)$$



(17)

From the graph of $g(x)$, then $g(x) < 0$, if $-4 < x < 4$

The range of $f(x)$ and $g(x)$ can be represented on the same number line (see chapter one) to obtain the range of values that satisfy (a) and (b) simultaneously. From the above we see that

the ranges of x that satisfy both a and b (this is where the lines overlap) are $-4 < x < -1$ and $3 < x < 4$.

Exercise 10

- Find the range of values of k for which $x^2 + (k - 3)x + k = 0$ has (a) real and distinct roots (b) roots of the same sign.
- If x is real and $x^2 + (2 - k)x + 1 - 2k = 0$, show that k cannot lie between certain limits and find these limits.
- Find the limitations required on the values of real number c in order that the equation $x^2 + 2cx - c + 2 = 0$ shall have real roots.
- Prove that if $x^2 > k(x + 1)$ for all real x then $-4 < k < 0$.
- Find the condition that must be satisfied by k in order that the expression $2x^2 + 6x + 1 + k(x^2 + 2)$ may be positive for all real values of x .
- Find the the range of values of x for which $x - 4 < x(x - 4) \leq 5$.
- (a) If $x = 2$ is a root of the equation $\alpha^2 x^2 + 2(2\alpha - 5)x + 8 = 0$, find the possible value(s) of α and the corresponding value(s) of x . (b) Find the range(s) of possible values of the real number α if $\alpha^2 x^2 + 2(2\alpha + 5)x + 8 = 0$ for all real values of x .

8. Determine for each of the expressions $f(x)$ and $g(x)$ the range(s) of values of x for which it is positive. Give your answer correct to 2 decimal places. Explain briefly the reason for your answers. (a) $f(x) = x^2 + 4x - 6$ (b) $g(x) = -x^2 - 8x + 2$.
9. (a) State the range of values of x for which $2x^2 + 5x - 12$ is negative. (b) The value of a constant is such that the quadratic function $f(x) = x^2 - 4x + a + 3$ is never negative. Determine the nature of roots of the function $f(x) = (x^2 + 1)(a - 1)$. Deduce the values of a for which the equation has equal roots.
10. By eliminating x and y from the equation $\frac{1}{x} + \frac{1}{y} = 1$, $x + y = a$ and $\frac{y}{x} = m$, where $a \neq 0$. Obtain a relationship between m and a given that a is real. Determine the ranges of values of a for which m is real.
11. If $a > 0$, prove that the quadratic expression $ax^2 + bx + c$ is positive for all real values of x , when $b^2 < 4ac$. Hence find the range of values of p for which the quadratic function $f(x) \equiv 4x^2 + 4px - (3p^2 + 4p - 3)$ is positive for all real values of x . Illustrate your result by sketching the graphs for each case $p = 0$, $p = 1$.
12. (a) If a is a positive constant, find the set of values of x for which $a(x^2 + 2x - 8)$ is negative. Find the value of a if this function has a minimum value of 11. (b) Find two quadratic functions of x which are 0 at $x = 1$, which take value 10 when $x = 0$ and has a maximum value of 18. Sketch the graph of these two functions.
13. Find the set of values of k for which $f(x) \equiv 3x^2 - 5x - k$ is greater than unity for all values of x . Show that for all k , the minimum value of $f(x)$ occurs when $x = \frac{5}{6}$. Find k if this minimum value is 0.
14. Prove that $3x^2 - 4x + 2 > 0$ for all real values of x .
15. Find the value of $k (\neq 1)$ so that the quadratic function $k(x + 2)^2 - (x - 1)(x - 2)$ is equal to 0. Find also (a) the range of k for which the function possesses a minimum value, (b) The range of values of k for which the value of the function never exceeds 12.5. Sketch the graph of the function for $k = \frac{1}{2}$ and for $k = 2\frac{1}{2}$.
16. Find the set of values of x for which $2x^2 + 3x + 2 > 0$
17. The root of the equation $9x^2 + 6x + 1 = 4kx$ where k is a constant are denoted by α and β
 (a) Show that the equation whose roots are $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ is $x^2 + 6x + 9 = 4kx$. (b) Find the set of values of k for which α and β are real. (c) Find also the values of k for which α and β are real and positive.
18. Define the function f as $f : x \rightarrow x^2 - 3x - 4$, for $x \in \mathbb{R}$. Find the range of f and the values of x for which $f(x) = 0$.

4.5 The Remainder and Factor Theorems

The remainder theorem enables us to find the remainder when certain algebraic expressions are divided by a given factor without necessarily performing the actual division. An immediate consequence of the remainder theorem is the factor theorem.

4.5.1 The Remainder Theorem

If $P(x)$ is a polynomial in x , then the remainder when $P(x)$ is divided by $(x - b)$ is $P(b)$.

Let $P(x)$ be a polynomial in x , $Q(x)$ be the quotient when $P(x)$ is divided

by $(x - b)$ and R be the remainder. Thus $P(x) = (x - b)Q(x) + R$.

$$\therefore P(b) = (b - b)Q(b) + R = 0Q(b) + R = R.$$

4.5.2 The Factor theorem

If $P(x)$ vanishes when $x = a$, then $(x - a)$ is a factor of $P(x)$. Thus, $P(a) = 0$ if $(x - a)$ is a factor of $P(x)$.

Example 4.14 : Find the remainder when $x^3 + 5x^2 - 2x - 1$ is divided by $(x + 3)$.

Solution: Let $f(x) = x^3 + 5x^2 - 2x - 1$. Applying the remainder theorem we have:

$$f(-3) = (-3)^3 + 5(-3)^2 - 2(-3) - 1 = -27 + 45 + 6 - 1 = 23.$$

$\therefore$ the remainder is 23.

Example 4.15 : Find the values of the constants g and h if $(x - 1)$ and $(3x - 1)$ are both factors of the polynomial $3x^4 + gx^3 - 6x^2 + hx - 1$.

Solution: Let $f(x) = 3x^4 + gx^3 - 6x^2 + hx - 1$;

if $(x - 1)$ is a factor, then $f(1) = 0$,

thus $f(1) = 3 + g - 6 + h - 1 = 0$, i.e. $g + h = 4$... (i)

If $(3x - 1)$ is a factor, then $f(\frac{1}{3}) = 0$,

$$\text{thus } f(\frac{1}{3}) = 3(\frac{1}{3})^4 + g(\frac{1}{3})^3 - 6(\frac{1}{3})^2 + h\frac{1}{3} - 1 = 0.$$

$$\Rightarrow \frac{1}{27} + \frac{g}{27} - \frac{2}{3} + \frac{h}{3} - 1 = 0,$$

$$\text{i.e. } \frac{g}{27} + \frac{h}{3} = \frac{44}{27} \text{ or } g + 9h = 44 \dots (ii).$$

From (i) and (ii) we have a system of simultaneous equations and solving we have $g = -1$ and $h = 5$.

Example 4.16 : Find the value of m if $(x - 2)$ is a factor of $x^3 + mx^2 - 4x + 4$. Find the other factors of the expression.

Solution: Let $f(x) = x^3 + mx^2 - 4x + 4$; Then $f(2) = 2^3 + 4m - 8 + 4 = 0 \implies 4m = -4$
 $m = -1$. The expression is $x^3 - x^2 - 4x + 4$.

Using factor theorem let's check whether $(x \pm 1)$ and $(x + 2)$ are factors:

$f(-1) = -1 - 1 + 4 + 4 = 6$, thus $(x + 1)$ is not a factor.

$f(1) = 1 - 1 - 4 + 4 = 0$, hence $(x - 1)$ is a factor.

$f(-2) = (-2)^3 - 4 - 4(-2) + 4 = -8 - 4 + 8 + 4 = 0$, giving $(x + 2)$ as a factor. Now,

$$(x - 2)(x - 1)(x + 2) = (x^2 - 3x + 2)(x + 2) = x^3 - x^2 - 4x + 4.$$

$\therefore$ the other factors are $(x - 1)$ and $(x + 2)$.

Example 4.17 : The equations $x^2 + 9x + 2 = 0$ and $x^2 + hx + 5 = 0$ have a common root. Find the quadratic equation giving the two possible values of h .

Solution: Let the common root be a . Then $(x - a)$ will be a factor of both $x^2 + 9x + 2$ and $x^2 + hx + 5$.

Thus $a^2 + 9a + 2 = 0$ and $a^2 + ha + 5 = 0$. i.e. $a^2 + 9a + 2 = a^2 + ha + 5$

i.e. $(9 - h)a = 3$. $\therefore a = \frac{3}{9-h}$, substituting this in $a^2 + 9a + 2 = 0$;

we get: $\frac{9}{(9-h)^2} + \frac{27}{9-h} + 2 = 0$. i.e. $9 + 27(9 - h) + 2(9 - h)^2 = 0$

i.e. $9 + 27 \cdot 9 - 27h + 2(9^2 - 18h + h^2) = 0$, or $9 + 27 \cdot 9 - 27h + 2 \cdot 9^2 - 36h + 2h^2 = 0$

$2h^2 - 63h + 414 = 0$ is the required quadratic equation for h .

Exercise 11

- Find the remainder when: **(i)** $x^4 - 6x^2 + 1$ is divided by $(x + 2)$ and **(ii)** $x^2 - 1$ is divided by $(x + 1)$.
- Find the value of the constant k if the remainder is 22 when the expression $3x^3 + kx^2 - 4x + 6$ is divided by $(x - 4)$.
- Find the values of m and n if $(x - 2)$ and $(x - 3)$ are both factors of $x^3 + mx^2 + nx + 6$.
- Find the value of the constant k for which the polynomial $x^4 + x^3 + kx^2 + 5x - 10$ has $(x + 2)$ as a factor.
- The expression $gx^2 + hx - 2$ has $(x + 2)$ as a factor. When the expression is divided by $(x - 1)$, the remainder is 6. Find the values of g and h .

6. When $x^2 + mx + n$ is divided by $x - h$ the remainder is the same as when the quadratic expression is divided by $x - 2h$. Find two possible values of h .

5 Trigonometry

5.1 Objectives

In this module we:

- Revisit the trigonometry ratios introduced by the study of the relationships between the sides and the angles of a triangle.
- Explore the various trigonometric ratios for some special angles between 0° and 360° .
- Revisit the relationship between degree and radian measure of an angle and do some conversions.
- Explore the special case of even and odd functions including the trigonometric ratios.
- Plot and sketch graphs of trigonometric functions.
- Establish the fundamental trigonometric identities, study multiple angles, the sine and cosine rules and then discuss the factor formulae.
- Investigate the general solutions of equations of the form $y = a \sin x + b \cos x$, equations involving factor formulae and parametric equations.

5.2 Learning Outcomes

At the end of this module students should be able to:

- Identify and use all the trigonometric ratios.
- Recall and routinely use the trigonometric ratios for some special angles between 0° and 360° .
- Do conversions from degree to radian and radian to degree.
- Manipulate the trigonometric functions, identify their properties and plot their graphs with ease.
- Establish the fundamental trigonometric identities, including those involving multiple angles.
- Use the sine and cosine rules, the factor formulae, and parametric equations.

5.3 Learning Activities

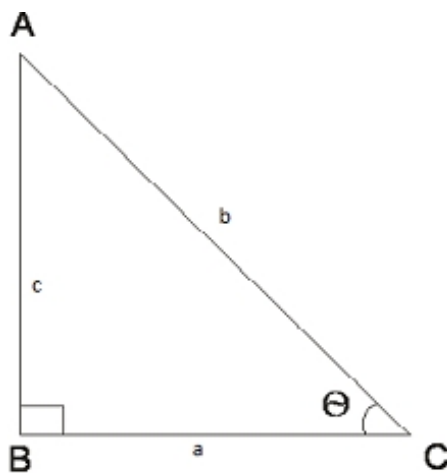
Students should:

- Explore notes, power point presentations and exercises, individually and in groups.
- Engage in discussions with lecturers and their peers using forums and other interactive facilities on the Moodle site.
- Take and pass online and other self assessment tests.
- Solve relevant questions from past MTH111 Examinations.

5.4 The Trigonometric Ratios

Introduction

Trigonometry has been introduced before as a branch of mathematics that deals with the study of the relationships between the sides and the angles of a triangle. These relationships are defined in terms of trigonometric ratios such as sine(sin), cosine(cos), and tangent(tan). Relative to a right-angled triangle ABC below, with the angle marked θ (theta) as the reference angle, the side AB is called the opposite side and the side BC is called the adjacent side. The side AC is called the hypotenuse and it is the side opposite the right angle.



(18)

We define the trigonometric ratios as follows, where a, b and c are the sides of a triangle, opposite angles A, B and C respectively.

$$\sin \theta = \frac{\text{Opposite side to angle } \theta}{\text{Hypotenuse}} = \frac{c}{b}, \quad \cos \theta = \frac{\text{Adjacent side to angle } \theta}{\text{Hypotenuse}} = \frac{a}{b},$$

From these 2 basic trigonometric ratios, we can define many others.

We first of all define $\tan \theta$ by

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{\text{Opposite side to angle } \theta}{\text{Adjacent side to angle } \theta} = \frac{a}{c}$$

From the right angled triangle ABC, angle $A = 90^\circ - \theta$, and we see that

$$\sin(90^\circ - \theta) = \frac{a}{b} = \cos \theta$$

Similarly

$$\cos(90^\circ - \theta) = \frac{c}{b} = \sin \theta$$

.

Note that angles θ and $(90^\circ - \theta)$ are complementary (that is, they add up to 90°) hence we conclude that the sine of an angle is always equal to the cosine of its complement.

Other trigonometric ratios such as secant, cosecant and cotangent can also be defined in terms of sine, cosine and tangent thus:

$$\sec \theta = \frac{1}{\cos \theta}, \quad \csc \theta = \frac{1}{\sin \theta}, \quad \cot \theta = \frac{1}{\tan \theta} = \frac{\cos \theta}{\sin \theta}$$

.

We derive from the figure above the following trigonometric identities:

$$\tan(90^\circ - \theta) = \frac{a}{c} = \cot \theta.$$

We also note that $(\sin \theta)^2 + (\cos \theta)^2 = \left(\frac{c}{b}\right)^2 + \left(\frac{a}{b}\right)^2 = \frac{a^2+c^2}{b^2} = \frac{b^2}{b^2} = 1$, (since by Pythagoras' theorem, $a^2 + b^2 = c^2$).

$$\therefore \sin^2 \theta + \cos^2 \theta = 1$$

Dividing through the equation above by $\sin^2 \theta$ and then by $\cos^2 \theta$, to obtain respectively,

$$1 + \cot^2 \theta = \csc^2 \theta$$

and

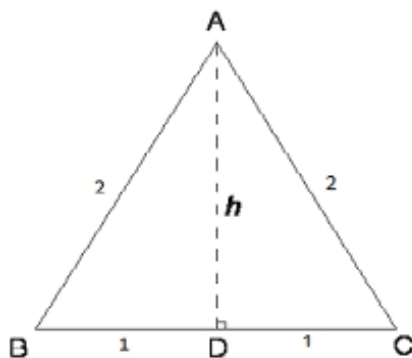
$$1 + \tan^2 \theta = \sec^2 \theta$$

These are the basic trigonometric identities.

5.4.1 Trigonometric Ratios of Some Special Angles

$$\{0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ\}$$

We consider an equilateral triangle ABC of side 2 units, with a perpendicular bisector AD of the base BC drawn from A to D which also bisects the vertical angle as shown below:



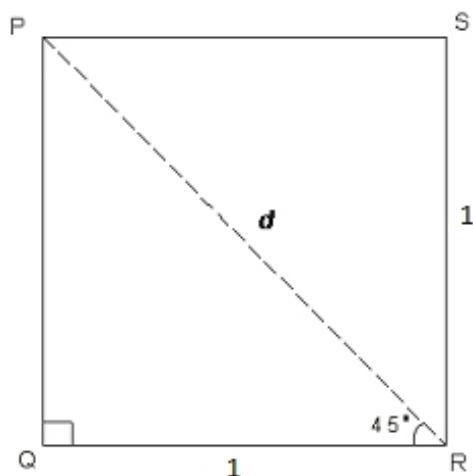
(19)

From the triangle above, the height h of the triangle $ABC = h = AD = \sqrt{2^2 - 1} = \sqrt{4 - 1} = \sqrt{3}$.

$$\therefore \sin 30^\circ = \frac{1}{2}, \cos 30^\circ = \frac{\sqrt{3}}{2}, \text{ and } \tan 30^\circ = \frac{1}{\sqrt{3}}.$$

$$\text{Similarly } \sin 60^\circ = \frac{\sqrt{3}}{2}, \cos 60^\circ = \frac{1}{2}, \text{ and } \tan 60^\circ = \sqrt{3}.$$

Now, consider the unit square $PQRS$ with the diagonal PR .



(20)

From the figure the diagonal $PR = d$ is given by:

$$d^2 = 1^2 + 1^2 = 2, \therefore d = \sqrt{2}. \text{ Hence:}$$

$$\sin 45^\circ = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}, \quad \cos 45^\circ = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2},$$

We deduce that $\tan 45^\circ = 1$.

For trigonometric ratios of angles between 0° and 90° , we can consider the right-angled triangle PQR , right angled at Q with angle $PRQ = \theta$

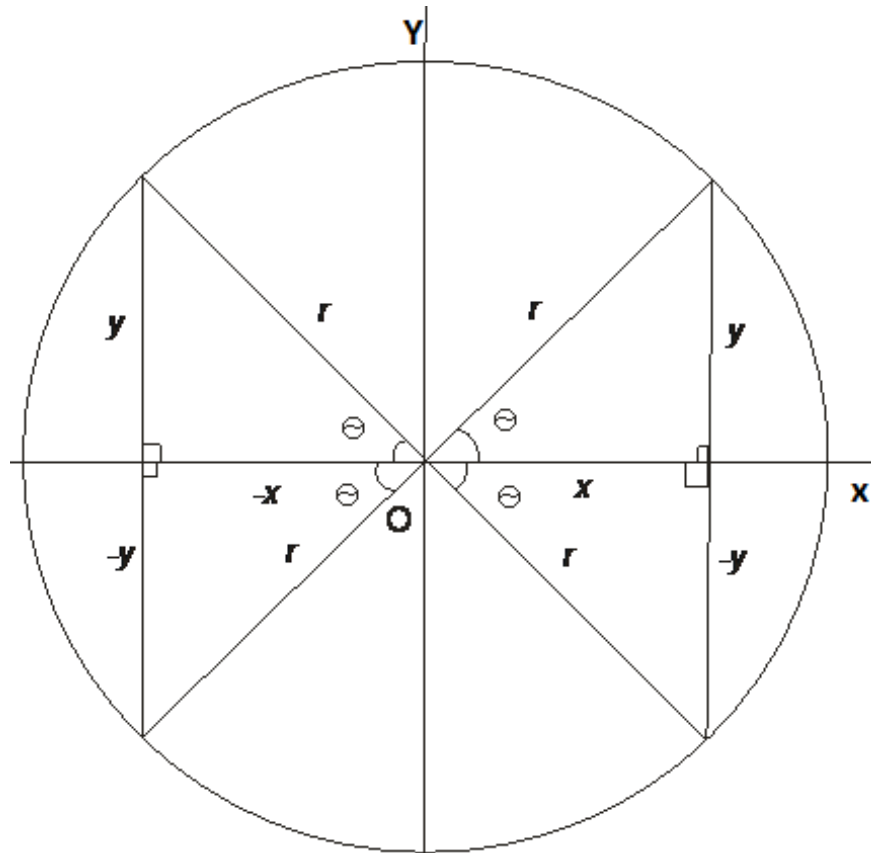
Then $\sin \theta = \frac{PQ}{PR}$. As P approaches Q , θ approaches 0° , i.e. as $P \rightarrow Q$, $\theta \rightarrow 0^\circ$, hence $PR \rightarrow QR$.

$\therefore \sin 0^\circ = \frac{0}{|QR|} = 0$, and thus $\cos 90^\circ = 0$, since the *sine* of an angle is always equal to the *cosine* of its complement.

Similarly, $\cos 0^\circ = \frac{QR}{QR} = 1$ (since $\theta \rightarrow 0^\circ$ implies $PR \rightarrow QR$).

5.4.2 Trigonometric Ratios of Angles between 0° and 360°

Consider the circle of radius r below



(21)

In the first Quadrant,

Second Quadrant

$$\sin \theta = \frac{y}{r} \text{ (positive)}$$

$$\cos \theta = \frac{x}{r} \text{ (positive)}$$

$$\tan \theta = \frac{y}{x} \text{ (positive)}$$

All the trigonometric ratios in first quadrant are positive.

.

Third Quadrant

$$\sin \theta = \frac{-y}{r} \text{ (negative)}$$

$$\cos \theta = \frac{-x}{r} \text{ (negative)}$$

$$\tan \theta = \frac{-y}{-x} = \frac{y}{x} \text{ (positive)}$$

In the third quadrant, only tangent is positive. In the fourth quadrant, only cosine has a positive value.

$$\sin \theta = \frac{y}{r} \text{ (positive)}$$

$$\cos \theta = \frac{-x}{r} \text{ (negative)}$$

$$\tan \theta = \frac{y}{-x} \text{ (negative)}$$

Only sine has a

positive value in

the second quadrant

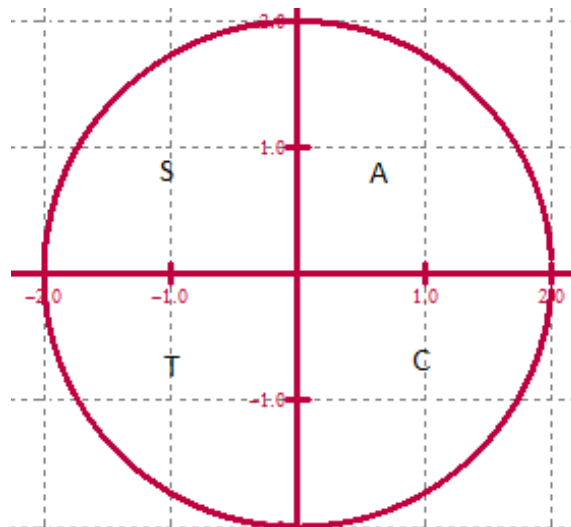
Fourth Quadrant

$$\sin \theta = \frac{-y}{r} \text{ (negative)}$$

$$\cos \theta = \frac{x}{r} \text{ (positive)}$$

$$\tan \theta = \frac{-y}{x} \text{ (negative)}$$

Some people use acronyms such as **CAST** or **ACTS** to enable them remember the signs of trigonometric ratios of angles of all magnitudes, see diagram below:



(22)

Starting from the fourth quadrant and moving anti-clockwise, we use **CAST**, while starting from the first quadrant and moving in the clockwise direction, we use **ACTS**.

We note that in the first quadrant,

$$\sin \theta = + \sin \theta, \cos \theta = + \cos \theta, \text{ and } \tan \theta = + \tan \theta$$

In the second quadrant,

$$\sin \theta = +\sin(180^\circ - \theta), \cos \theta = -\cos(180^\circ - \theta), \text{ and } \tan \theta = -\tan(180^\circ - \theta)$$

The third quadrant has

$$\sin \theta = -\sin(\theta - 180^\circ), \cos \theta = -\cos(\theta - 180^\circ), \text{ and } \tan \theta = +\tan(\theta - 180^\circ)$$

and the fourth quadrant has

$$\sin \theta = -\sin(360^\circ - \theta), \cos \theta = +\cos(360^\circ - \theta), \text{ and } \tan \theta = -\tan(360^\circ - \theta)$$

Example 5.1 : Express in terms of the trigonometric ratios of positive acute angles: (a) $\sin 160^\circ$ (b) $\cos(400^\circ)$ (c) $\sin(-200^\circ)$ (d) $\cos(-300^\circ)$ (e) $\tan(-900^\circ)$.

Solution: 1 (a) $\sin 160^\circ = \sin(180^\circ - 160^\circ) = \sin 20^\circ$ (b) $\cos 400^\circ = \cos(400^\circ - 360^\circ) = \cos 40^\circ$ (c) $\sin(-200^\circ) = \sin(-200^\circ + 360^\circ) = \sin 160^\circ = \sin(180^\circ - 160^\circ) = \sin 20^\circ$, or $\sin(-200^\circ) = -\sin 200^\circ = -(-\sin(200^\circ - 180^\circ)) = \sin 20^\circ$ (d) $\cos(-300^\circ) = \cos(-300^\circ + 360^\circ) = \cos 60^\circ$ (e) $\tan(-900^\circ) = \tan(-900^\circ + 360^\circ) = \tan(-540^\circ) = \tan(-540^\circ + 360^\circ) = \tan(-180^\circ) = \tan(-180^\circ + 360^\circ) = \tan(180^\circ) = -\tan(180^\circ - 180^\circ) = \tan 0^\circ$.

Example 5.2 : Evaluate without using tables:

$$(a) \cos 210^\circ \quad (b) \cos^2 330^\circ \quad (c) 2 \sin 120^\circ \cos 120^\circ \quad (d) \cos^2 300^\circ - \sin^2 300^\circ$$

Solution: (a) $\cos 210^\circ = -\cos(210^\circ - 180^\circ) = -\cos 30^\circ = -\frac{\sqrt{3}}{2}$ (b) $\cos^2 330^\circ = \cos^2(360^\circ - 330^\circ) = \cos^2 30^\circ = (\frac{\sqrt{3}}{2})^2 = \frac{3}{4}$. (c) $2 \sin 120^\circ \cos 120^\circ = 2 \sin(180^\circ - 120^\circ)(-\cos(180^\circ - 120^\circ)) = -2 \sin 60^\circ \cos 60^\circ = -2 \times \frac{\sqrt{3}}{2} \times \frac{1}{2} = -\frac{\sqrt{3}}{2}$. (d) $\cos^2 300^\circ - \sin^2 300^\circ = \cos^2(360^\circ - 300^\circ) - (-\sin(360^\circ - 300^\circ))^2 = \cos^2 60^\circ - \sin^2 60^\circ = (\frac{1}{2})^2 - (\frac{\sqrt{3}}{2})^2 = -\frac{1}{2}$.

Exercise 12 Find in its simplest form, the value of $\frac{\sin 3\theta + \tan \theta}{1 + \cos 2\theta}$ if $\theta = 60^\circ$.

Example 5.3 : Given that $\sin A = \frac{5}{13}$ and A is an obtuse angle, find without using tables, the value of $\frac{2 \sin A - \cos A}{\frac{1}{2} - \tan A}$.

Solution: Note that $x^2 + y^2 = r^2$. Therefore, $x = \sqrt{13^2 - 5^2} = \sqrt{144} = 12$ units.

$$\therefore \cos A = -\frac{12}{13}, \tan A = -\frac{5}{12}.$$

$$\therefore \frac{2 \sin A - \cos A}{\frac{1}{2} - \tan A} = \frac{2 \times \frac{5}{13} - (-\frac{12}{13})}{\frac{1}{2} - (-\frac{5}{12})} = \frac{\frac{10}{13} + \frac{12}{13}}{\frac{1}{2} + \frac{5}{12}} = \frac{24}{13}.$$

Example 5.4 : Show that $\frac{\sin \alpha + \tan \alpha}{\cot \alpha + \sec \alpha} = \sin \alpha \tan \alpha$.

Solution: LHS = $\frac{\sin \alpha + \tan \alpha}{\frac{1}{\tan \alpha} + \frac{1}{\sin \alpha}} = \frac{\sin \alpha + \tan \alpha}{\frac{\sin \alpha + \tan \alpha}{\sin \alpha \tan \alpha}} = (\sin \alpha + \tan \alpha) \left(\frac{\sin \alpha \tan \alpha}{\sin \alpha + \tan \alpha} \right) = \sin \alpha \tan \alpha = \text{RHS}.$

Exercise 13

- Express in terms of the trigonometric ratios of positive acute angles: (i) $\sin 140^\circ$ (ii) $\sin(-945^\circ)$ (iii) $\cos 210^\circ$
- Evaluate without using tables, giving your answer in simplified surd form where necessary: (i) $\sin 135^\circ$ (ii) $\sin^2 330^\circ$ (iii) $\cos^2 210^\circ - \sin^2 210^\circ$ (iv) $\frac{1 + \cos 210^\circ}{1 - \cos 210^\circ}$ (v) $\frac{\tan 330^\circ + \tan 240^\circ}{1 - \tan 330^\circ \tan 240^\circ}$
- If $\sin \theta = p$ and θ is obtuse, find expressions for $\cos \theta$ and $\tan \theta$ in terms of p .
- If $\tan A = \frac{4}{3}$, where A is an acute angle, find the value of $\frac{4 \cos A - 3 \sin A}{4 \cos A + 3 \sin A}$.
- Show that $\frac{\cos \beta}{1 + \sin \beta} = \sec \beta - \tan \beta$.
- Show that $(1 + \tan \theta)^2 + (1 - \tan \theta)^2 = \frac{2}{\cos^2 \theta}$.

5.4.3 Degree and Radian Measure of an Angle

Two kinds of units commonly used in measuring angles are the degree and the radian. The degree is a more familiar and more commonly used unit of angular measure in elementary mathematics. A degree ($^\circ$) is defined as $\frac{1}{360}$ of a complete rotation. It has sub-divisions such as the minute ($'$) which is $\frac{1}{60}$ of a degree and the second ($''$) which is $\frac{1}{60}$ of a minute.

The radian measure is employed almost exclusively in advanced mathematics and in many branches of science and engineering. In this section, we introduce the concept of radian and examine the relationship between degrees and radians.

A radian is the angle subtended at the centre of a circle, by an arc equal in length to the radius of the circle. Let us consider a circle with centre O and radius r and the arc PQ be equal in length to r and θ is the angle subtended at the centre of the circle, and $\theta = 1$ rad.

Example 5.5 : Express each of the given angles in degrees: (i) $\frac{2\pi}{3}$ (ii) $\frac{\pi}{30}$ (iii) $-\frac{5\pi}{9}$ (iv) $\frac{3\pi}{8}$.

Solution: Note that $1\text{rad} = \frac{180^\circ}{\pi}$.

$$\therefore (i) \frac{2\pi}{3} = \frac{2\pi}{3} \times \frac{180^\circ}{\pi} = 120^\circ \quad (ii) \frac{\pi}{30} = \frac{\pi}{30} \times \frac{180^\circ}{\pi} = 6^\circ$$

$$(iii) -\frac{5\pi}{9} = -\frac{5\pi}{9} \times \frac{180^\circ}{\pi} = -100^\circ \quad (iv) \frac{3\pi}{8} = \frac{3\pi}{8} \times \frac{180^\circ}{\pi} = 67\frac{1}{2}^\circ.$$

Exercise 14

1. Express the following in degrees: (i) $\frac{\pi}{7}\text{rad}$ (ii) $\frac{4\pi}{7}\text{rad}$ (iii) $\frac{8\pi}{9}\text{rad}$ (iv) 0.45rad (v) $\frac{\pi}{9}\text{rad}$.
2. Express the following in radians: (i) 36° (ii) $112^\circ 40'$ (iii) 135° (iv) $1'$ (v) 270° (vi) $394^\circ 32'$.

5.4.4 Even and Odd Functions

Definition 5.1 : *Even function:* A function f is even if $f(-x) = f(x)$, for all x in its domain of definition. The function is odd if $f(-x) = -f(x)$.

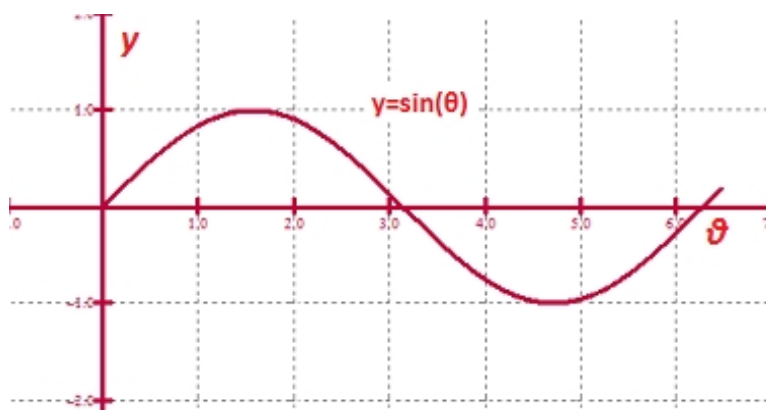
$f(x) = x^2$, $f(\alpha) = \cos \alpha$, are even functions while $f(x) = x^3$, $f(\alpha) = \sin \alpha$ are odd functions. Trigonometric ratios can be even or odd functions.

Definition 5.2 : *Periodic Function:* A function f is said to be periodic if there exists $T > 0$ such that $f(x + T) = f(x)$. The function has period T , if T is the smallest value for which the relation holds. A function involving trigonometric ratios is called a trigonometric function. All trigonometric functions are periodic.

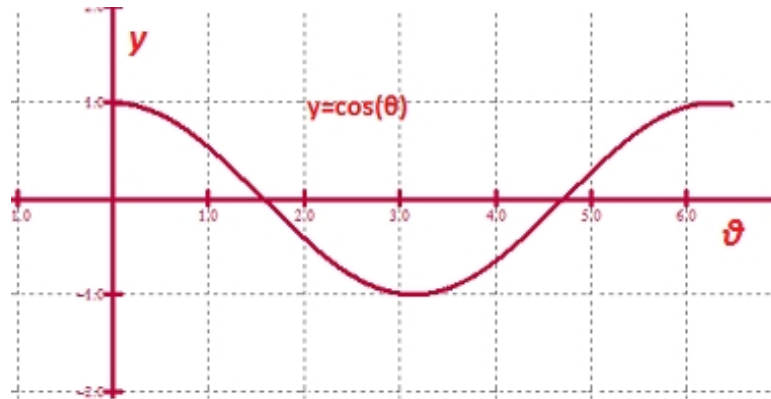
$f(x) = \sin x$, $g(x) = \cos x$ are periodic with period $T = 2\pi$ rads.

5.4.5 Graphs of Trigonometric Functions

The graphs of $y = \sin \theta$ $y = \cos \theta$, for $0 \leq \theta \leq 2\pi$. $\cos \theta$, $0 \leq \theta \leq 2\pi$.



(23)



(24)

5.4.6 Fundamental Trigonometric Identities

We study in this subsection, some more important identities involving trigonometric functions. These identities can be employed to simplify or change the form of trigonometric expressions.

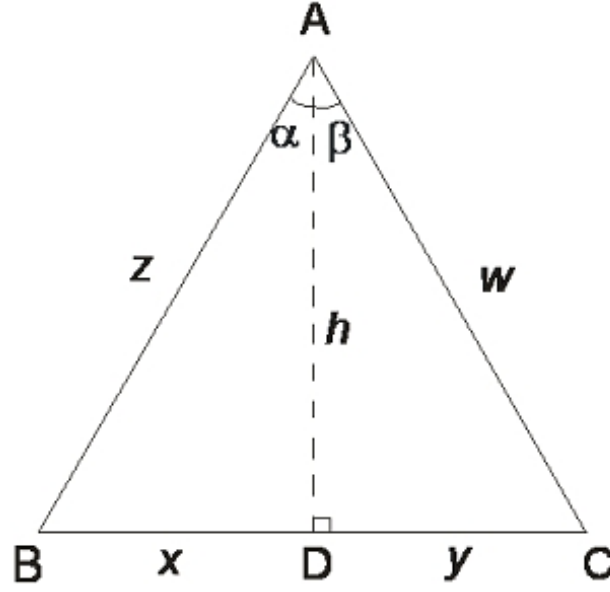
5.4.7 Addition Formulae

The following identities are often referred to as addition formulae or addition theorems in trigonometry:

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

Consider the triangle ABC and construct a perpendicular bisector AD of BC .



(25)

Using the sine rule, we obtain $\frac{\sin \angle BAC}{|BC|} = \frac{\sin \angle ABC}{|AC|}$

$$\therefore \frac{\sin(\alpha + \beta)}{x + y} = \frac{\sin(90^\circ - \alpha)}{w} = \frac{\frac{h}{z}}{w} = \frac{h}{wz}.$$

$$\therefore \sin(\alpha + \beta) = \frac{h(x + y)}{wz} = \frac{hx}{wz} + \frac{hy}{wz} = \left(\frac{x}{z}\right)\left(\frac{h}{w}\right) + \left(\frac{h}{z}\right)\left(\frac{y}{w}\right) = \sin \alpha \cos \beta + \cos \alpha \sin \beta.$$

Using cosine rule, we obtain in a similar manner

$$\cos \angle BAC = \frac{|AC|^2 + |AB|^2 - |BC|^2}{2|AC||AB|} = \frac{1}{2} \left[\frac{|AC|}{|AB|} + \frac{|AB|}{|AC|} - \frac{|BC|}{|AC|} \cdot \frac{|BC|}{|AB|} \right]$$

$$\begin{aligned} \text{This implies } \cos(\alpha + \beta) &= \frac{1}{2} \left[\frac{w}{z} + \frac{z}{w} - \frac{(x + y)}{w} \cdot \frac{(x + y)}{z} \right] = \frac{1}{2} \left[\frac{w^2 + z^2}{wz} - \frac{x^2 + 2xy + y^2}{wz} \right] \\ &= \frac{1}{2} \left[\frac{w^2 + z^2 - x^2 - 2xy - y^2}{wz} \right] = \frac{1}{2} \left[\frac{2h^2 - 2xy}{wz} \right], \text{ since } w^2 - y^2 = z^2 - x^2 = h^2. \end{aligned}$$

$$= \frac{h}{z} \cdot \frac{h}{w} - \frac{x}{z} \cdot \frac{y}{w} = \cos \alpha \cos \beta - \sin \alpha \sin \beta.$$

Thus,

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta, \quad \cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

Putting $\beta = -\beta$ in the above formulae, we obtain

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta, \quad \cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\text{Thus, } \tan(\alpha + \beta) = \frac{\sin(\alpha + \beta)}{\cos(\alpha + \beta)} = \frac{\sin \alpha \cos \beta + \cos \alpha \sin \beta}{\cos \alpha \cos \beta - \sin \alpha \sin \beta} = \frac{\frac{\sin \alpha \cos \beta}{\cos \alpha \cos \beta} + \frac{\cos \alpha \sin \beta}{\cos \alpha \cos \beta}}{\frac{\cos \alpha \cos \beta}{\cos \alpha \cos \beta} - \frac{\sin \alpha \sin \beta}{\cos \alpha \cos \beta}} = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}.$$

$$\therefore \tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

Similarly,

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

Example 5.6 : Find without using tables the value of (i) $\sin \frac{5\pi}{12}$ (ii) $\cos \frac{\pi}{12}$

Solution: (i) $\sin \frac{5\pi}{12} = \sin(\frac{\pi}{4} + \frac{\pi}{6}) = \sin \frac{\pi}{4} \cos \frac{\pi}{6} + \cos \frac{\pi}{4} \sin \frac{\pi}{6} = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{2}}{2} (\sqrt{3} + 1)$

(ii) $\cos \frac{\pi}{12} = \cos(\frac{\pi}{4} - \frac{\pi}{6}) = \cos \frac{\pi}{4} \cos \frac{\pi}{6} + \sin \frac{\pi}{4} \sin \frac{\pi}{6} = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2}$
 $= \frac{\sqrt{2}}{2} (\sqrt{3} + 1).$

Note that $\frac{\pi}{12}$ and $\frac{5\pi}{12}$ are complementary angles, hence $\sin \frac{5\pi}{12} = \cos \frac{\pi}{12}$.

Example 5.7 : If $\sin \alpha = \frac{5}{13}$, and $\cos \beta = \frac{1}{5}$, find the possible values of $\sin(\alpha + \beta)$.

Solution: $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$. If α and β are acute angles, then $\cos \alpha = \frac{12}{13}$, $\sin \beta = \frac{\sqrt{24}}{5}$,

hence, $\sin(\alpha + \beta) = \frac{5}{13} \times \frac{1}{5} + \frac{12}{13} \times \frac{\sqrt{24}}{5} = \frac{5+24\sqrt{6}}{65}$,

But if α is acute and $\frac{3\pi}{2} \leq \beta \leq 2\pi$, then $\sin \beta = -\frac{\sqrt{24}}{5}$. $\therefore \sin(\alpha + \beta) = \frac{5}{13} \times \frac{1}{5} + \frac{12}{13} \times (-\frac{\sqrt{24}}{5}) = \frac{5-24\sqrt{6}}{65}$.

Example 5.8 : Find without using tables the value of $\sin 15^\circ$.

Solution: $\sin 15^\circ = \sin(45^\circ - 30^\circ) = \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ$

$$= \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{2}}{4} (\sqrt{3} - 1).$$

Exercise

1. Use the addition formulae to prove the following identities: (i) $\sin(180^\circ + \theta) = -\sin \theta$ (ii) $\tan(180^\circ + \theta) = -\tan \theta$. (iii) $\sin(\frac{\pi}{2} - \theta) = \cos \theta$ (iv) $\cos(300^\circ - \theta) = \frac{\cos \theta - \sqrt{3} \sin \theta}{2}$. (v) $\tan(225^\circ - \theta) = \frac{1 - \tan \theta}{1 + \tan \theta}$ (vi) $\cos 3x \cos 2x + \sin 3x \sin 2x = \cos x$.
2. Find without using tables the value of $\tan 15^\circ$.
3. Show that $\sin^2 A + \sin^2(A + 120^\circ) + \sin^2(A - 120^\circ) = \frac{3}{2}$.
4. If $\sin \theta = \frac{2}{3}$ and $\cos \phi = -\frac{2}{7}$, find the possible values of $\cos(\theta + \phi)$.

5.4.8 Multiple Angles

By putting $\beta = \alpha$ in the addition formulae, we obtain $\sin(\alpha + \alpha) = \sin 2\alpha = 2 \sin \alpha \cos \alpha$,

$$\cos(\alpha + \alpha) = \cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha = 2 \cos^2 \alpha - 1 = 1 - \sin^2 \alpha.$$

$$\therefore \tan(\alpha + \alpha) = \tan 2\alpha = \frac{\tan \alpha + \tan \alpha}{1 - \tan \alpha \tan \alpha} = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}.$$

Using the above formulae, we can express $\tan \alpha$ in terms of $\tan \frac{\alpha}{2}$ thus,

$$\tan \alpha = \tan 2\left(\frac{\alpha}{2}\right) = \frac{\tan \frac{\alpha}{2} + \tan \frac{\alpha}{2}}{1 - \tan^2 \frac{\alpha}{2}} = \frac{2 \tan \frac{\alpha}{2}}{1 - \tan^2 \frac{\alpha}{2}}.$$

If we let $t = \tan \frac{\alpha}{2}$, then,

$$\tan \alpha = \frac{2t}{1 - t^2}$$

$$\text{Similarly, } \sin \alpha = 2 \sin \frac{\alpha}{2} \cos \frac{\alpha}{2} = \frac{2 \sin \frac{\alpha}{2} \cos \frac{\alpha}{2}}{\cos^2 \frac{\alpha}{2} + \sin^2 \frac{\alpha}{2}} = \frac{2 \tan \frac{\alpha}{2}}{1 + \tan^2 \frac{\alpha}{2}} = \frac{2t}{1 + t^2}.$$

$$\therefore \sin \alpha = \frac{2t}{1 + t^2}$$

$$\text{and } \cos \alpha = \cos^2 \frac{\alpha}{2} - \sin^2 \frac{\alpha}{2} = \frac{\cos^2 \frac{\alpha}{2} - \sin^2 \frac{\alpha}{2}}{\cos^2 \frac{\alpha}{2} + \sin^2 \frac{\alpha}{2}} = \frac{1 - \tan^2 \frac{\alpha}{2}}{1 + \tan^2 \frac{\alpha}{2}} = \frac{1 - t^2}{1 + t^2}.$$

Hence

$$\cos \alpha = \frac{1 - t^2}{1 + t^2}.$$

$$\text{Since } \cos \alpha = \cos^2 \frac{\alpha}{2} - \sin^2 \frac{\alpha}{2} = 2 \cos^2 \frac{\alpha}{2} - 1,$$

$$\Rightarrow 2 \cos^2 \frac{\alpha}{2} = 1 + \cos \alpha$$

$$\therefore \cos \frac{\alpha}{2} = \sqrt{\frac{1 + \cos \alpha}{2}}.$$

$$\text{Similarly, } \cos \alpha = 1 - 2 \sin^2 \frac{\alpha}{2},$$

$$\therefore \sin \frac{\alpha}{2} = \sqrt{\frac{1 - \cos \alpha}{2}}$$

Also,

$$\tan \frac{\alpha}{2} = \frac{\sin \frac{\alpha}{2}}{\cos \frac{\alpha}{2}} = \frac{\sqrt{\frac{1 - \cos \alpha}{2}}}{\sqrt{\frac{1 + \cos \alpha}{2}}} = \sqrt{\frac{1 - \cos \alpha}{1 + \cos \alpha}}.$$

$$\text{From the cosine rule, } \cos A = \frac{b^2 + c^2 - a^2}{2bc}, \Rightarrow \cos A = 2 \cos^2 \frac{A}{2} - 1 = \frac{b^2 + c^2 - a^2}{2bc}.$$

$$\Rightarrow 2 \cos^2 \frac{A}{2} = \frac{(b+c)^2 - a^2}{2bc} = \frac{(a+b+c)(b+c-a)}{2bc}.$$

Let $a + b + c = 2s$, where s is the semi-perimeter of triangle ABC , i.e $s = \frac{a+b+c}{2}$.

$$\therefore 2 \cos^2 \frac{A}{2} = \frac{4s(s-a)}{2bc} = \frac{2s(s-a)}{bc}.$$

$$\therefore \cos \frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}.$$

In the same vein,

$$\cos \frac{B}{2} = \sqrt{\frac{s(s-b)}{ac}}, \cos \frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}}.$$

$$\text{Also, } \cos A = 1 - 2 \sin^2 \frac{A}{2} = \frac{b^2 + c^2 - a^2}{2bc}.$$

$$\therefore 2 \sin \frac{A}{2} = \frac{(a+b-c)(a-b+c)}{2bc} = \frac{4(s-c)(s-b)}{2bc} = \frac{2(s-b)(s-c)}{bc}.$$

$$\therefore \sin \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}.$$

Hence,

$$\sin \frac{B}{2} = \sqrt{\frac{(s-a)(s-c)}{ac}}, \sin \frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{ab}}.$$

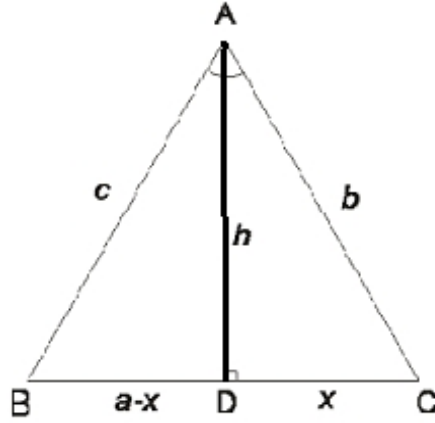
$$\text{Similarly, } \tan \frac{A}{2} = \frac{\sin \frac{A}{2}}{\cos \frac{A}{2}} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}.$$

$$\therefore \tan \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}, \tan \frac{B}{2} = \sqrt{\frac{(s-a)(s-c)}{s(s-b)}}, \tan \frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{s(s-c)}}.$$

5.4.9 Sine and Cosine Rules

5.4.10 Sine rule

The sine rule states that in any triangle ABC ,



(26)

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

Consider a triangle ABC , and construct a perpendicular from A to BC .

Construct a perpendicular AD to meet BC at D . From the figure above $\sin B = \frac{h}{c} \Rightarrow h = c \sin B$, and similarly, $\sin C = \frac{h}{b} \Rightarrow h = b \sin C$.

$$\therefore h = c \sin B = b \sin C, \Rightarrow \frac{b}{\sin B} = \frac{c}{\sin C}.$$

Similarly $\frac{a}{\sin A} = \frac{c}{\sin C}$.

$$\therefore \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

The formula also holds for obtuse angled triangles.

5.4.11 Cosine Rule:

The cosine rule states that in any triangle ABC ,

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C.$$

Proof: from the triangle above,

$$c^2 = (a - x)^2 + h^2 = a^2 - 2ax + x^2 + h^2, \text{ by Pythagoras' rule.}$$

$$\therefore c^2 = a^2 - 2ax + b^2, \text{ since } b^2 = h^2 + x^2.$$

$$\text{But } x = b \cos C, \Rightarrow c^2 = a^2 + b^2 - 2ab \cos C.$$

In the same vein, $a^2 = b^2 + c^2 - 2bc \cos A$, and $b^2 = a^2 + c^2 - 2ac \cos B$.

$$\therefore a^2 = b^2 + c^2 - 2bc \cos A, \quad b^2 = a^2 + c^2 - 2ac \cos B, \quad c^2 = a^2 + b^2 - 2ab \cos C.$$

Example 5.9 : Given that in triangle ABC , $a = 4, b = 5, c = 6$. Find the angles A, B , and C .

Solution: Using cosine rule, $4^2 = 5^2 + 6^2 - 2 \times 5 \times 6 \cos A \Rightarrow \cos A = 0.75, \therefore A = \cos^{-1} 0.75 \simeq 41.4^\circ$.

Also, $5^2 = 4^2 + 6^2 - 2 \times 4 \times 6 \cos B \Rightarrow \cos B = 0.5625 \Rightarrow B = \cos^{-1} 0.5625 \simeq 55.8^\circ$.

$$\cos A = \frac{c^2 + b^2 - a^2}{2bc} = \frac{36 + 25 - 16}{60} = \frac{45}{60} = 0.75, \Rightarrow A = \cos^{-1}(0.75) = 41.41^\circ$$

Similarly,

$$\begin{aligned} \cos B &= \frac{c^2 + a^2 - b^2}{2ac} = \frac{36 + 16 - 25}{48} = \frac{27}{48} \\ \Rightarrow B &= \cos^{-1}\left(\frac{27}{48}\right) = \cos^{-1}\left(\frac{9}{16}\right) = 55.77^\circ \end{aligned}$$

Also,

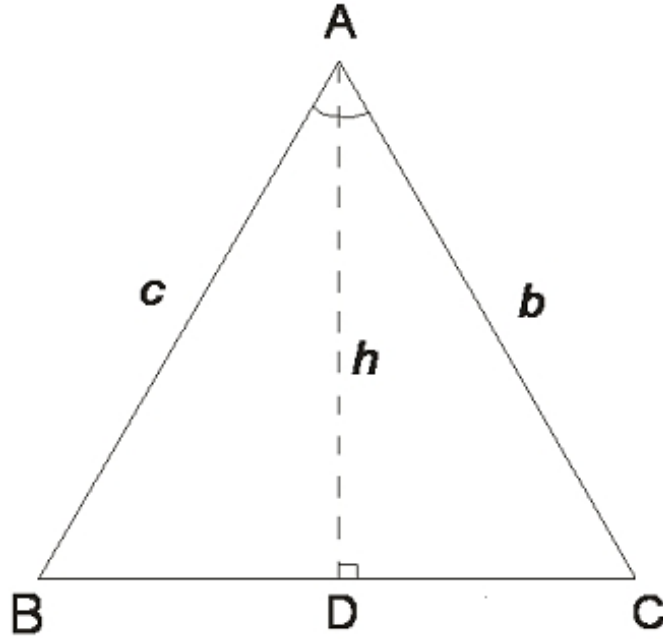
$$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{16 + 25 - 36}{40} = \frac{5}{40}, \Rightarrow C = \cos^{-1}\left(\frac{5}{40}\right) = 82.8^\circ.$$

Exercise 15

1. Solve each of the following triangles, given that (i) $a = 12, b = 10, c = 8$ (ii) $A = 123^\circ, B = 62^\circ, c = 0.34$ (iii) $a = 26.35, A = 37^\circ 23', B = 43^\circ 31'$.
2. Show that $a^2 + bc = b^2 + c^2$, if $A = 60^\circ$.
3. Prove that $a^2 + b^2 + c^2 = 2bc \cos A + 2ac \cos B + 2ab \cos C$.

5.4.12 The Area of a Triangle

Consider the triangle ABC , with height, h , $\sin C = \frac{h}{b} \Rightarrow h = b \sin C$.



(27)

$$\therefore \text{Area, } A = \frac{1}{2}ab \sin C = \frac{1}{2}ab \cdot 2 \sin \frac{C}{2} \cos \frac{C}{2} = ab \sin \frac{C}{2} \cos \frac{C}{2}.$$

$$= ab \sqrt{\frac{(s-a)(s-b)}{ab}} \times \sqrt{\frac{s(s-c)}{ab}}.$$

$$\therefore A = \sqrt{s(s-a)(s-b)(s-c)}.$$

The formula above is usually referred to as **Hero's formula**.

Example 5.10 : Express $\tan 3A$ in terms of $\tan A$.

$$\tan A = \tan(2A + A) = \frac{\tan 2A + \tan A}{1 - \tan 2A \tan A} = \frac{\frac{2 \tan A}{1 - \tan^2 A} + \tan A}{1 - \frac{2 \tan A}{1 - \tan^2 A} \tan A} = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}.$$

Example 5.11 : Show that $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$.

$$\begin{aligned} \text{Solution: } \cos 3\theta &= \cos(2\theta + \theta) = \cos 2\theta \cos \theta - \sin 2\theta \sin \theta = (2 \cos^2 \theta - 1) \cos \theta - 2 \sin \theta \cos \theta \sin \theta \\ &= 2 \cos^3 \theta - \cos \theta - 2 \cos \theta \sin^2 \theta = 2 \cos^3 \theta - \cos \theta - 2 \cos \theta (1 - \cos^2 \theta) = 4 \cos^3 \theta - 3 \cos \theta. \end{aligned}$$

Exercise 16

- Express (i) $\sin 4\theta$ in terms of $\sin \theta$, (ii) $\tan 4\theta$ in terms of $\tan \theta$.

2. Show that $\sin \frac{\pi}{12} \cos \frac{5\pi}{12} = \frac{2-\sqrt{3}}{4}$.
3. Given that $\sin \frac{\pi}{10} = \frac{1}{4}(\sqrt{5} - 1)$, find the exact value of $\cos \frac{\pi}{5}$.
4. Find without using tables the values of (i) $\cos 22\frac{1}{2}^\circ$ (ii) $\tan 22\frac{1}{2}^\circ$.
5. Prove that $\frac{1+\sin \phi - \cos \phi}{1+\sin \phi + \cos \phi} = \tan \frac{\phi}{2}$.
6. Without using tables, determine the value of $\tan 7\frac{1}{2}^\circ$.
7. Using the formulae $\sin 3\beta = 3 \sin \beta - 4 \sin^3 \beta$, and $\cos 3\beta = 4 \cos^3 \beta - 3 \cos \beta$, express $\cos 5\beta$ in terms of $\cos \beta$. Hence, prove that $\cos 18^\circ = \frac{1}{4}\sqrt{(10 + 2\sqrt{5})}$, and evaluate $\cos 54^\circ$, leaving your result in surd form.
8. Find the area of the triangle with the following dimensions: (i) $a = 623\text{cm}, b = 913\text{cm}, A = 34^\circ$ (ii) $A = 57.9, b = 126.3, c = 137.2$.
9. The sides of a triangle are $13k, 20k$, and $21k$ units, and the area of the triangle is 14 square units. Find the numerical value of k .

5.4.13 Factor Formulae

Recall $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$,

and $\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$.

Adding corresponding members of these equations, we obtain

$$\sin(\alpha + \beta) + \sin(\alpha - \beta) = 2 \sin \alpha \cos \beta,$$

$$\Rightarrow \sin \alpha \cos \beta = \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)].$$

Similarly, $\cos(\alpha + \beta) + \cos(\alpha - \beta) = 2 \cos \alpha \cos \beta$,

$$\Rightarrow \cos \alpha \cos \beta = \frac{1}{2} [\cos(\alpha + \beta) + \cos(\alpha - \beta)].$$

By subtracting the expressions, we obtain

$$\sin(\alpha + \beta) - \sin(\alpha - \beta) = 2 \cos \alpha \sin \beta,$$

$$\Rightarrow \cos \alpha \sin \beta = \frac{1}{2} [\sin(\alpha + \beta) - \sin(\alpha - \beta)].$$

and $\cos(\alpha + \beta) - \cos(\alpha - \beta) = -2 \sin \alpha \sin \beta$.

$$\Rightarrow \sin \alpha \sin \beta = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)].$$

If in these identities, we put $\alpha + \beta = A$ and $\alpha - \beta = B$, then

$\alpha = \frac{1}{2}(A + B)$, and $\beta = \frac{1}{2}(A - B)$, we obtain

$$\sin A + \sin B = 2 \sin \frac{1}{2}(A + B) \cos \frac{1}{2}(A - B), \quad \cos A + \cos B = 2 \cos \frac{1}{2}(A + B) \cos \frac{1}{2}(A - B).$$

and

$$\sin A - \sin B = 2 \cos \frac{1}{2}(A + B) \sin \frac{1}{2}(A - B), \quad \cos A - \cos B = -2 \sin \frac{1}{2}(A + B) \sin \frac{1}{2}(A - B).$$

Example 5.12 :

- Express these quantities as products: (i) $\cos 50^\circ + \cos 12^\circ$ (ii) $\sin 40^\circ + \sin 20^\circ$ (iii) $\sin 152^\circ - \sin 30^\circ$. (iv) $\cos 2\theta + \cos 4\theta$.
- Prove that $\cos A + 2 \cos 3A + \cos 5A = 4 \cos^2 A \cos 3A$.
- If A, B and C are the angles of triangle ABC , show that $\cos A + \cos B + \cos C = 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$.
- Prove that $\frac{\sin 4\omega - \sin 2\omega}{\cos 4\omega + \cos 2\omega} = \tan \omega$.

Solution:

- (i) $\cos 50^\circ + \cos 12^\circ = 2 \cos \frac{1}{2}(50 + 12) \cos \frac{1}{2}(50 - 12) = 2 \cos 31^\circ \cos 19^\circ$. (ii) $\sin 40^\circ + \sin 20^\circ = 2 \sin \frac{1}{2}(40^\circ + 20^\circ) \cos \frac{1}{2}(40^\circ - 20^\circ) = 2 \sin 30^\circ \cos 10^\circ$. (iii) $\sin 152^\circ - \sin 30^\circ = 2 \cos \frac{1}{2}(152^\circ + 30^\circ) \sin \frac{1}{2}(152^\circ - 30^\circ) = 2 \cos 91^\circ \sin 61^\circ$. (iv) $\cos 2\theta + \cos 4\theta = 2 \cos \frac{1}{2}(2\theta + 4\theta) \cos \frac{1}{2}(2\theta - 4\theta) = 2 \cos 3\theta \cos \theta$.
- $LHS = (\cos A + 2 \cos 3A) + (\cos 3A + \cos 5A) = 2 \cos \frac{1}{2}(A + 3A) \cos \frac{1}{2}(A - 3A) + 2 \cos \frac{1}{2}(3A + 5A) \cos \frac{1}{2}(3A - 5A) = 2 \cos 2A \cos(-A) + 2 \cos 4A \cos(-A) = 2 \cos 2A \cos A + 2 \cos 4A \cos A = 2 \cos A (\cos 2A + \cos 4A) = 2 \cos A [2 \cos \frac{1}{2}(2A + 4A) \cos \frac{1}{2}(2A - 4A)] = 2 \cos A (2 \cos 3A \cos(-A)) = 4 \cos^2 A \cos 3A = RHS$.
- $\cos A + \cos B + \cos C = \cos A + 2 \cos \frac{1}{2}(B + C) \cos \frac{1}{2}(B - C) = 1 - 2 \sin^2 \frac{A}{2} + 2 \sin \frac{A}{2} \cos \frac{1}{2}(B - C) = 1 + 2 \sin \frac{A}{2} [\cos \frac{1}{2}(B - C) - \sin \frac{A}{2}] = 1 + 2 \sin \frac{A}{2} [\cos \frac{1}{2}(B - C) - \cos \frac{1}{2}(B + C)] = 1 + 2 \sin \frac{A}{2} (2 \sin \frac{B}{2} \sin \frac{C}{2}) = 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$.
- $\frac{\sin 4\omega - \sin 2\omega}{\cos 4\omega + \cos 2\omega} = \frac{2 \cos \frac{1}{2}(4\omega + 2\omega) \sin \frac{1}{2}(4\omega - 2\omega)}{2 \cos \frac{1}{2}(4\omega + 2\omega) \cos \frac{1}{2}(4\omega - 2\omega)} = \frac{2 \cos 3\omega \sin \omega}{2 \cos 3\omega \cos \omega} = \frac{\sin \omega}{\cos \omega} = \tan \omega$

Exercise 17

- Express as a product: (i) $\sin 40^\circ + \sin 20^\circ$ (ii) $\cos 50^\circ + \cos 30^\circ$ (iii) $\cos 8\theta - \cos 2\theta$.
- Express as sums or differences: (i) $\sin 30^\circ \cos 50^\circ$ (ii) $\cos 3x \cos 4x$ (iii) $\sin 4x \cos 2x$.
- Prove that (i) $\frac{\sin 6x - \sin 4x}{\cos 5x} = 2 \sin x$ (ii) $\frac{\sin \theta - \sin 2\theta + \sin 3\theta}{\cos \theta - \cos 2\theta + \cos 3\theta} = \tan 2\theta$.
- If A, B and C are angles of a triangle, show that $\sin A + \sin B + \sin C = 4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$.
- Prove that $\frac{\cos \alpha + \cos \beta}{\sin \alpha - \sin \beta} = \cot \frac{\alpha - \beta}{2}$.

5.4.14 Trigonometric Equations

To solve any equation involving trigonometric functions, appropriate trigonometric identities may be employed to simplify such equations before solutions are obtained. We illustrate this by using the following examples:

Example 5.13 : Solve the following equations for values of θ between 0° and 360° :
(i) $\sin \theta = \cos 55^\circ$ (ii) $2 \sin^2 \theta + 2 \cos 2\theta = 1$

Solution: (i) $\sin \theta = \cos 55^\circ \Rightarrow \theta = \sin^{-1}(\cos 55^\circ) = 35^\circ$

$$(ii) 2 \sin^2 \theta + 2 \cos 2\theta = 1 \Rightarrow 2 \sin^2 \theta + 2(1 - 2 \sin^2 \theta) = 1$$

$$\Rightarrow 2 - 2 \sin^2 \theta = 1, \therefore \sin^2 \theta = \frac{1}{2}. \text{Hence, } \sin \theta = \pm \frac{1}{\sqrt{2}}.$$

$$\therefore \sin \theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ, 135^\circ.$$

$$\text{and } \sin \theta = -\frac{1}{\sqrt{2}} \Rightarrow \theta = 225^\circ, 315^\circ.$$

$$\therefore \text{the solution is } \theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ.$$

5.4.15 The General Solution of $\sin \theta = c$

If $\sin \theta = c$, with c a constant such that $-1 \leq c \leq 1$, we see that there is always a solution of the equation between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$. Such a solution is usually referred to as the principal solution. Suppose the principal solution is α , then the general expression for the solution of the equation is given by $\theta = k\pi + (-1)^{k+1}\alpha$, where $k \in Z$, is called the general solution of the equation $\sin \theta = c$.

Example 5.14 : Find the general solution of the equation $\cos 2x + 3 \sin x + 1 = 0$.

Solution: $\cos^2 x - \sin^2 x + 3 \sin x + 1 = 2 - 2 \sin^2 x + 3 \sin x$, $\therefore 2 + 3 \sin x - 2 \sin^2 x = 0$,
 $\Rightarrow 2 + 4 \sin x - \sin x - 2 \sin x = 0$, $\therefore (2 - \sin x)(1 + 2 \sin x) = 0$
 $\therefore 2 - \sin x = 0$ or $1 + 2 \sin x = 0$, i.e $\sin x = 2$ or $\sin x = -\frac{1}{2}$.

But $\sin x = 2$ is impossible (why)?

$\therefore \sin x = -\frac{1}{2} \Rightarrow x = -\frac{\pi}{6}$ is the principal solution.

$\therefore$ the general solution is $x = k\pi + (-1)^{k+1}(\frac{\pi}{6})$.

Similarly, the general solution of $\cos \theta = c$, $-1 \leq c \leq 1$, is given by $\theta = 2n\pi \pm \alpha$, $n \in Z$ and $0 < \alpha < \pi$ is the principal solution.

For the equation $\tan \theta = c$, the principal solution α lies between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$ and the general solution is given by $\theta = n\pi \pm \alpha$, $n \in Z$.

Example 5.15 : Find the general solution of the equation $2 \cos^2 \theta - 3 \cos \theta - 2 = 0$.

Solution: $\therefore (\cos \theta - 2)(2 \cos \theta + 1) = 0 \Rightarrow \cos \theta = -\frac{1}{2}$ or $\cos \theta = 2$ (which is impossible!)
 $\therefore$ the principal solution is $\theta = \frac{2\pi}{3}$. Thus the general solution is $\theta = 2n\pi \pm \frac{2\pi}{3}$.

Example 5.16 : Find the general solution of the equation $3 \tan^2 x - 1 = 0$.

Solution: If $\tan x = \frac{1}{\sqrt{3}}$, the principal solution is $x = \frac{\pi}{6}$ and if $\tan x = -\frac{1}{\sqrt{3}}$, the principal solution is $x = -\frac{\pi}{6}$. Hence the general solution is $x = n\pi \pm \frac{\pi}{6}$.

5.4.16 The Function $y = a \sin x + b \cos x$

The expression $a \sin x + b \cos x$, where a and b are constants can be written as $R \sin(x + \alpha) \equiv R \sin x \cos \alpha + R \cos x \sin \alpha$,

provided that $R \cos \alpha = a$ and $R \sin \alpha = b$

$\therefore (R \cos \alpha)^2 + (R \sin \alpha)^2 = a^2 + b^2 \Rightarrow R^2 = a^2 + b^2$. $\therefore R = \sqrt{a^2 + b^2}$.

$\therefore y = a \sin x + b \cos x \equiv R \sin(x + \alpha)$, has a maximum value R , when $x + \alpha = \frac{\pi}{2}$,

and a minimum value $-R$, when $x + \alpha = \frac{3\pi}{2}$.

5.4.17 The Equation $a \sin x + b \cos x = c$

To solve the equation $a \sin x + b \cos x = c$, where a , b and c are constants, we express $a \sin x + b \cos x \equiv R \sin(x \pm \alpha)$ or $a \sin x + b \cos x \equiv R \cos(x \pm \alpha)$, where $R > 0$ and $0 \leq \alpha \leq 2\pi$, R and α are to be determined.

For example, $a \sin x + b \cos x \equiv R \sin(x - \alpha) = R \sin x \cos \alpha - R \cos x \sin \alpha$,

where $R \cos \alpha = a$ and $R \sin \alpha = b$ such that $R = \sqrt{a^2 + b^2}$, and $\tan \alpha = \frac{b}{a}$.

Hence the original equation becomes $R \sin(x - \alpha) = c$.

The signs of a and b determine the quadrant in which α lies. For instance, if $a < 0$ and $b > 0$, α lies in the second quadrant.

Example 5.17 : Solve the equation $4 \cos x + 3 \sin x = 2$, for values of x between 0° and 360° .

Solution: $\therefore R \cos \alpha = 4$ and $R \sin \alpha = 3. \therefore R = \sqrt{4^2 + 3^2} = \sqrt{16 + 9} = \sqrt{25} = 5$

and $\tan \alpha = \frac{3}{4} \Rightarrow \alpha = \tan^{-1} \frac{3}{4} \simeq 36.9^\circ, \therefore$ the equation becomes $5 \cos(x - 36.9^\circ) = 2 \Rightarrow \cos(x - 36.9^\circ) = 0.4$.

$\therefore x - 36.9^\circ = 66.4^\circ, 293.6^\circ \Rightarrow x = 103.3^\circ, 330.5^\circ$.

Example 5.18 : Express the following functions in the form $A \sin(x + \alpha)$, and hence find the maximum and minimum values of each: (a) $\sin x + \cos x$ (b) $4 \sin x - 3 \cos x$.

Solution: $\therefore A \sin \alpha = 1$, and $A \cos \alpha = 1, \Rightarrow A^2 = 2$, hence, $A = \sqrt{2}$, and $\tan \alpha = 1, \Rightarrow \alpha = \tan^{-1} 1 = \frac{\pi}{4}$.

(a) $\sin x + \cos x \equiv \sqrt{2} \sin(x + \frac{\pi}{4})$. Hence the maximum value is $\sqrt{2}$, and it is attained when $x + \frac{\pi}{4} = \frac{\pi}{2}$, i.e $x = \frac{\pi}{4}$, while a minimum value of $-\sqrt{2}$ is attained when $x + \frac{\pi}{4} = \frac{3\pi}{2}$, i.e $x = \frac{5\pi}{4}$.

(b) $4 \sin x - 3 \cos x \equiv A \sin(x - \alpha) = A \sin x \cos \alpha - A \cos x \sin \alpha$.

$\therefore A \sin \alpha = 4$, and $A \cos \alpha = -3, \therefore A = 5$, and $\tan \alpha = -0.75, \Rightarrow \alpha = \tan^{-1} -0.75 \simeq 143.1^\circ$.

$\Rightarrow 4 \sin x - 3 \cos x \equiv 5 \sin(x - 143.1^\circ)$. $\therefore$ maximum value is 5, and minimum value -5 .

Example 5.19 : Solve the equation $4 \sin x - 3 \cos x = 2$ by means of the substitution $t = \tan \frac{x}{2}$.

Solution: Substituting for $\sin x$ and $\cos x$ in the equation, we obtain $4(\frac{2t}{1+t^2}) - 3(\frac{1-t^2}{1+t^2}) = 2$

$\Rightarrow 8t - 3(1 - t^2) = 2(1 + t^2), \Rightarrow 8t - 3 + 3t^2 = 2 + 2t^2$.

$\therefore t^2 + 8t - 5 = 0, \Rightarrow t = \frac{-8 \pm \sqrt{64+20}}{2} = \frac{-8 \pm \sqrt{84}}{2} = \frac{-8 \pm 9.165}{2}$.

$\therefore t = 0.5825$ or $t = -8.5825$.

Thus, $\tan \frac{x}{2} = -8.5825, \Rightarrow \frac{x}{2} = \tan^{-1} -8.5825 = 96.65^\circ, 276.65^\circ$,

$\Rightarrow x = 193.3^\circ, 553.3^\circ$.

or $\tan \frac{x}{2} = 0.5825, \Rightarrow \frac{x}{2} = \tan^{-1} 0.5825 = 30.22^\circ, 210.22^\circ$,

$\Rightarrow x = 60.44^\circ, 420.44^\circ. \therefore t = 60.44^\circ, 193.3^\circ, 420.44^\circ, 553.3^\circ$.

5.4.18 Equations Involving Factor Formulae

Example 5.20 : Solve the equation $\sin 3x + \sin 5x = \sin 4x, 0 \leq x \leq 2\pi$.

Solution: $\sin 3x + \sin 5x = 2 \sin \frac{1}{2}(3x + 5x) \cos \frac{1}{2}(3x - 5x) = \sin 4x$

$$\therefore 2 \sin 4x \cos x = \sin 4x, \Rightarrow 2 \sin 4x \cos x - \sin 4x = 0.$$

$$\Rightarrow \sin 4x(2 \cos x - 1) = 0, \Rightarrow \sin 4x = 0 \text{ or } 2 \cos x - 1 = 0,$$

$$\therefore 4x = \pi, 2\pi, 3\pi, 4\pi, 5\pi, 6\pi, 7\pi, 8\pi, \Rightarrow x = \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi, \frac{5\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4}, 2\pi$$

$$\text{or } \cos x = 0.5, \Rightarrow x = \cos^{-1} 0.5 = \frac{\pi}{3}, \frac{5\pi}{3}.$$

Hence the complete solution is $x = \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi, \frac{5\pi}{4}, \frac{5\pi}{3}, \frac{3\pi}{2}, \frac{7\pi}{4}, 2\pi$.

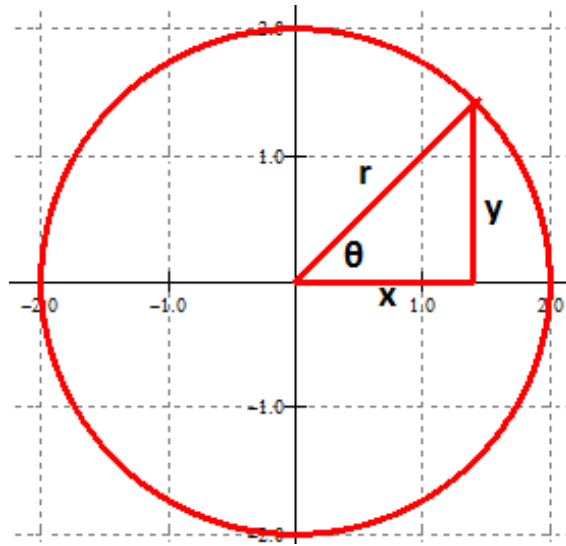
Exercise 18

1. Solve the following equations for values of θ between 0° and 360° (i) $\cos \theta = -\frac{1}{2}$ (ii) $\sin \theta = \frac{\sqrt{2}}{2}$ (iii) $\sin^2 \theta - \sin \theta - 2 = 0$ (iv) $4 \sin^2 \theta \tan \theta - \tan \theta = 0$ (v) $3 \tan^2 \theta - 5 \sec \theta + 1 = 0$.
2. Find the general solution of each of the following equations: (i) $\sin x = \frac{\sqrt{3}}{2}$ (ii) $3 \tan^2 x = 1$ (iii) $2 \sin^2 x + 3 \cos x = 0$ (iv) $\sin 5\theta + \sin 3\theta + \cos \theta = 0$ (v) $\cos \theta - \sin 2\theta = \cos 5\theta$.
3. Express each of the following in the form $R \sin(x + \alpha)$, hence find its maximum and minimum values: (i) $\sin x + \cos x$ (ii) $2 \sin x + 3 \sin x$ (iii) $4 \sin x - 3 \cos x$ (iv) $2 \cos x + 3 \sin x$.
4. Find the values of x between 0 and 2π for which (i) $\sin x + \cos x = 1$ (ii) $2 \cos x + 3 \sin x = 3$ (iii) $\sin x - \sqrt{3} \cos x = 1$.
5. Find all the angles between 0° and 360° inclusive for which $\sin^2 \theta + 2 \cos 2\theta = 2 \cos \theta$.
6. Prove that $\sin x + \sin 2x + \cos x + \cos 2x = 2\sqrt{2} \cos \frac{1}{2}x \sin(\frac{3x}{2} + \frac{\pi}{4})$. Hence, or otherwise, determine all values of x between 0 and 2π for which $\sin x + \sin 2x + \cos x + \cos 2x = 0$.
7. Solve the equation $\frac{\tan 2\theta + \tan \theta}{1 - \tan 2\theta \tan \theta} = \sqrt{3}$, for $0 \leq \theta \leq 2\pi$.
8. Solve by means of the substitution $t = \tan \frac{x}{2}$, the equation $2 \sin x + 3 \cos x = -1$, for $0 \leq x \leq 2\pi$.

5.4.19 Parametric Functions in Trigonometry

A function of the form $X = x(t)$; $Y = y(t)$ is said to be a parametric function with t as a parameter.

Consider the diagram below:



(28)

From the diagram,

$$\sin \theta = \frac{y}{r}, \Rightarrow y = r \sin \theta, \text{ and } \cos \theta = \frac{x}{r}, \Rightarrow x = r \cos \theta$$

Note that $x^2 + y^2 = r^2 \sin^2 \theta + r^2 \cos^2 \theta = r^2(\sin^2 \theta + \cos^2 \theta) = r^2$.

$\therefore x^2 + y^2 = r^2$, which is the equation of a circle of radius r with centre at the origin. Observe that a cartesian equation can be obtained from a parametric equation by eliminating the parameter.

Example 5.21 : (a) Eliminate t from the equations $x = 2 + 3 \cos t$; $y = 1 + 3 \sin t$. (b) Eliminate θ from the equations $x = \cos \theta + \sin \theta$; $y = 1 - \cos \theta$ (c) If $x = \tan \theta - \sin \theta$; $y = \tan \theta + \sin \theta$, prove that $(x^2 - y^2)^2 = 16xy$.

Solution (i) $\frac{x-2}{3} = \cos t$; $\frac{y-1}{3} = \sin t \Rightarrow (x-2)^2 + (y-1)^2 = 9$

(ii) Students to solve this.

(iii) $x^2 - y^2 = (\tan \theta - \sin \theta + \tan \theta + \sin \theta)(\tan \theta - \sin \theta - \tan \theta - \sin \theta) = (2 \tan \theta)(-2 \sin \theta)$.

$$\therefore x^2 - y^2 = -4 \tan \theta \sin \theta$$

$$\Rightarrow (x^2 - y^2)^2 = (-4 \tan \theta \sin \theta)^2 = 16 \tan^2 \theta \sin^2 \theta = 16 [\tan^2 \theta (1 - \cos^2 \theta)]$$

$$= 16(\tan^2 \theta - \sin^2 \theta) = 16(\tan \theta + \sin \theta)(\tan \theta - \sin \theta) = 16xy.$$

$$\therefore (x^2 - y^2)^2 = 16xy.$$

Exercise 19

1. Eliminate θ from each of the following equations: (i) $x = \sin \theta$, $y = \cos \theta$. (ii) $x = \cos \theta + \sin \theta$, $y = 1 - \cos \theta$ (iii) $x = \sin \theta$, $y = \tan 2\theta$ (iv) $x = \cos 2\theta$, $y = \cos \theta$.

2. If $x = 2 \cos \theta - \cos 2\theta$ and $y = 2 \sin \theta - \sin 2\theta$, show that $(x^2 + y^2)^2 - 6(x^2 + y^2) + 8x - 3 = 0$.
3. Express as a cartesian equation the pair of parametric equations $x = a \cos^3 t$, $y = b \sin^3 t$.
4. If $u = x \cos \alpha + y \sin \alpha$, and $v = x \sin \alpha - y \cos \alpha$, Show that $u^2 + v^2 = x^2 + y^2$.

6 Complex Numbers

6.1 Learning Objectives

In this module we:

- Introduce the complex numbers as an extension to the real numbers.
- Examine properties of complex numbers and discuss different ways of representing them.
- Discuss De Moivre's theorem and the roots of complex numbers

6.2 Learning Outcomes

At the end of this module students should be able to:

- Manipulate complex numbers, be aware of the similarities with real numbers and the differences between them.
- Identify situations in which it is advantageous to use the algebraic or geometric representation of complex numbers.
- Use De Moivre's theorem to establish trigonometric identities and explore the complex roots of numbers.

6.3 Learning Activities

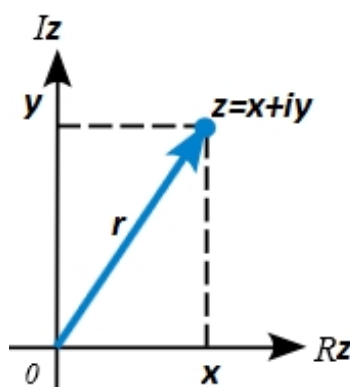
Students should:

- Explore notes, power point presentations and exercises, individually and in groups.
- Engage in discussions with lecturers and their peers using forums and other interactive facilities on the Moodle site.
- Take and pass online and other self assessment tests.
- Solve relevant questions from past MTH111 Examinations.

6.4 Introduction to complex numbers

Complex numbers originated from the inability to find real roots or solutions to quadratic equations of the form $ax^2 + bx + c = 0$ for which the discriminant $b^2 - 4ac < 0$.

Definition 6.1 : ***Complex Numbers:** Let x and y be two real numbers, then the ordered pair (x, y) of real numbers is called a complex number. If z denotes a complex number (x, y) , then we can write $z = (x, y)$, where the real numbers x and y are called the real and imaginary parts respectively of z , i.e. $\Re z = x$ and $\Im z = y$.*



(29)

For example, the real and imaginary parts of the complex number $z = (3, 4)$ are 3 and 4 respectively. If the real part of a complex number is zero, i.e. $z = (0, y)$ then the complex number z is said to be purely imaginary. For instance, $(0, 5)$ is purely imaginary. Suppose the imaginary part of a complex number z is zero, then we can write $z = (x, 0)$ as x , i.e. $z =$

$(x, 0) = x$. We say such a complex number is real. Hence the real numbers are a subset of the complex numbers.

Equality of complex numbers: Given two complex numbers $z_1 = (x_1, y_1)$ and $z_2 = (x_2, y_2)$ then $z_1 = z_2$ if and only if their component parts are equal, i.e. iff $x_1 = x_2$ and $y_1 = y_2$.

Sum and product of complex numbers

Let $z_1 = (x_1, y_1)$ and $z_2 = (x_2, y_2)$ be two complex numbers. Then their sum is defined by $z_1 + z_2 = (x_1 + x_2, y_1 + y_2)$

The product of z_1 and z_2 is given by $z_1 z_2 = (x_1 x_2 - y_1 y_2, x_1 y_2 + x_2 y_1)$.

For instance, if $z_1 = (3, 4)$ and $z_2 = (-5, 2)$ then $z_1 + z_2 = (3 + (-5), 4 + 2) = (-2, 6)$ while $z_1 z_2 = (3 \times (-5) - 4 \times 2, 3 \times 2 + (-5) \times 4) = (-7, -14)$

Let i denotes the purely imaginary number $(0, 1)$ and $x = (x, 0)$ then using the definition of sum and product above, we can write a complex number $z = (x, y) = (x, 0) + (0, 1)y$ in Cartesian form as $z = x + iy$ or $z = x + yi$ where $i = (0, 1)$ with the property that $i^2 = (-1, 0) = -1$. With i so defined, we can also write the sum and product of two complex numbers as $z_1 + z_2 = (x_1 + iy_1) + (x_2 + iy_2) = x_1 + x_2 + i(y_1 + y_2)$ and $z_1 z_2 = x_1 x_2 - y_1 y_2 + i(x_1 y_2 + x_2 y_1)$ respectively. The properties of the sum and product of complex numbers are the same as for real numbers.

The **commutative laws for sum and product are respectively**, $z_1 + z_2 = z_2 + z_1$, and $z_1 z_2 = z_2 z_1$;

The **associative laws are given by**: $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$, and $(z_1 z_2) z_3 = z_1 (z_2 z_3)$. These properties follow easily from addition and multiplication of real numbers.

Thus if $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, then $z_1 + z_2 = (x_1 + iy_1) + (x_2 + iy_2) = x_1 + x_2 + i(y_1 + y_2) = x_2 + x_1 + i(y_2 + y_1) = (x_2 + iy_2) + (x_1 + iy_1) = z_2 + z_1$.

Similarly, $z_1 z_2 = (x_1 + iy_1)(x_2 + iy_2) = x_1 x_2 - y_1 y_2 + i(x_1 y_2 + x_2 y_1) = x_2 x_1 - y_2 y_1 + i(x_2 y_1 + x_1 y_2) = (x_2 + iy_2)(x_1 + iy_1) = z_2 z_1$. ative

The proof of the distributive law $z_1 (z_2 + z_3) = z_1 z_2 + z_1 z_3$ is left as an exercise.

The numbers $0 = 0 + i0$ and $1 = 1 + i0$ are respectively called the additive and multiplicative identities of the complex numbers. That is for any complex number z , then $z + 0 = z$ and $z \times 1 = z$. Note that 0 and 1 are the only numbers with such properties.

Subtraction: Given any complex number $z = x + iy$, we define $-z = -x - iy$ as its additive inverse which satisfies the equation $z + (-z) = 0$. Moreover, for a given complex number z , there is only one additive inverse namely $-z$. Additive inverses are used to define subtraction: $z_1 - z_2 = z_1 + (-z_2)$. So if $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ then $z_1 - z_2 = x_1 - x_2 + i(y_1 - y_2)$.

Complex Conjugate: Given a complex number $z = x + iy$ then the complex number $\bar{z} = x - iy$ is called the complex conjugate of z . For example, if $z = 3 + 4i$ then its complex conjugate is $\bar{z} = 3 - 4i$. This number $\bar{z}$ represented by the point $(x, -y)$, is the reflection of the point (x, y) about the real axis. We can easily verify that for any complex number z , $\overline{\bar{z}} = z$.

We note that although z and $\bar{z}$ are complex, it is always true that $z \cdot \bar{z} = z \times \bar{z}$ is always real. In fact it is given by $z \cdot \bar{z} = x^2 + y^2$ if $z = x + iy$. We call $\sqrt{z \cdot \bar{z}} = \sqrt{x^2 + y^2}$ the modulus of z and it is denoted by $|z|$.

For any nonzero complex number $z = x + iy$, there is a number z^{-1} such that $z z^{-1} = 1$. This number z^{-1} is called the multiplicative inverse of z . To find the inverse $w = u + iv$ of $z = x + iy$, we need to solve the equation $(x + iy)(u + iv) = 1$ for u and v in terms of x and y . This gives $u = \frac{x}{x^2 + y^2}$ and $v = \frac{-y}{x^2 + y^2}$. Therefore the multiplicative inverse of z is

$$w = u + iv = \frac{x - iy}{x^2 + y^2}. \text{ We can get the inverse by using the complex conjugate as follows:}$$

$$\frac{1}{z} = \frac{1}{z \cdot \bar{z}} = \frac{\bar{z}}{|z|^2} = \frac{x - iy}{x^2 + y^2}.$$

Division: Given two complex numbers z_1 and $z_2 \neq 0$ then the quotient $\frac{z_1}{z_2}$ is a complex number defined by $\frac{z_1}{z_2} = z_1 z_2^{-1} = \frac{x_1 x_2 + y_1 y_2 + i(y_1 x_2 - y_2 x_1)}{x_2^2 + y_2^2} = \frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2} + i \frac{y_1 x_2 - y_2 x_1}{x_2^2 + y_2^2}$, where $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, $z_2 \neq 0$. This quotient is not defined when $z_2 = 0$; as $z_2 = 0$ means that $x_2^2 + y_2^2 = 0$.

Example 6.1 :

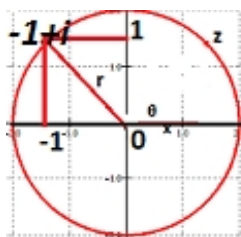
1. Evaluate the following sums: (i) $(3 + i) + (1 + 2i)$ (ii) $(-5 + 3i) + (1 - i)$.
2. Perform the following differences (i) $(2 - 3i) - (1 + 2i)$ (ii) $(-5 + 3i) - (3 + i)$.
3. Find the following products (i) $(2 - 3i)(4 + 5i)$ (ii) $(3 - 4i)(3 + 4i)$, hence evaluate (iii) $(p + iq)(p - iq)$.
4. Simplify the following quotients (i) $\frac{3i-2}{1+2i}$ (ii) $(\frac{1}{2-3i})(\frac{1}{1+i})$ (iii) $\frac{\cos \theta + i \sin \theta}{\cos \phi - i \sin \phi}$.

Solution:

1. (i) $(3 + i) + (1 + 2i) = 3 + 1 + i(1 + 2) = 4 + 3i$ and (ii) $(-5 + 3i) + (1 - i) = -5 + 1 + 3i - i = -4 + 2i$.
2. (i) $(2 - 3i) - (1 + 2i) = 2 - 1 - 3i - 2i = 1 - 5i$. and (ii) $(-5 + 3i) - (3 + i) = -5 - 3 + 3i - i = -8 + 2i$.
3. (i) $(2 - 3i)(4 + 5i) = 2 \cdot 4 + 2 \cdot 5i - 3i \cdot 4 - 3i \cdot 5i = 8 + 10i - 12i - 15i^2 = 8 + 15 - 2i = 23 - 2i$
(ii) $(3 - 4i)(3 + 4i) = 3 \cdot 3 + 3 \cdot 4i - 4i \cdot 3 - 4i \cdot 4i = 9 + 12i - 12i - 16i^2 = 9 + 16 = 25$ (iii)
From the second example we can conclude that $(p + iq)(p - iq) = p^2 + q^2$.
4. (i) $\frac{3i-2}{1+2i} = \frac{3i-2}{1+2i} \cdot \frac{1-2i}{1-2i} = \frac{3i \cdot 1 + 3i(-2i) - 2 \cdot 1 - 2(-2i)}{1 \cdot 1 + 1(-2i) + 2i \cdot 1 + 2i(-2i)} = \frac{3i - 6i^2 - 2 + 4i}{1 - 2i + 2i - 4i^2} = \frac{3i + 6 - 2 + 4i}{1 + 4} = \frac{4 + 7i}{5} = \frac{4}{5} + \frac{7i}{5}$.
(ii) $(\frac{1}{2-3i})(\frac{1}{1+i}) = \frac{1 \cdot 1}{2 \cdot 1 + 2 \cdot i - 3i \cdot 1 - 3i \cdot i} = \frac{1}{2 + 2i - 3i - 3i^2} = \frac{1}{2 - i + 3} = \frac{1}{5 - i} = \frac{1 \cdot 5 + i}{5 - i(5 + i)} = \frac{5 + i}{5 \cdot 5 + 5 \cdot i - i \cdot 5 - i \cdot i} = \frac{5 + i}{25 + 5i - 5i + 1} = \frac{5 + i}{26} = \frac{5}{26} + \frac{i}{26}$.
(iii) $\frac{\cos \theta + i \sin \theta}{\cos \phi - i \sin \phi} = \frac{\cos \theta + i \sin \theta}{\cos \phi - i \sin \phi} \cdot \frac{\cos \phi + i \sin \phi}{\cos \phi + i \sin \phi} = \frac{\cos \theta \cos \phi + i \cos \theta \sin \phi + i \sin \theta \cos \phi + i^2 \sin \theta \sin \phi}{\cos \phi \cos \phi + i \cos \phi \sin \phi - i \sin \phi \cos \phi - i^2 \sin \phi \sin \phi}$

$$\begin{aligned}
&= \frac{\cos \theta \cos \phi + i(\cos \theta \sin \phi + \sin \theta \cos \phi) - \sin \theta \sin \phi}{\cos \phi \cos \phi + i \cos \phi \sin \phi - i \sin \phi \cos \phi + \sin \phi \sin \phi} = \frac{\cos \theta \cos \phi - \sin \theta \sin \phi + i(\cos \theta \sin \phi + \sin \theta \cos \phi)}{\cos^2 \phi + \sin^2 \phi} = \frac{\cos(\theta - \phi) + i \sin(\theta - \phi)}{1} \\
&= \cos(\theta - \phi) + i \sin(\theta - \phi). \text{ This follows from the fact that } \cos(\theta - \phi) = \cos \theta \cos \phi + \sin \theta \sin \phi \\
&\text{and } \sin(\theta - \phi) = \sin \theta \cos \phi - \cos \theta \sin \phi.
\end{aligned}$$

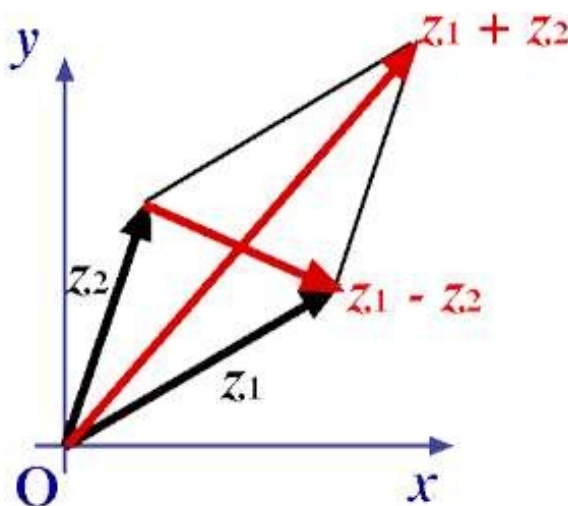
It is natural to associate the complex number $z = x + iy$ with a point in the plane whose cartesian coordinates is (x, y) . Each complex number corresponds to just one point, and conversely. The number $-1 + i$, for instance, is represented by the point $(-1, 1)$ in the figure below.



(30)

The number z can also be thought of as a directed line segment, or vector, from the origin to the point (x, y) . In fact, we often refer to a complex number z as the point z or the vector z . When used for the purpose of displaying the numbers $z = x + iy$ geometrically, the xy -plane is called the complex plane or the z -plane. The x -axis is called the real axis, and the y -axis is known as the imaginary axis.

Given any two complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ the sum $z_1 + z_2$ corresponds to the point $(x_1 + x_2, y_1 + y_2)$. It also corresponds to a vector with those coordinates as its components. Hence $z_1 + z_2$ may be obtained vectorially as shown in the figure below.



(31)

The difference $z_1 - z_2 = z_1 + (-z_2)$ corresponds to the sum of the vectors for z_1 and $-z_2$ (See the figure). Note that the number $z_1 - z_2$ can be interpreted as the directed line segment from the point (x_2, y_2) to the point (x_1, y_1) just as we can do with vectors.

Although the product of two complex numbers z_1 and z_2 is itself a complex number represented by a vector, that vector lies in the same plane as the vectors for z_1 and z_2 . Evidently, then, this product is neither the scalar nor the vector product used in ordinary vector analysis. We shall discuss the geometrical interpretation of the product of z_1 and z_2 later.

Absolute Value: Given a complex number $z = x + iy$, its modulus or magnitude or absolute value $|z|$ is defined as the nonnegative real number $|z| = \sqrt{x^2 + y^2} = (x^2 + y^2)^{\frac{1}{2}}$. Geometrically, the number $|z|$ is the distance between the point (x, y) and the origin $(0, 0)$ or the length of the vector representing z . It reduces to the usual absolute value in the real number system when $y = 0$. It should be noted that there is no ordering in the complex number system i.e. it is meaningless to say $z_2 < z_1$ unless both z_1 and z_2 are real. But the statement $|z_2| < |z_1|$ means that the point z_2 is closer to the origin than the point z_1 .

1. Find the modulus of (i) $3 - 4i$ (ii) $-1 + i$ (iii) $1 - i\sqrt{3}$.
2. Which of these points $-3 + i2$ and $1 + 4i$ is closer to the origin?

Solution:

1. (i) $|z| = |x + iy| = (x^2 + y^2)^{\frac{1}{2}} = (3^2 + (-4)^2)^{\frac{1}{2}} = (9 + 16)^{\frac{1}{2}} = (25)^{\frac{1}{2}} = 5$. (ii) $|z| = |x + iy| = (x^2 + y^2)^{\frac{1}{2}} = ((-1)^2 + 1^2)^{\frac{1}{2}} = \sqrt{1 + 1} = \sqrt{2}$. (iii) $|z| = |x + iy| = (x^2 + y^2)^{\frac{1}{2}} = [1^2 + (\sqrt{-3})^2]^{\frac{1}{2}} = \sqrt{1 + 3} = \sqrt{4} = 2$
- 2 $|z_1| = |x + iy| = (x_1^2 + y_1^2)^{\frac{1}{2}} = \sqrt{(-3)^2 + 2^2} = \sqrt{9 + 4} = \sqrt{13}$ and $|z_2| = |x_2 + iy_2| = (x_2^2 + y_2^2)^{\frac{1}{2}} = \sqrt{1^2 + 4^2} = \sqrt{1 + 16} = \sqrt{17}$

Therefore z_1 is closer to the origin than z_2 is.

The distance between two points represented by the complex numbers

$z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ is given by $|z_1 - z_2| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = |z_2 - z_1|$. The complex numbers z corresponding to the points lying on the circle with centre z_0 and radius r thus satisfy the equation $|z - z_0| = r$, or $|z - z_0| = \sqrt{(x - x_0)^2 + (y - y_0)^2} = r$ and conversely.

Example 6.2 : Show that the equation $|z - 1 + 3i| = 2$ represents a circle.

Solution: Let $z = x + iy$, then $|z - 1 + 3i| = 2 \implies |x + iy - 1 + 3i| = |(x - 1) + (y + 3)i| = \sqrt{(x - 1)^2 + (y + 3)^2} = 2$

$\implies (x - 1)^2 + (y + 3)^2 = 2^2 \implies (x - 1)^2 + (y + 3)^2 = 4$. This last expression is the equation of a circle centred at $(1, -3)$ with radius of 2 units.

Recall that if $z = x + iy$ then $x = \Re z$ and $y = \Im z$ and we can from the definition of absolute value write $|z|^2 = x^2 + y^2 = (\Re z)^2 + (\Im z)^2$ and also state that $\Re z \leq |\Re z| \leq |z|$ and

$$\Im z \leq |\Im z| \leq |z|, \text{ and } |\bar{z}| = |z|.$$

Example 6.3 : Given two complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ we can show that (i) $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$; the conjugate of a sum is equal to the sum of the conjugates.

(ii) $\overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2$; the conjugate of a difference is equal to the difference of the conjugates. (iii) $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$; the conjugate of a product is equal to the product of the conjugates. (iv) $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}$, $z_2 \neq 0$; the conjugate of a quotient is equal to the quotient of the conjugates.

Solution: (i) $\overline{z_1 + z_2} = \overline{(x_1 + iy_1) + (x_2 + iy_2)} = \overline{x_1 + x_2 + i(y_1 + y_2)} = x_1 + x_2 - i(y_1 + y_2) = x_1 + x_2 - iy_1 - iy_2 = x_1 - iy_1 + x_2 - iy_2 = \bar{z}_1 + \bar{z}_2$

(ii)- (iv) are left as exercises.

The sum $z + \bar{z}$ of a complex number z and its conjugate $\bar{z}$ is the real number $2x = 2\Re z$.

The difference $z - \bar{z}$ is the purely imaginary number $2iy = 2i\Im z$.

Hence we have the two identities: $\Re z = \frac{z + \bar{z}}{2}$ and $\Im z = \frac{z - \bar{z}}{2i}$.

An important identity relating the conjugate of a complex number $z = x + iy$ to its modulus is $z\bar{z} = |z|^2 = x^2 + y^2$. This provides another way of simplifying the quotient $\frac{z_1}{z_2}$. The procedure is to rationalise the denominator by multiplying both numerator and denominator by the conjugate $\bar{z}_2$ of the denominator z_2 so that the denominator becomes the real number $|z_2|^2 = x_2^2 + y_2^2$.

Example 6.4 : Simplify the quotient $\frac{-1+3i}{2-i}$.

Solution: $\frac{-1+3i}{2-i} = \frac{(-1+3i)(2+i)}{(2-i)(2+i)} = \frac{-5+5i}{|2-i|^2} = \frac{-5+5i}{5} = -1 + i$

We shall now state and prove some of the following results:

- (i) $|z_1 z_2| = |z_1| |z_2|$ i.e. absolute value of product is equal to product of absolute values.
- (ii) $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$, $z_2 \neq 0$ i.e. absolute value of quotient is equal to quotient of absolute values.
- (iii) $|z_1 + z_2| \leq |z_1| + |z_2|$. This is called the triangular inequality.

Proof: (i) $|z_1 z_2|^2 = (z_1 z_2)(\overline{z_1 z_2})$ ($\because |z|^2 = z\bar{z}$)
 $= (z_1 z_2)(\bar{z}_1 \bar{z}_2) = z_1 z_2 \bar{z}_1 \bar{z}_2$ ($\because \overline{\bar{z}_1 z_2} = \bar{z}_1 \bar{z}_2$)
 $= z_1 \bar{z}_1 z_2 \bar{z}_2 = |z_1|^2 |z_2|^2$ [$\because |z|^2 = z\bar{z}$].

Hence $|z_1 z_2|^2 = |z_1|^2 |z_2|^2$. Taking square roots of both sides yields the results: $|z_1 z_2| = |z_1| |z_2|$.

(ii) We follow the steps used in (i) above to verify the second result.

$$\begin{aligned}
\text{(iii) } |z_1 + z_2|^2 &= (z_1 + z_2)(\overline{z_1 + z_2}) \quad (\because |z|^2 = z\bar{z}) \\
&= (z_1 + z_2)(\bar{z}_1 + \bar{z}_2) \quad (\because \overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2) \\
&= z_1\bar{z}_1 + z_1\bar{z}_2 + z_2\bar{z}_1 + z_2\bar{z}_2 \\
&= |z_1|^2 + 2\Re(z_1\bar{z}_2) + |z_2|^2 \quad (\because z_1\bar{z}_2 + z_2\bar{z}_1 = 2\Re(z_1\bar{z}_2) = 2\Re(\bar{z}_1 z_2)) \\
&\leq |z_1|^2 + 2|\Re(z_1\bar{z}_2)| + |z_2|^2 \quad (\because \Re z_1 \leq |\Re z_1|) \\
&\leq |z_1|^2 + 2|z_1\bar{z}_2| + |z_2|^2 \quad (\because |\Re z_1| \leq |z_1|) \\
&\leq |z_1|^2 + 2|z_1||\bar{z}_2| + |z_2|^2 \quad (\because |z_1 z_2| = |z_1||z_2|) \\
&\leq |z_1|^2 + 2|z_1||z_2| + |z_2|^2 \quad (\because |z| = |\bar{z}|)
\end{aligned}$$

$$\text{i.e. } |z_1 + z_2|^2 \leq (|z_1| + |z_2|)^2$$

Taking the square root of both sides to obtain the moduli (nonnegative), we have the result:
 $|z_1 + z_2| \leq |z_1| + |z_2|$.

We can generalise the triangular inequality to any finite number of terms. That is, one can write that

$$|z_1 + z_2 + \dots + z_n| \leq (|z_1| + |z_2| + \dots + |z_n|) \quad \text{or} \quad \left| \sum_{i=1}^n z_i \right| \leq \sum_{i=1}^n |z_i|.$$

This can be verified by means of mathematical induction.

6.4.1 Polar representation of complex numbers

Let (r, θ) be the polar coordinates of a point (x, y) corresponding to the complex number $z = x + iy$. Since $x = r \cos \theta$ and $y = r \sin \theta$ then z can be written as $z = r(\cos \theta + i \sin \theta)$. For instance, $1 + i = \sqrt{2}(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}) = \sqrt{2}(\cos \frac{-7\pi}{4} + i \sin \frac{-7\pi}{4})$. The number or angle θ is called the argument or amplitude of the complex number $z = x + iy$ and it is given by the formula $\theta = \arctan\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{y}{x}\right)$.

Geometrically, $\arg z$ is the angle (measured in radians) that z makes with the positive real axis when z is interpreted as a directed line segment from the origin. The argument is not unique since if θ is $\arg z$ so is $\theta + 2\pi n$, $n = 0, \pm 1, \pm 2, \dots$. Thus for any nonzero complex number z , the value of $\arg z$ that lies in the range $-\pi < \theta \leq \pi$ is called the principal value denoted by $\text{Arg } z$. This principal value is unique.

If $z = 0$, θ is not defined and so, we shall always assume that the complex number z is nonzero when using polar form.

If $z \neq z_0$, the representation $z - z_0 = \rho(\cos \phi + i \sin \phi)$ of $z - z_0$ in polar form can be interpreted geometrically as below.

$\rho = |z - z_0|$ is the distance between the points z and z_0 while $\arg(z - z_0) = \phi$ is the angle of inclination of the vector representing $z - z_0$.

6.4.2 Identities involving Arguments

Theorem 5 Given two complex numbers z_1 and z_2 , then (i) $\arg(z_1 z_2) = \arg z_1 + \arg z_2$ and (ii) $\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2$ provided $z_2 \neq 0$.

Proof: (i) Let $z_1 = r_1 (\cos \theta_1 + i \sin \theta_1)$ and $z_2 = r_2 (\cos \theta_2 + i \sin \theta_2)$ then $\arg z_1 = \theta_1$, and $\arg z_2 = \theta_2$ so $z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$ which implies that $\arg(z_1 z_2) = \theta_1 + \theta_2 = \arg z_1 + \arg z_2$. Hence the result. Note that these statements about the argument is not true for the principal value. For example, if $z_1 = -1$ and $z_2 = i$ then $\arg z_1 = \pi$, and $\arg z_2 = \frac{\pi}{2}$ but $\arg z_1 z_2 = -\frac{\pi}{2}$ and not $\frac{3\pi}{2}$.

(ii) Let $z_1 = r_1 (\cos \theta_1 + i \sin \theta_1)$ and $z_2 = r_2 (\cos \theta_2 + i \sin \theta_2)$ then $\frac{z_1}{z_2} = \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]$ which shows that $\arg\left(\frac{z_1}{z_2}\right) = \theta_1 - \theta_2 = \arg z_1 - \arg z_2$, concluding the proof.

6.4.3 Exponential form of complex numbers

It is sometimes more convenient to express $e^{i\theta}$ as $\cos \theta + i \sin \theta$. This is known as Euler's (pronounced Oiler's) Formula, i.e. $e^{i\theta} = \cos \theta + i \sin \theta$.

If $z_1 = e^{i\theta_1}$ and $z_2 = e^{i\theta_2}$ then using Euler's Formula we can show that $z_1 z_2 = e^{i\theta_1} e^{i\theta_2} = e^{i(\theta_1 + \theta_2)}$. Similarly we can also show that $\frac{z_1}{z_2} = \frac{e^{i\theta_1}}{e^{i\theta_2}} = e^{i(\theta_1 - \theta_2)}$. This is equivalent to the case of e^x for real x . Thus if $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$ then $z_1 z_2 = r_1 r_2 e^{i\theta_1} e^{i\theta_2} = r_1 r_2 e^{i(\theta_1 + \theta_2)}$.

6.4.4 Powers and roots of unity of complex numbers

The integral powers of a nonzero complex number $z = r e^{i\theta}$ is given by $z^n = r^n e^{in\theta}$, which is valid for all $n = 0, \pm 1, \pm 2, \dots$. For $r = 1$, $(e^{i\theta})^n = e^{in\theta}$, $n = 0, \pm 1, \pm 2, \dots$ or $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$. This is known as De Moivre's theorem. The theorem is useful in the computation of the roots of nonzero complex numbers.

Example 6.5 : Solve the equation $z^n = 1$. This can be restated as find the n th roots of unity.

Solution: For non zero $z = r e^{i\theta}$, we look for r and θ such that $(r e^{i\theta})^n = 1$ or $r^n e^{in\theta} = 1 e^{i(0+2\pi k)}$. Hence $r^n = 1$ and $n\theta = 0 + 2\pi k$, $k = 0, \pm 1, \pm 2, \dots$. Consequently, $r = 1$, and $\theta = \frac{2\pi k}{n}$ and so we have n distinct solutions given by $c_k = e^{i\left(\frac{2\pi k}{n}\right)} = \cos \frac{2\pi k}{n} + i \sin \frac{2\pi k}{n}$, $k = 0, 1, \dots, n-1$. It should be noted that for values of k greater than $n-1$, the roots are repeated.

This fact demonstrate the Fundamental Theorem of Algebra which states that a polynomial equation of n degree has exactly n (roots) solutions. The complex numbers, $c_k = e^{i(\frac{2\pi k}{n})} = \cos \frac{2\pi k}{n} + i \sin \frac{2\pi k}{n}$, $k = 0, 1, \dots, n-1$ are the n th distinct roots of unity.

Geometrically, the n roots of unity represent the vertices of a regular polygon with n sides inscribed in a unit circle centred at the origin and one of the roots is $c_0 = 1$.

Remark 2 Note that if $c \neq 1$ is one of the root of unity, then it is easy to show that $1 + c + c^2 + c^3 + \dots + c^{n-1} = 0$ using the sum of n terms of a GP.

Example 6.6 : Solve the equation $z^3 = 1$ and discuss the roots.

Solution: $c_k = e^{i(\frac{2\pi k}{3})} = \cos \frac{2\pi k}{3} + i \sin \frac{2\pi k}{3}$ are the solutions of the given equation i.e. $c_0 = 1$, $c_2 = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$, $c_3 = \cos \frac{2\pi 2}{3} + i \sin \frac{2\pi 2}{3} = \cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$. The roots represent the vertices of an equilateral triangle inscribed in a unit circle centred at the origin. Using either c_2 or c_3 from the above given example, we can easily verify that $1 + c_2 + c_2^2 = 1 + c_3 + c_3^2 = 0$.

Exercise 22 Solve the equation $z^4 = 1$ and discuss the roots.

6.4.5 Roots of complex numbers other than unity

The concept of roots of unity can be used to find the roots of the nonzero complex number $z_0 = r_0 \exp i\theta_0$. The roots are obtained by solving the equation $z^n = z_0$ for z , and these roots are given by the numbers

$$c_k = \sqrt[n]{r_0} \exp \left[i \left(\frac{\theta_0}{n} + \frac{2k\pi}{n} \right) \right], \quad k = 0, 1, 2, \dots, n-1. (*)$$

where the number $\sqrt[n]{r_0}$ is the length of each radius vector representing the n roots. An argument of the first root, c_0 is $\frac{\theta_0}{n}$ and the other roots are obtained by adding integral multiples of $\frac{2\pi}{n}$. Consequently, as the case with the n th root of unity, the roots when $n = 2$, always lie at the opposite ends of a diameter of a circle, one root being the negative of the other; and when $n \geq 3$, they lie at the vertices of a regular polygon of n sides. If c is any particular n th root of z_0 , the set of all other n th roots can be written as $c, cw_n, cw_n^2, cw_n^3, \dots, cw_n^{n-1}$ where $w_n = \exp i(\frac{2\pi}{n})$. This is true because multiplication of any nonzero complex number by w_n corresponds to increasing the argument of that number by $\frac{2\pi}{n}$. We shall denote the set of n th roots of a nonzero complex number z_0 by $z_0^{\frac{1}{n}}$. If in particular z_0 is a positive real number r_0 , then the symbol $r_0^{\frac{1}{n}}$ denotes a set of roots while $\sqrt[n]{r_0}$ is reserved for the one positive root. When the value of θ_0 that is used in the expression for the roots, $c_k (*)$ is the principal value of the $\arg z_0$ ($-\pi < \theta \leq \pi$), then the number c_0 is often referred to as the principal n th root of z_0 . Thus, when z_0 is a positive real number r_0 , its principal root is $\sqrt[n]{r_0}$. If $z_0 = 0$, then $z^n = z_0$ has only one solution $z = 0$. Hence the only n th root of zero is zero.

It is sometimes more convenient to write z_0 in its general exponential form $z_0 = r_0 \exp [i (\theta_0 + 2k\pi)]$, $k = 0, \pm 1, \pm 2, \dots$ and then we write $z_0^{\frac{1}{n}} = \sqrt[n]{r_0} \exp [i (\frac{\theta_0 + 2k\pi}{n})]$, where $k = 0, 1, 2, \dots, n-1$. This formula is valid where $n = 1, 2, 3, \dots$

Example 6.7 : Find all the values (roots) of $(-8i)^{\frac{1}{3}}$ (cube roots of $-8i$)

Solution: $r_0 = |-8i| = \sqrt{(-8)^2} = \sqrt{64} = 8$, $\theta_0 = -\frac{\pi}{2}$ as is the case with all negative purely imaginary numbers. Therefore $-8i = 8 \exp [i (\frac{-\pi}{2} + 2k\pi)]$, where $k = 0, \pm 1, \pm 2, \dots$ so that the desired distinct roots are $c_k = 2 \exp [i (\frac{-\pi}{6} + \frac{2\pi k}{3})]$, where $k = 0, 1, 2$. In cartesian coordinates, the roots are

$$c_0 = 2 \exp [i (\frac{-\pi}{6})] = 2 (\cos (\frac{-\pi}{6}) + i \sin (\frac{-\pi}{6})) = \sqrt{3} - i,$$

$$c_1 = 2 \exp [i (\frac{-\pi}{6} + \frac{2\pi}{3})] = 2 \exp [i (\frac{3\pi}{6})] = 2 (\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}) = 2i,$$

and

$c_2 = 2 \exp [i (\frac{-\pi}{6} + \frac{2\pi 2}{3})] = 2 \exp [i (\frac{11\pi}{6})] = 2 (\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6}) = -\sqrt{3} - i$. These roots lie at the vertices of an equilateral triangle inscribed in a circle of radius 2 units centred at the origin. The principal root is $c_0 = \sqrt{3} - i$.